



SOICT

HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Multimodal Machine Learning

Lecture 3.1: Multimodal Representation Fusion

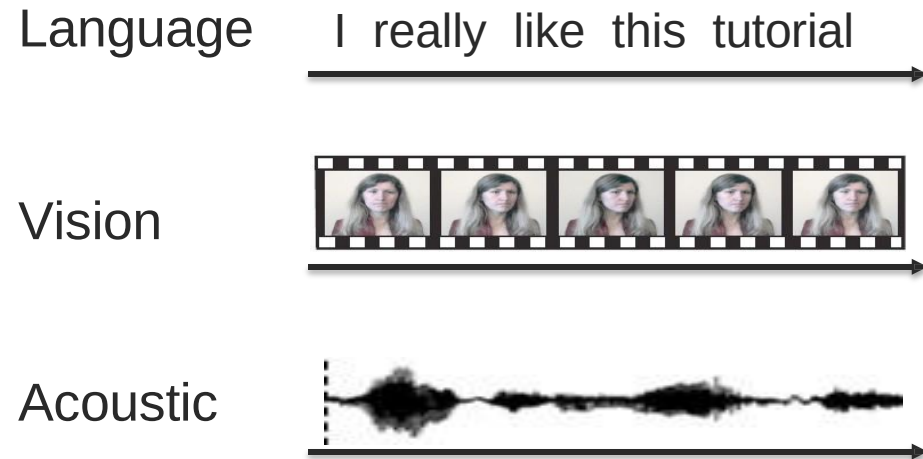
Dam Quang Tuan

Lecture Objectives

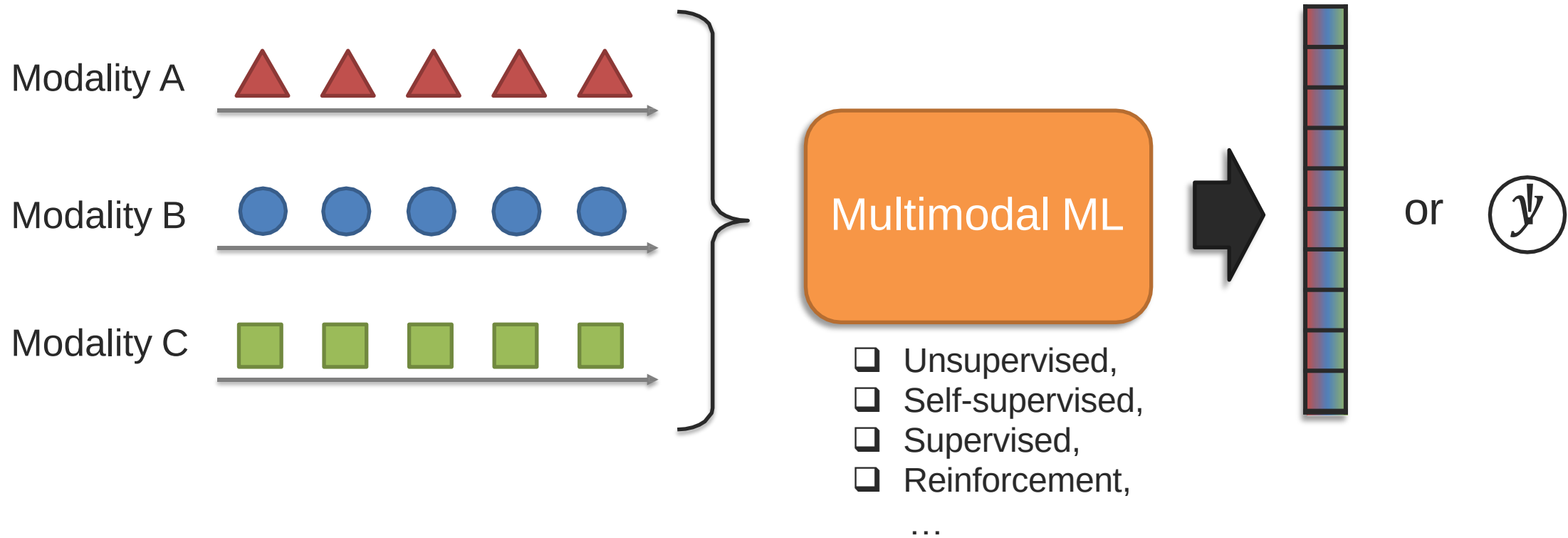
- Multimodal representations
 - Cross-modal interactions
- Representation fusion
 - Additive and multiplicative fusion
 - Tensor and polynomial fusion
 - Gated fusion
 - Modality-shift fusion
 - Dynamic fusion
 - Fusion on raw modalities
 - Heterogeneity-aware fusion
- Measuring non-additive interactions

Multimodal Representation

Multimodal Machine Learning



Multimodal Machine Learning

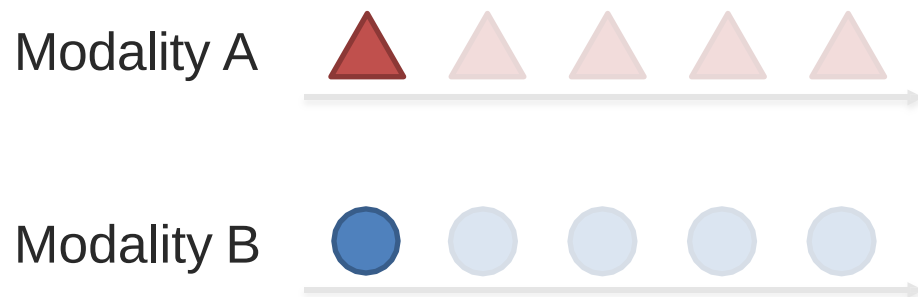


Challenge 1: Representation

Definition: Learning representations that reflect cross-modal interactions between individual elements, across different modalities

➡ This is a core building block for most multimodal modeling problems!

Individual elements:



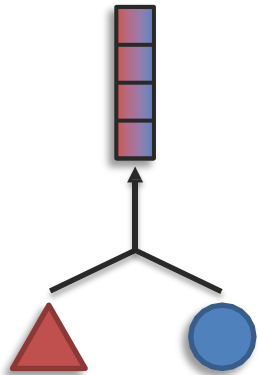
*It can be seen as a “local” representation
or
representation using holistic features*

Challenge 1: Representation

Definition: Learning representations that reflect cross-modal interactions between individual elements, across different modalities

Sub-challenges:

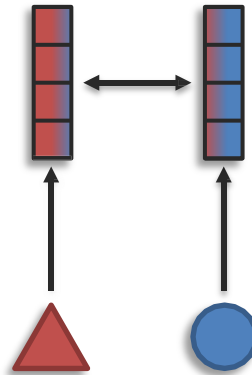
Fusion



modalities $>$ # representations

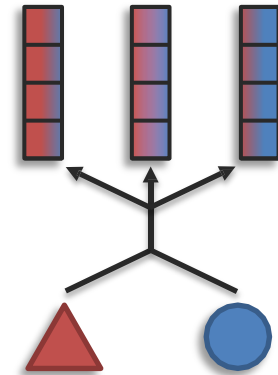
Today

Coordination



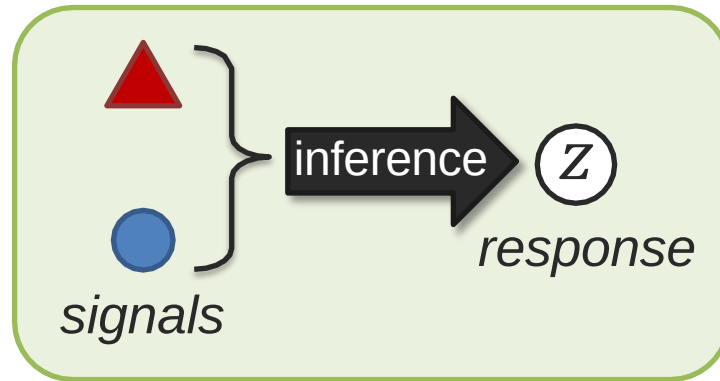
modalities $=$ # representations

Fission



modalities $<$ # representations

Cross-modal Interactions

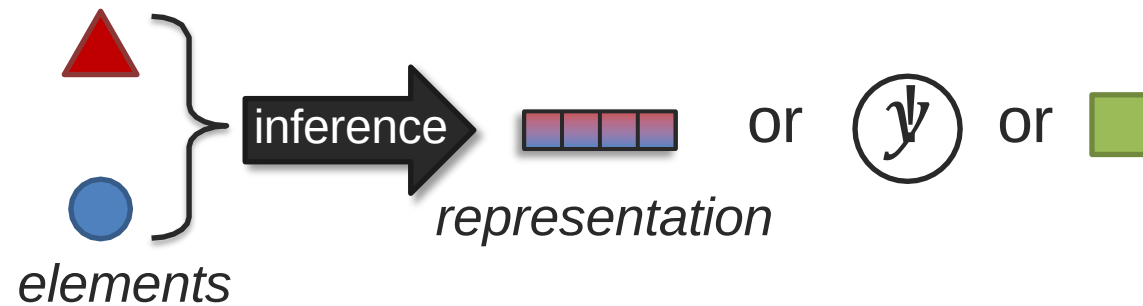


Interactions happen
during inference!



“Inference” examples:

- Representation fusion
- Prediction task
- Modality translation



Interacting Modalities

Interactions: How multimodal information changes when modalities are combined for a response.

*Is this
indoors?*



*A teacup on the right of a
laptop in a clean room.*

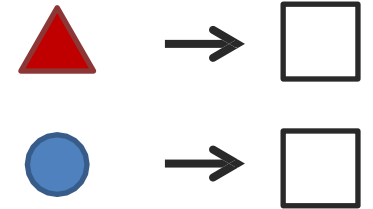


Yes!



Yes!

Redundant



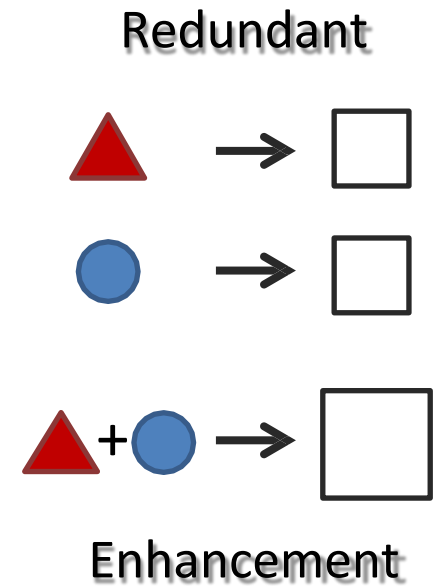
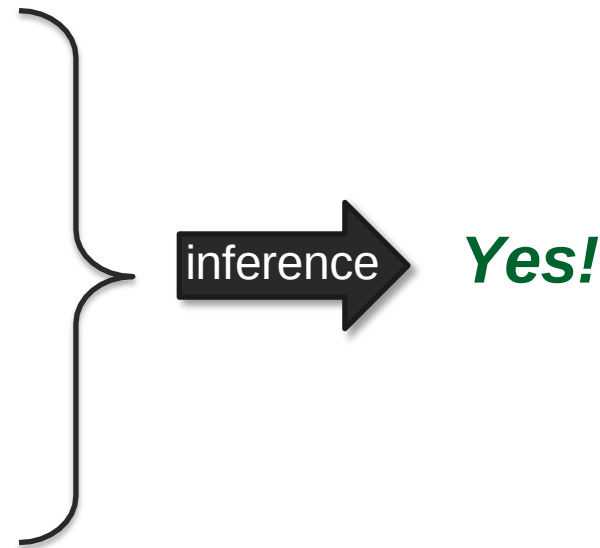
Interacting Modalities

Interactions: How multimodal information changes when modalities are combined for a response.

*Is this
indoors?*



*A teacup on the right of a
laptop in a clean room.*



Interacting Modalities

Interactions: How multimodal information changes when modalities are combined for a response.

*Is this a
living
room?*



*A teacup on the right of a
laptop in a clean room.*

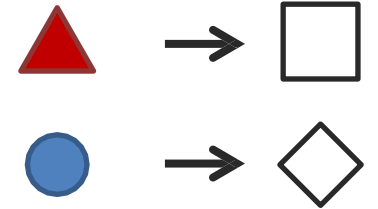


Yes!



*No, probably
study room.*

Non-redundant



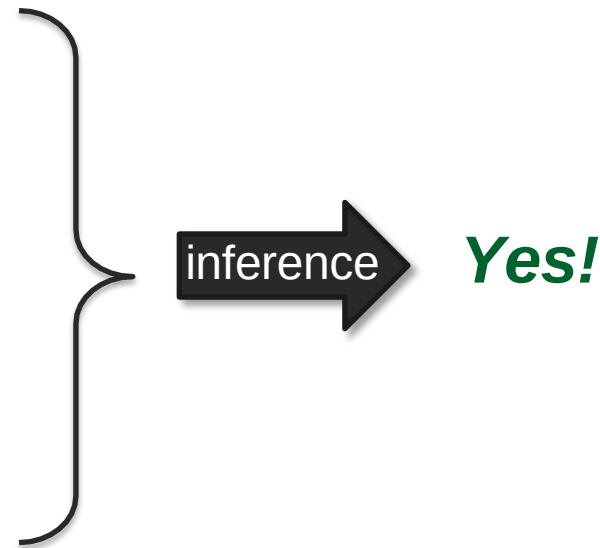
Interacting Modalities

Interactions: How multimodal information changes when modalities are combined for a response.

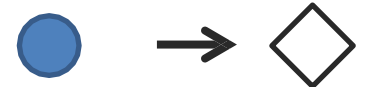
*Is this a
living
room?*



*A teacup on the right of a
laptop in a clean room.*



Non-redundant



Dominance

Interacting Modalities

Interactions: How multimodal information changes when modalities are combined for a response.

*Should I
work here?*



*A teacup on the right of a
laptop in a clean room.*

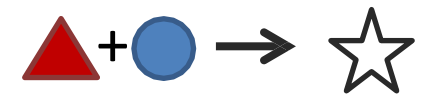
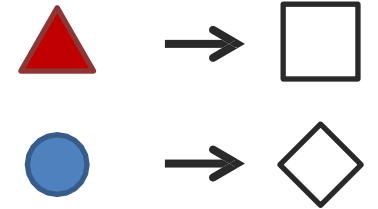


*Maybe? Comfy
sofa but table's
too small.*



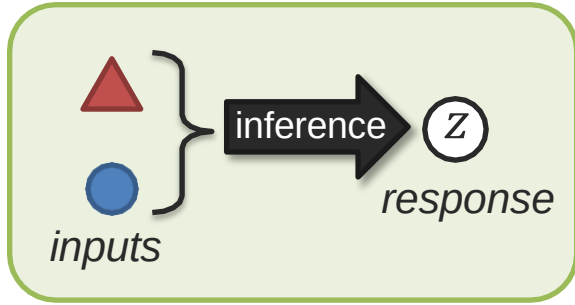
*Maybe? Clean
and there's tea.*

Non-redundant



Emergence

Taxonomy of Interaction Responses – A Behavioral Science View



Multimodal Communication



Redundancy

signal

response

a. → □

b. → □

signal

response

a+b → □

a+b → □

Equivalence

Enhancement

Nonredundancy

a → □

b → ○

a+b → □ and ○

a+b → □

a+b → □ (or □)

a+b → △

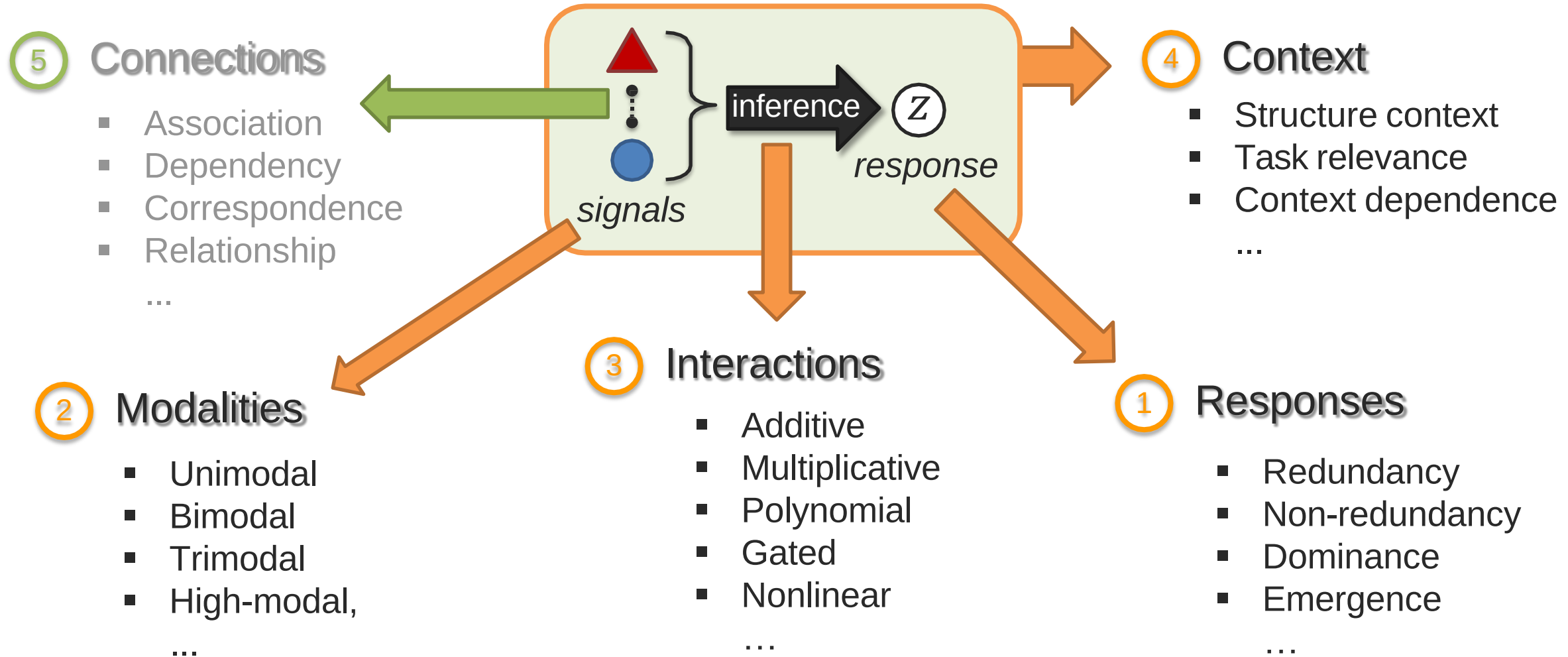
Independence

Dominance

Modulation

Emergence

Cross-modal Interactions – A Taxonomy



Cross-modal Interactions – Representation Fusion

Next week

5

Connections

- Association
- Dependency
- Correspondence
- Relationship

...

2

Modalities

- Unimodal
- Bimodal
- Trimodal
- High-modal,

...

3

Interactions

- Additive
- Multiplicative
- Polynomial
- Gated
- Nonlinear

...

Today

4

Context

- Structure context
- Task relevance
- Context dependence

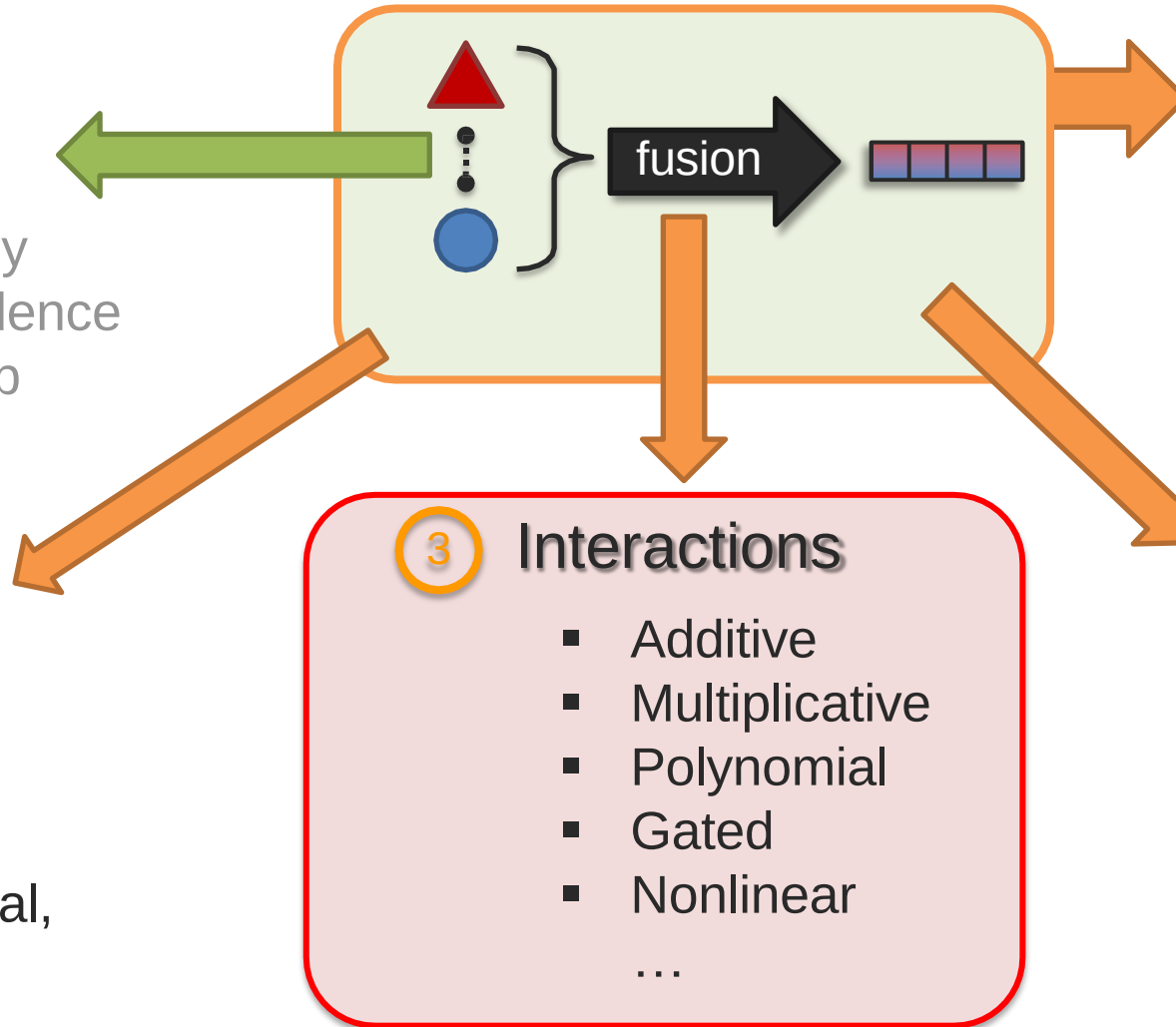
...

1

Responses

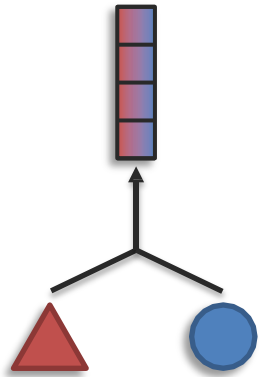
- Redundancy
- Non-redundancy
- Dominance
- Emergence

...



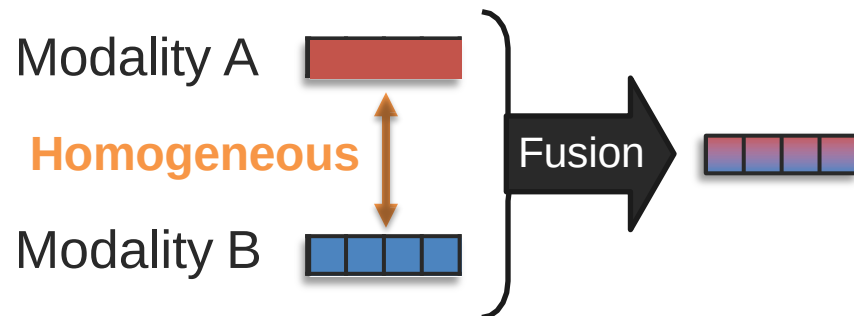
Representation Fusion

Sub-Challenge 1a: Representation Fusion

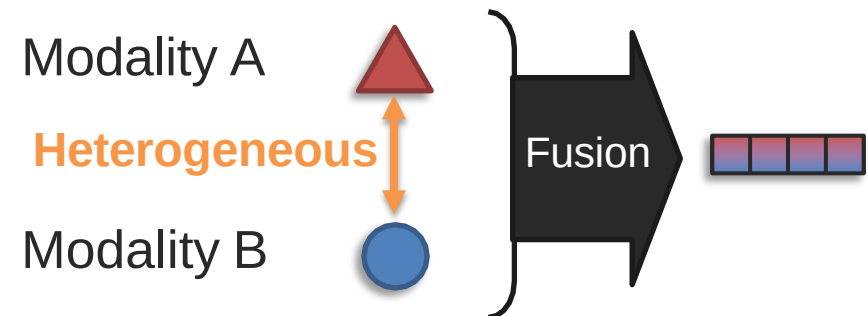


Definition: Learn a joint representation that models cross-modal interactions between individual elements of different modalities

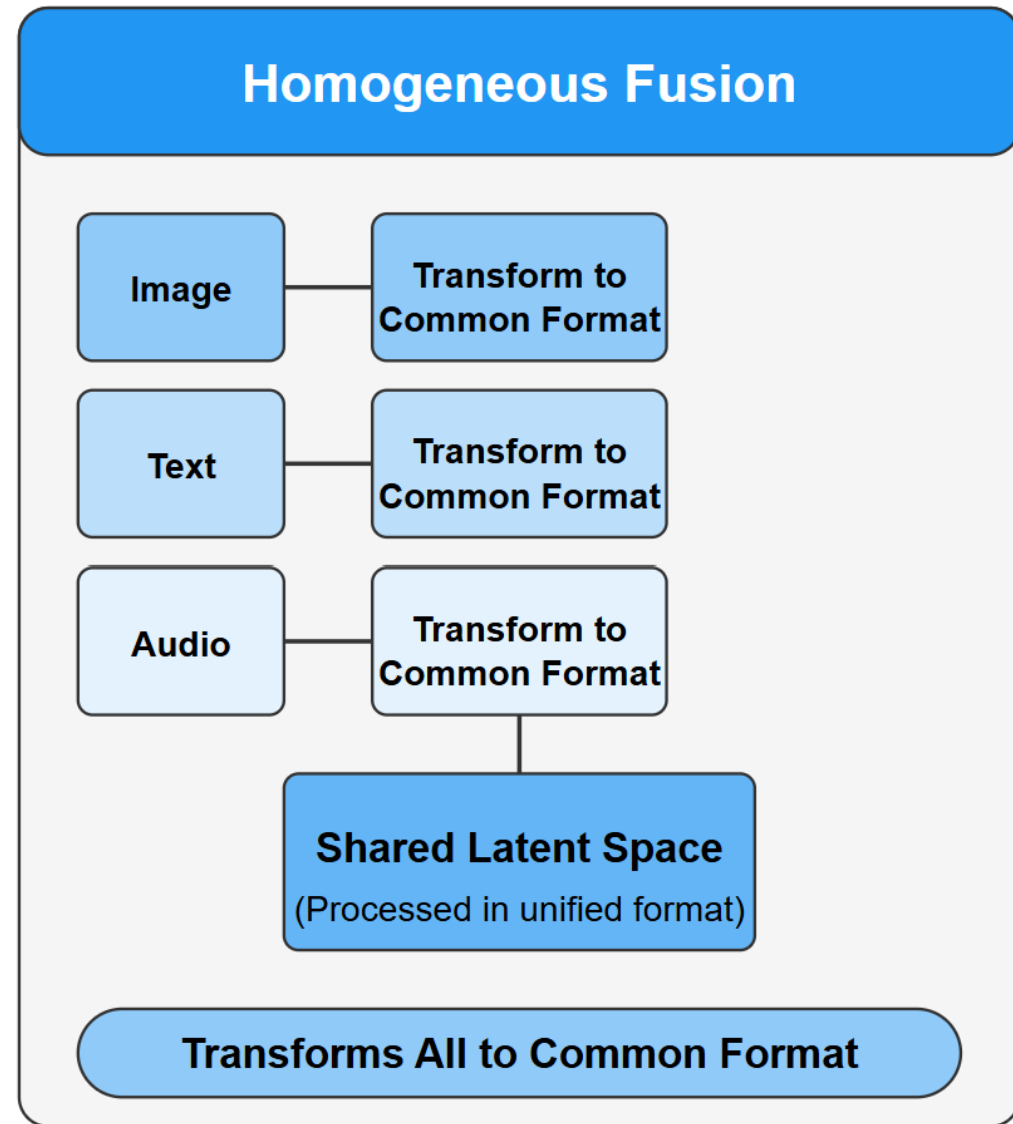
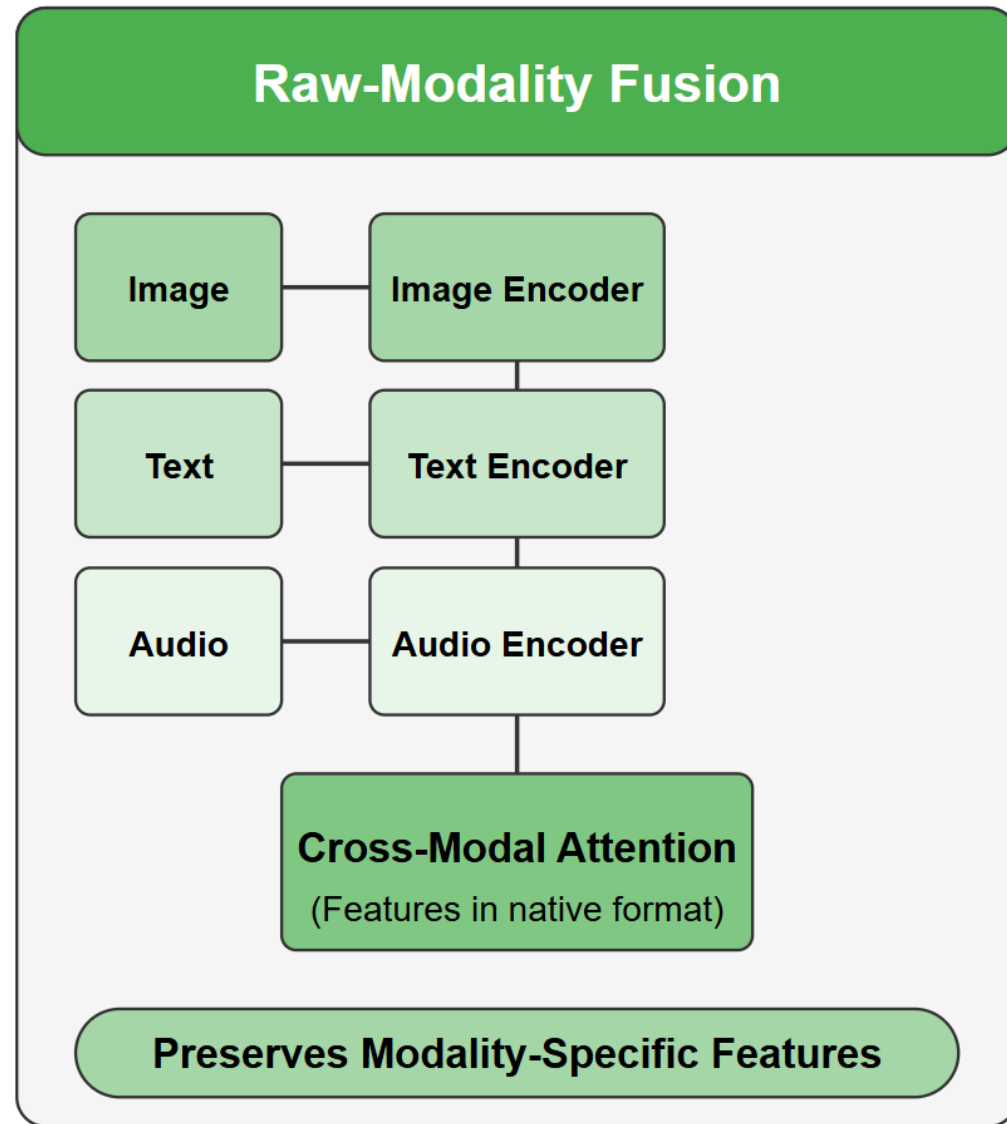
Basic fusion:



Raw-modality fusion:

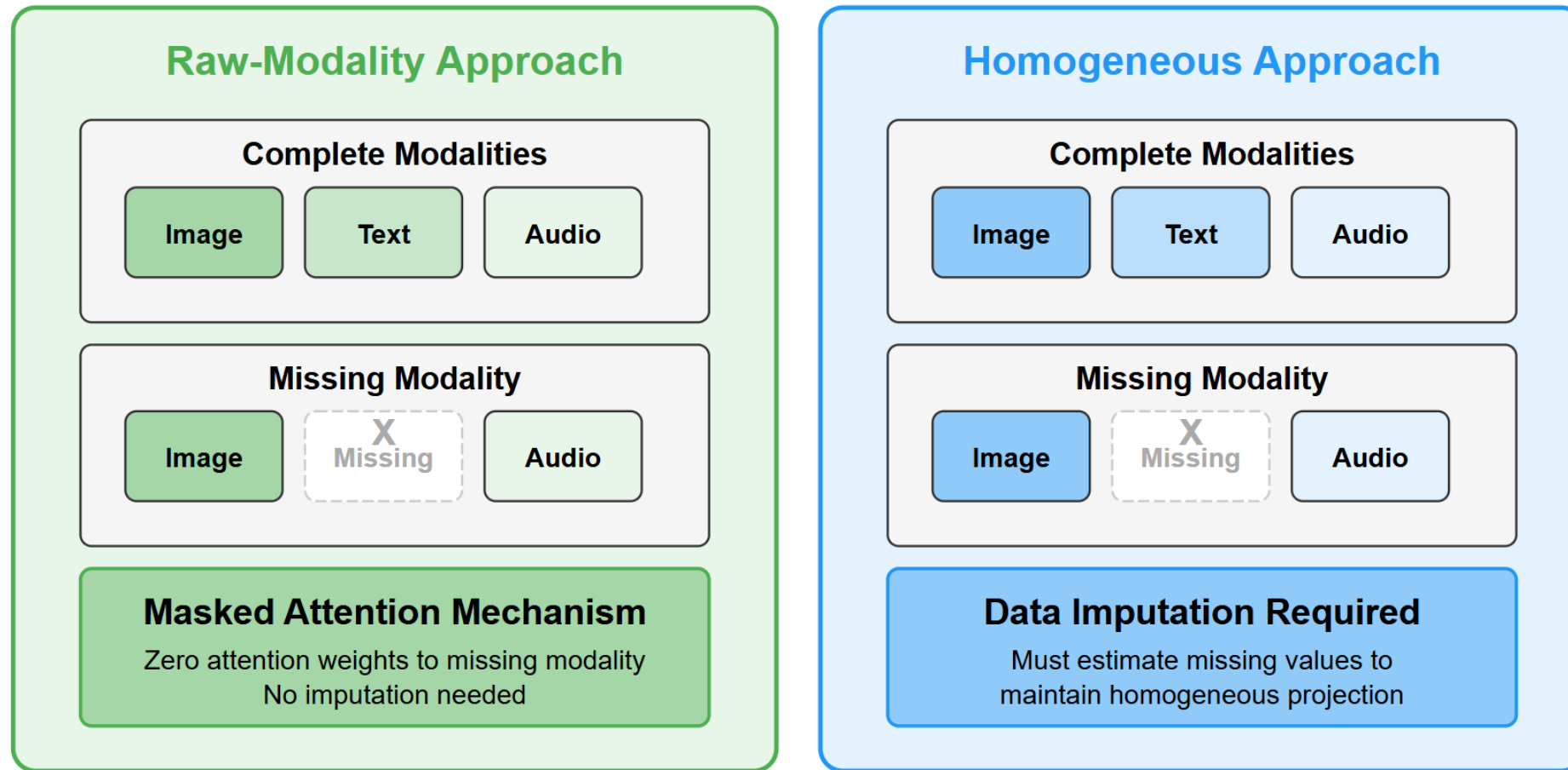


Raw-Modality vs. Homogeneous Fusion Approaches



Adaptability to Missing Modalities

Raw-Modality Fusion vs. Homogeneous Fusion

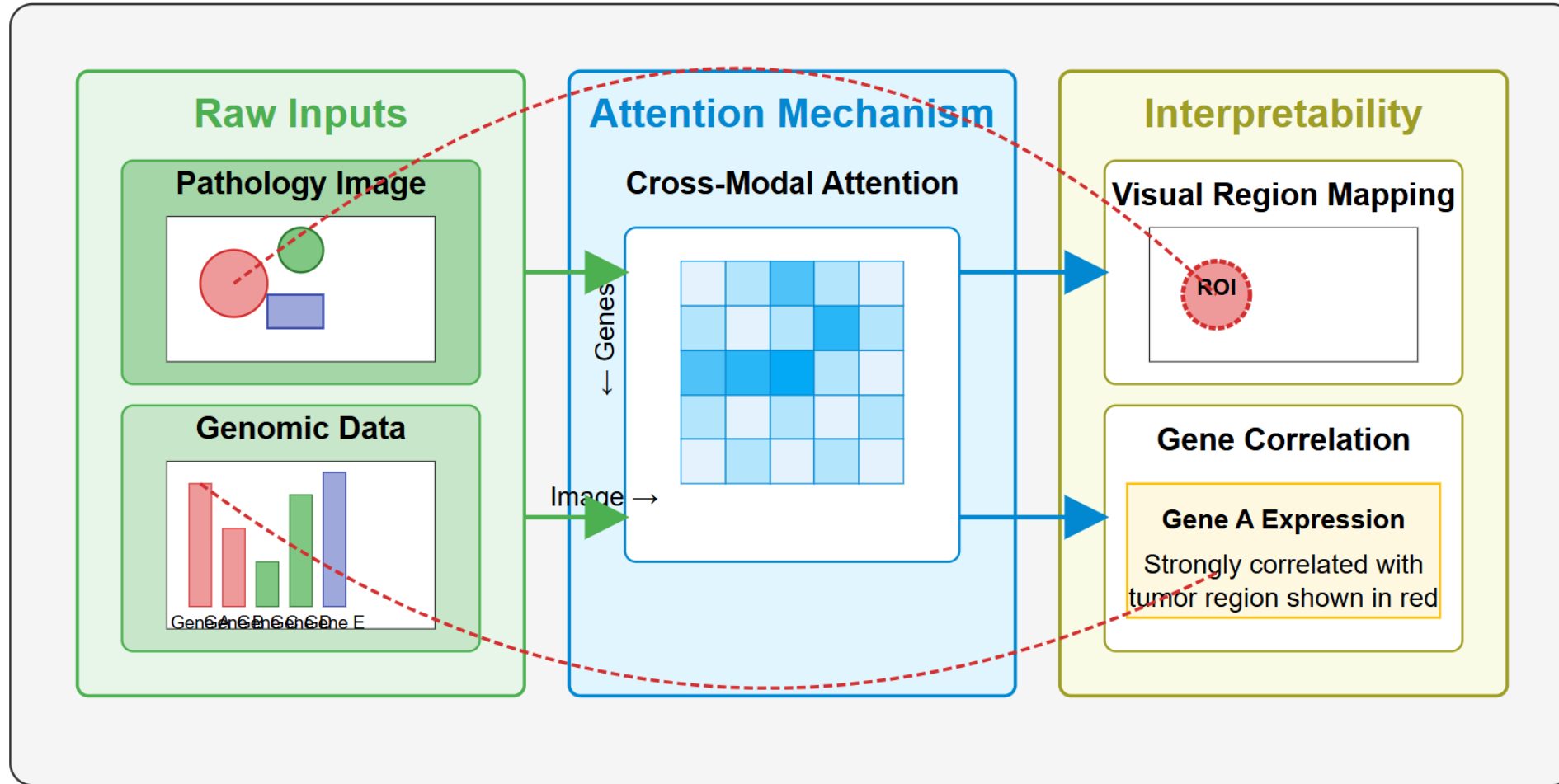


Raw-Modality Fusion: Maintains >90% accuracy with 40% missing data

Enhanced Interpretability

HEALNet: Multimodal Fusion for Heterogeneous Biomedical Data

Preserves Original Feature Spaces



Direct visualization of how specific image regions relate to genomic features

When to Use Raw-Modality Fusion

Summary of Key Scenarios and Benefits

Scenario	Reason to Choose Raw-Modality Fusion
 Heterogeneous Data Formats	Handles data with different structures (e.g., text + image) more effectively
 Preservation of Features	Retains modality-specific details (critical in medical imaging, genomics, etc.)
 Missing Data	Flexible handling of incomplete inputs via masking or attention mechanisms
 Interpretability	Clear insights into cross-modal interactions at the raw feature level
 Low Computational Resources	Avoids heavy transformations or shared latent space overhead

Raw-Modality Fusion excels when flexibility and preserving original data characteristics are paramount

When to Use Homogeneous Fusion

Key Scenarios and Benefits of Homogeneous Approaches

Scenario	Reason to Choose Homogeneous Fusion
 Unified Representation	When a common representation space is beneficial for downstream tasks
 Similar Modality Structures	When modalities share similar underlying structures (e.g., multiple image types)
 Pre-trained Embeddings	When leveraging pre-trained embeddings that already map into vector spaces
 Complete Data Availability	When all modalities are consistently available with little to no missing data
 Sufficient Computational Resources	When transformation overhead is not a concern and advanced projection models are available

Homogeneous Fusion excels when creating a unified representation is more important than preserving modality-specific features

Sub-Challenge 1a: Representation Fusion

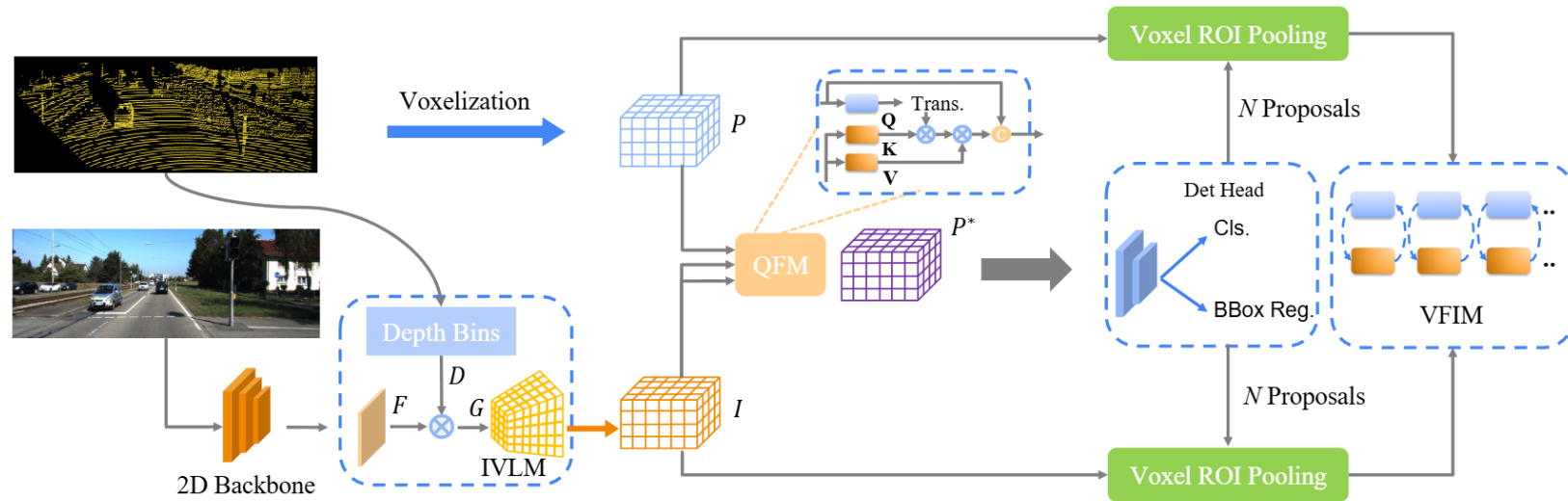


Fig. 2: The architecture of HMFI. Each image is processed by a 2D backbone network and fed into an image voxel lifter module (IVLM) to produce a homogeneous structure based on the depth bins transformed by the point cloud. Then, the processed homogeneous image and point cloud features are fused by the query fusion mechanism (QFM). Next, a voxel-based object detector is employed on fused features to produce 3D detection results. Finally, the voxel feature interaction module (VFIM) conducts feature interaction at object-level based on the detection results to improve semantic consistency in these two homogeneous cross-modal features.

Sub-Challenge 1a: Representation Fusion

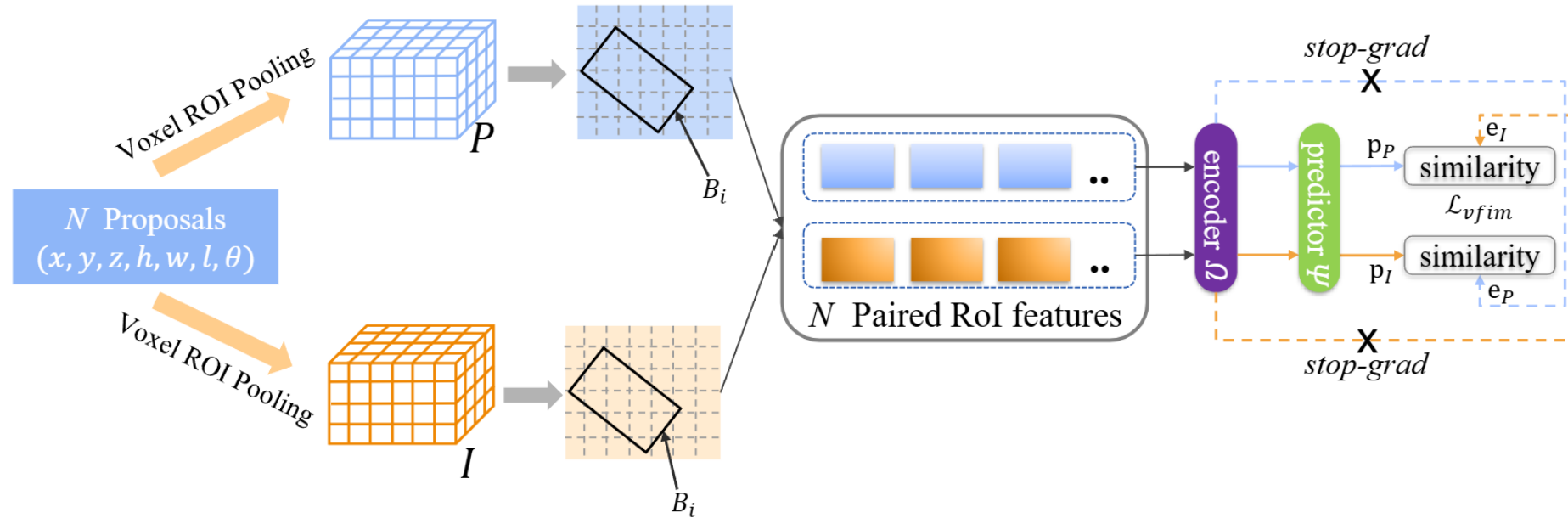
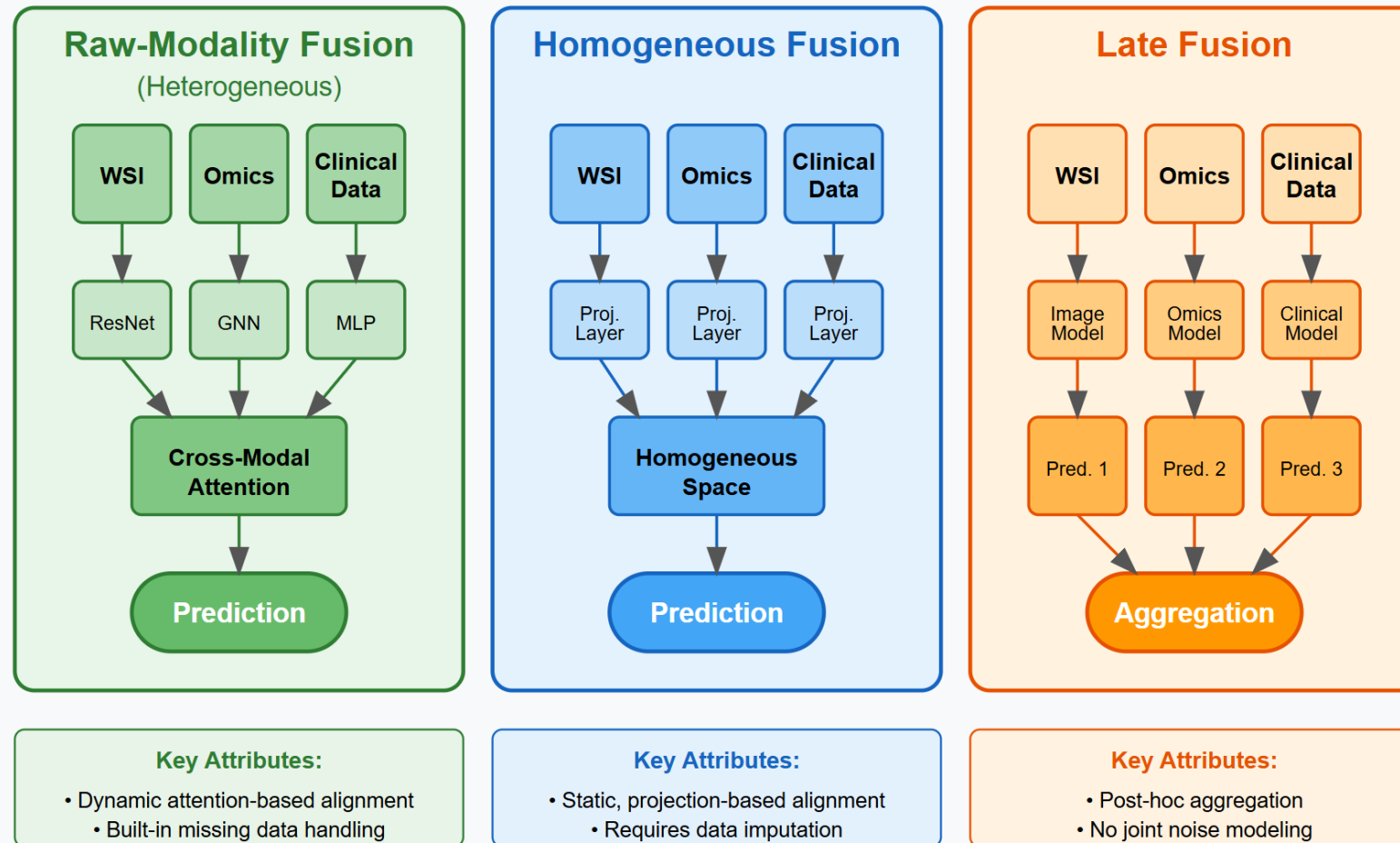


Fig. 4: The voxel feature interaction module. We use the voxel RoI pooling to extract features in these two homogeneous features according to the predicted proposals and form the paired RoI feature set. Then we adopt feature interaction between each pair of RoI features based on the symmetry similarity constraint loss to improve the object-level semantic consistency between two homogeneous representations

Sub-Challenge 1a: Representation Fusion

Raw-Modality Fusion vs. Other Approaches

Comparing Different Multimodal Fusion Strategies

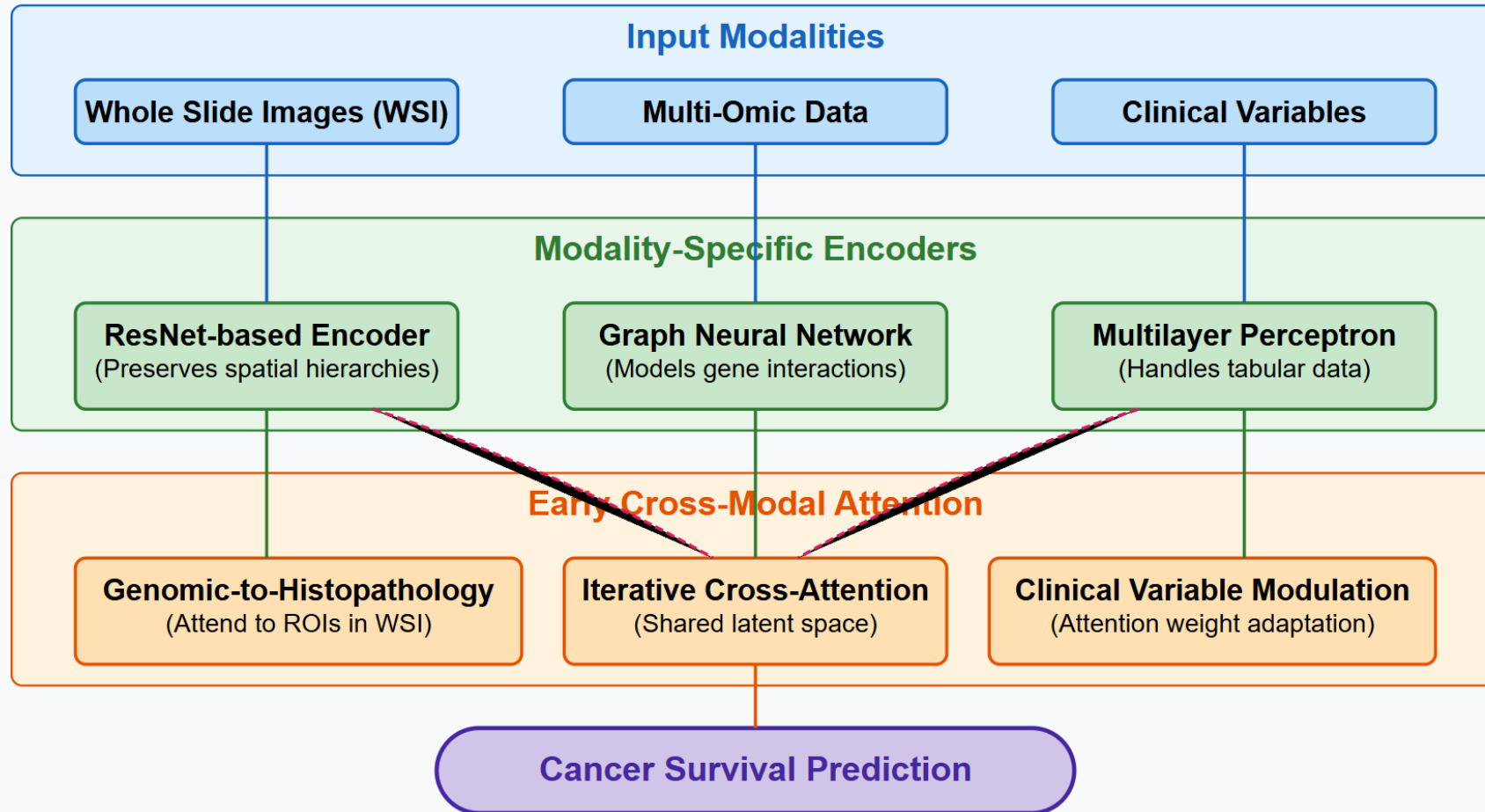


Sub-Challenge 1a: Representation Fusion

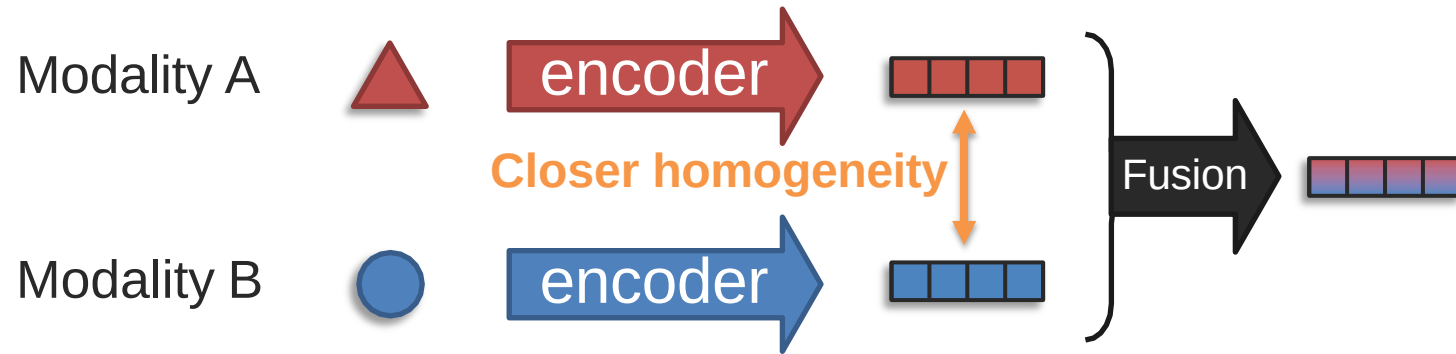
**HEALNet: Multimodal Fusion for
Heterogeneous Biomedical Data**

HEALNet Architecture

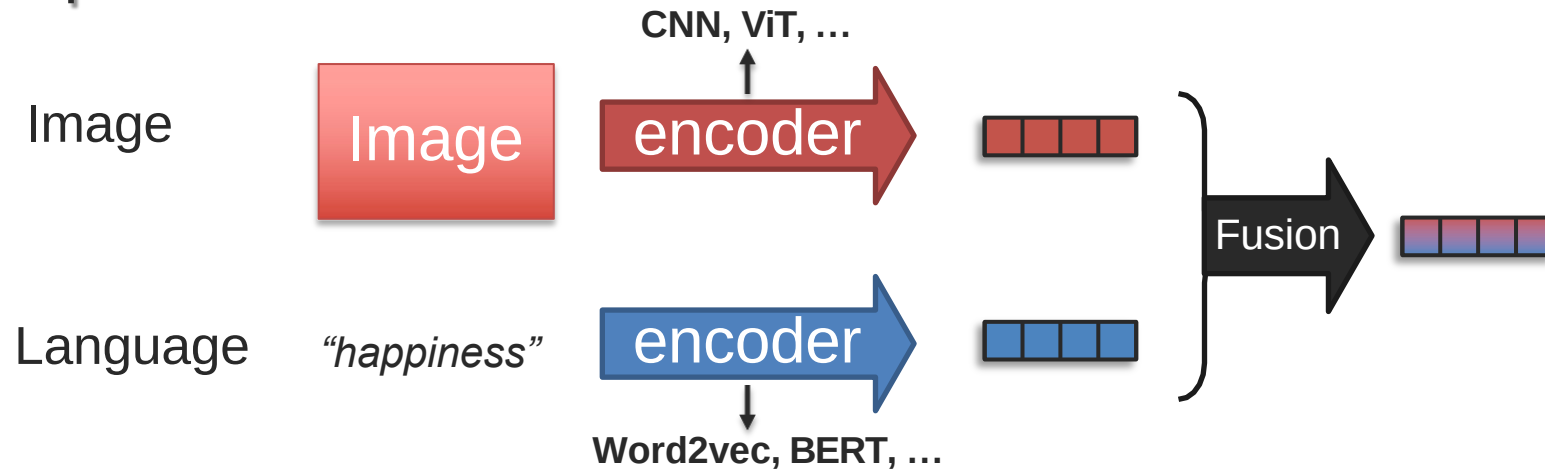
Hybrid Early-fusion Attention Learning Network



Fusion with Unimodal Encoders



Example:



➡ Unimodal encoders can be jointly learned with fusion network, or pre-trained

Example paper

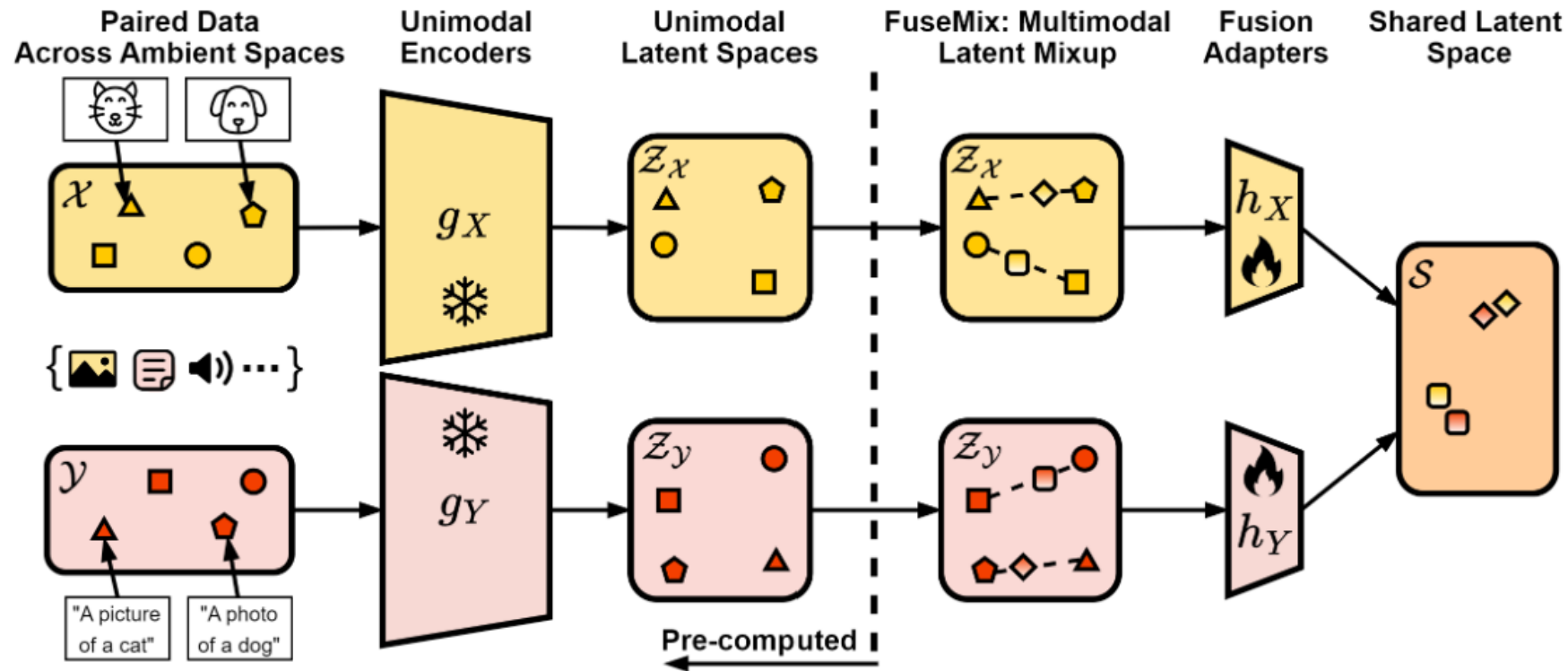


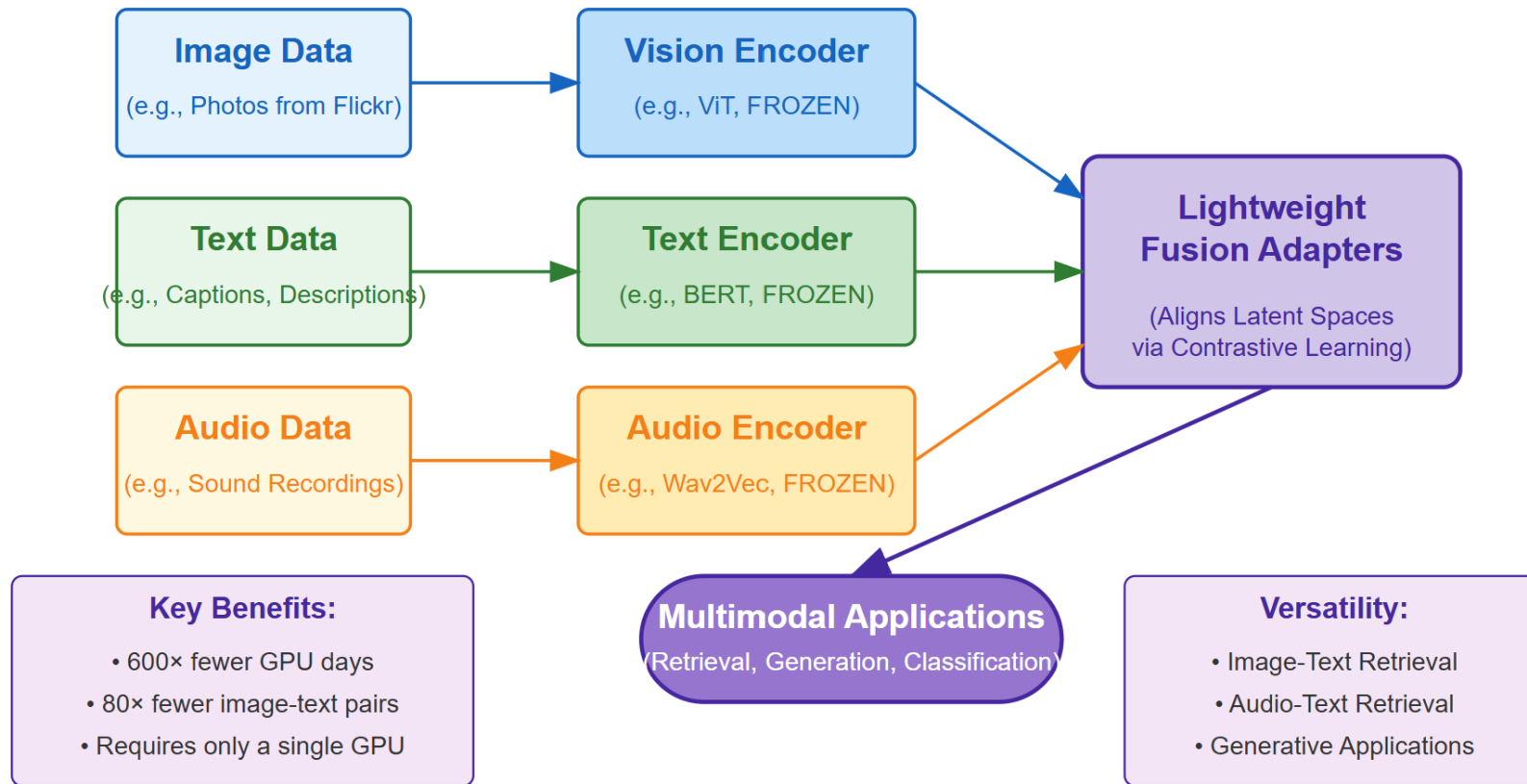
Figure 2. A schematic of our proposed fusion framework to align the latent spaces of pre-trained unimodal encoders using a minimal set of paired data. The unimodal encoders are kept frozen, and their latent encodings are pre-computed only once. FuseMix applies mixup on each latent space, importantly sharing the mixing coefficient across modalities, and is used as a modality-agnostic data augmentation. Then, the lightweight fusion adapters are trained to align the resulting augmented latents into a shared latent space.

Example paper

FuseMix Framework Overview

Data-Efficient Multimodal Fusion on a Single GPU

Ideal Example Paper: "***Data-Efficient Multimodal Fusion on a Single GPU***"

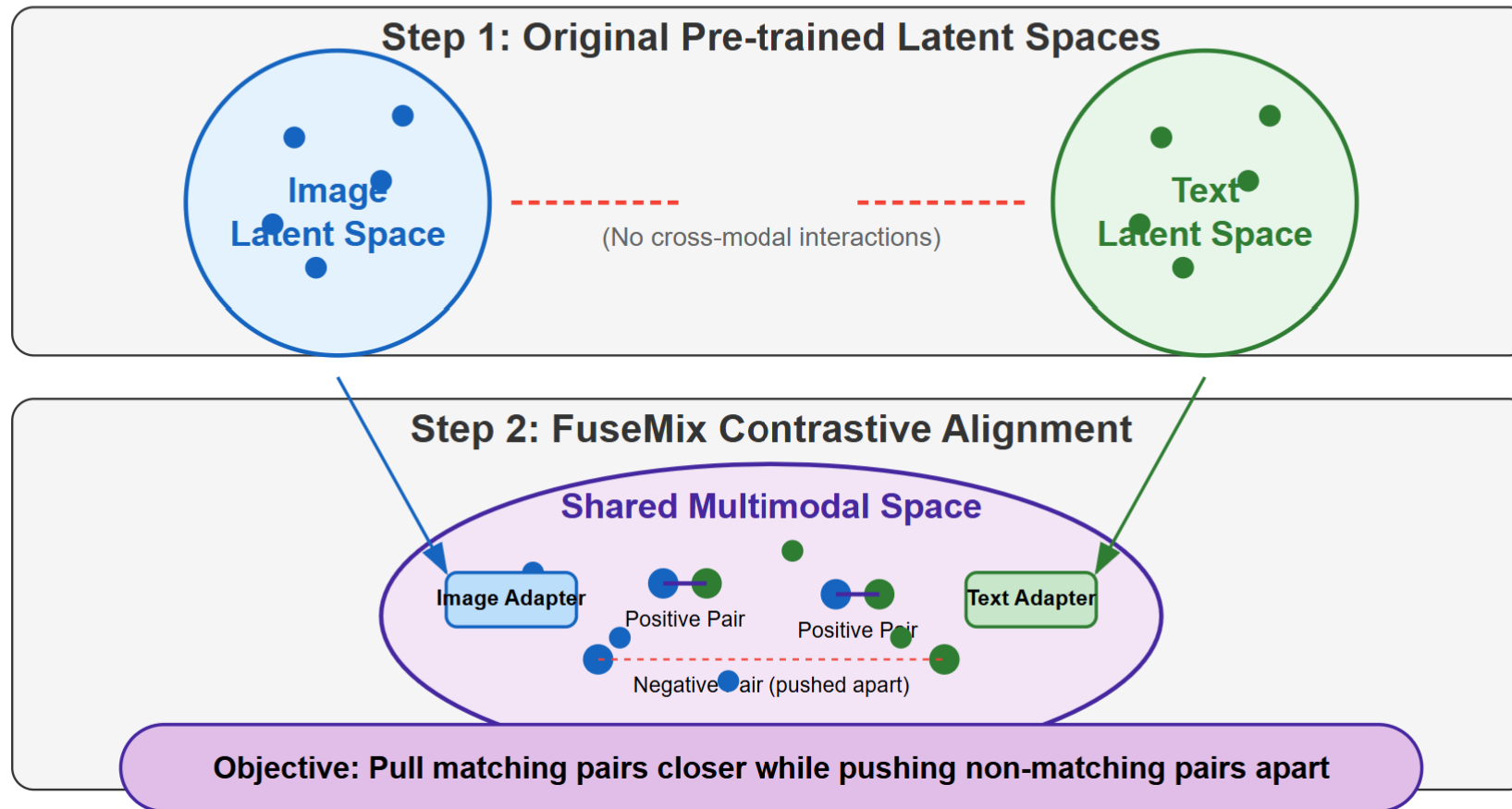


Efficient multimodal fusion using frozen pre-trained encoders and lightweight adapters

Example paper

Contrastive Latent Space Alignment

How FuseMix Aligns Modality-Specific Representations

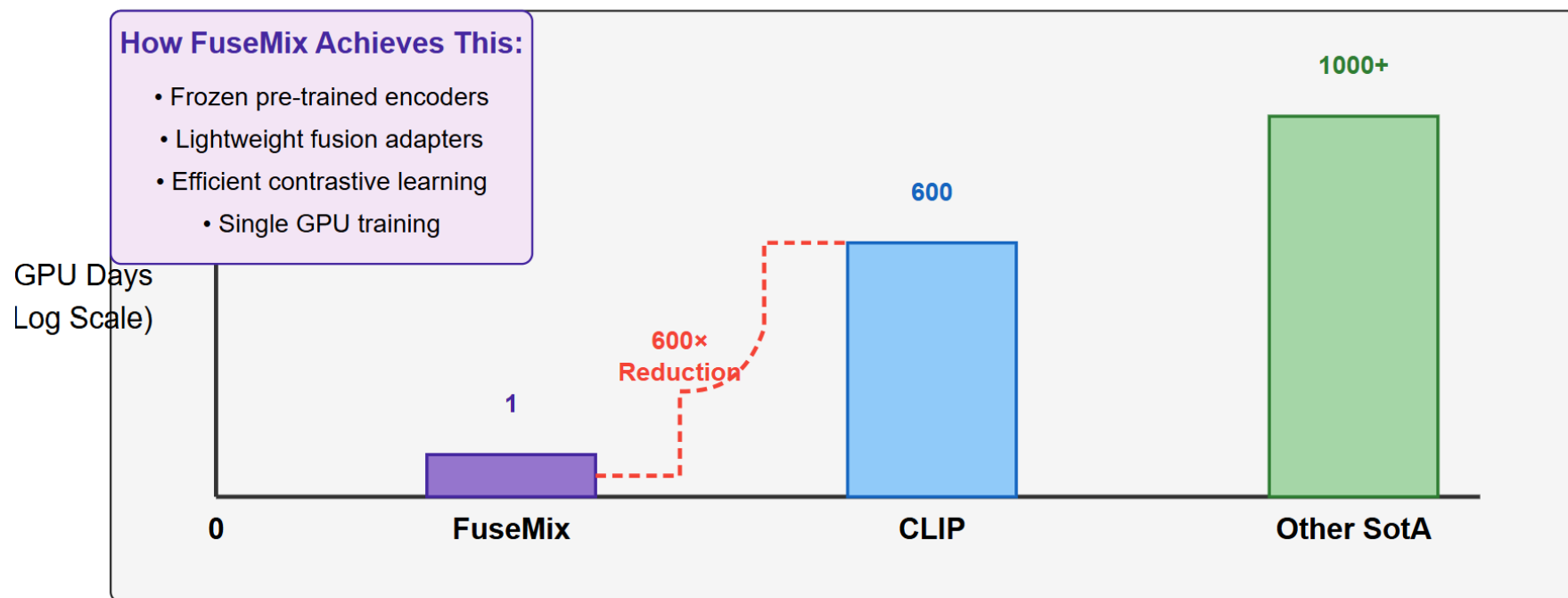


Ideal Example Paper: "Data-Efficient Multimodal Fusion on a Single GPU"

Example paper

Computational Efficiency of FuseMix

Comparison with State-of-the-Art Methods



Ideal Example Paper: "Data-Efficient Multimodal Fusion on a Single GPU"

Training Data Requirements

FuseMix: ~50K pairs

CLIP: ~400M pairs (80× more)

Example paper

Application Versatility of FuseMix

Plug-and-Play Framework for Multiple Modalities

Plug-and-Play Design: Supports arbitrary modality combinations

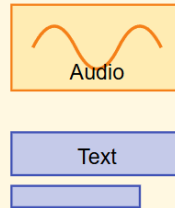
Image-Text Retrieval



Flickr30K Results:

Outperforms CLIP
with significantly
less training data

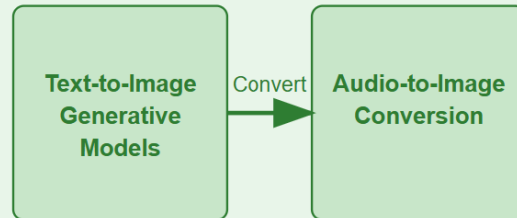
Audio-Text Retrieval



Easy Extension:

Replace image encoder
with audio encoder
while keeping architecture

Generative Applications

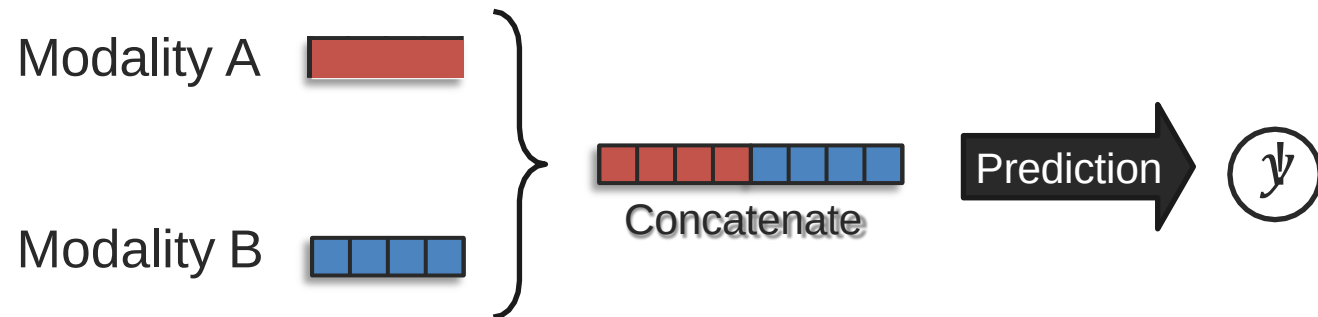


Seamless adaptation across multiple modality combinations with minimal adjustments

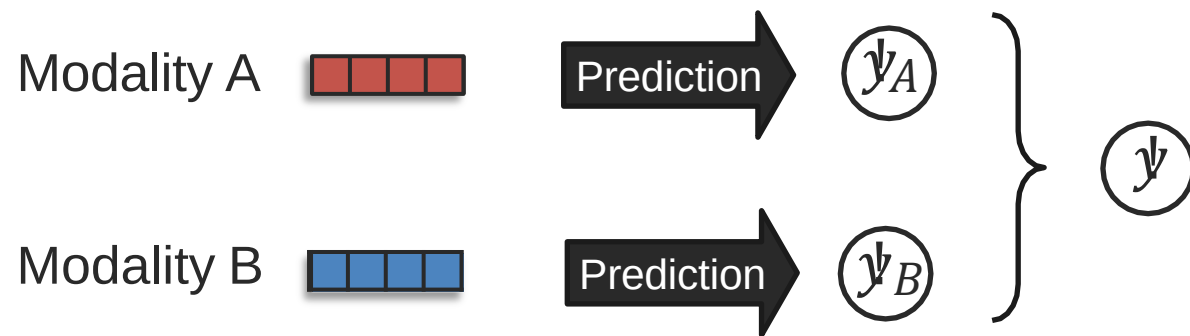
Ideal Example Paper: "Data-Efficient Multimodal Fusion on a Single GPU"

Early and Late Fusion – A historical View

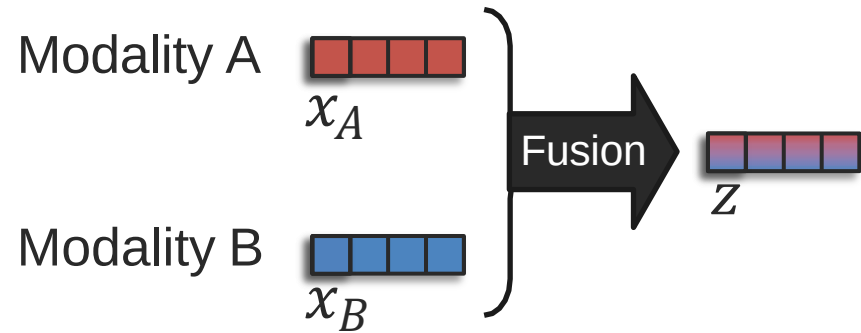
Early fusion:



Late fusion:



Basic Concepts for Representation Fusion (aka, Basic Fusion)



Goal: Model *cross-modal interactions* between the multimodal elements

Let's study the univariate case first
↳ (only 1-dimensional features)

Linear regression:

$$z = w_0 + w_1x_A + w_2x_B + w_3(x_A \times x_B) + \epsilon$$

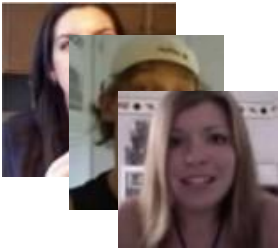
Diagram illustrating the components of the linear regression equation:

- w_0 : intercept (bias term)
- $w_1x_A + w_2x_B$: Additive terms
- $w_3(x_A \times x_B)$: Multiplicative term
- ϵ : error (residual term)

Linear Regression

Linear regression is used to test research hypotheses, over a whole dataset

300 book reviews



y : audience score

x_A : percentage of smiling

x_B : professional status
(0=non-critic, 1=critic)

H1: Does smiling reveal what the audience score was?

H2: Does the effect of smiling depend on professional status?

Linear regression:

$$y = w_0 + w_1x_A + w_2x_B + w_3(x_A \times x_B) + \epsilon$$

Diagram illustrating the components of the linear regression equation:

- w_0 : intercept (bias term)
- $w_1x_A + w_2x_B$: Additive terms
- $w_3(x_A \times x_B)$: Multiplicative term
- ϵ : error (residual term)

w_0 : average score when x_A and x_B are zero

w_1 : effect from x_A variable only

w_2 : effect from x_B variable only

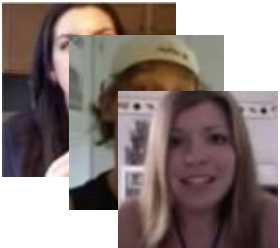
w_3 : effect from x_A and x_B interaction only

ϵ : residual not modeled by w_0 , w_1 , w_2 or w_3

Linear Regression

Linear regression is used to test research hypotheses, over a whole dataset

300 book reviews



y : audience score

x_A : percentage of smiling

x_B : professional status
(0=non-critic, 1=critic)

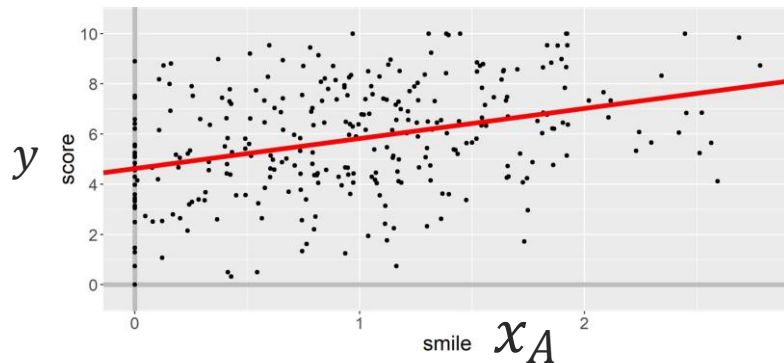
H1: Does smiling reveal what the audience score was?

H2: Does the effect of smiling depend on professional status?

Linear regression:

$$z = w_0 + \boxed{w_1} x_A + \epsilon$$

slope



Confidence interval: “95% confident that w parameter is contained within this interval”

	Estimate	95% CI
w_0	4.63	[4.20, 5.06]
w_1	1.20	[0.83, 1.57]

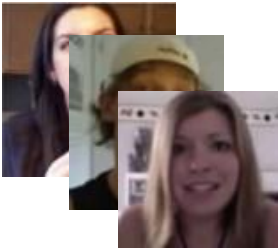
Confidence interval does not contain 0, so effect is significant

p-values would be another way to test hypothesis

Linear Regression

Linear regression is used to test research hypotheses, over a whole dataset

300 book reviews



y : audience score

x_A : percentage of smiling

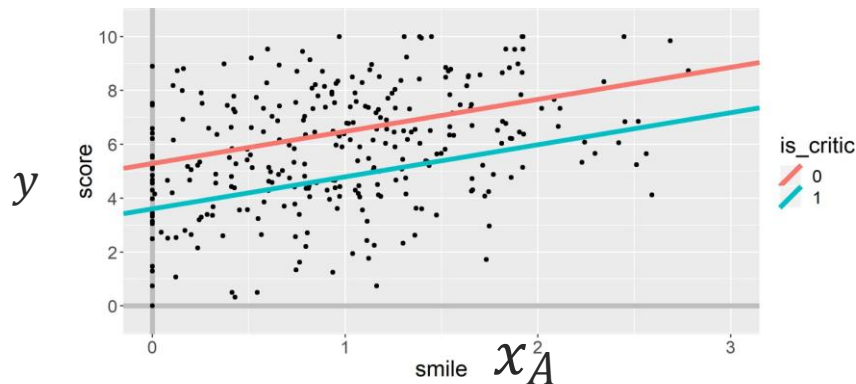
x_B : professional status
(0=non-critic, 1=critic)

H1: Does smiling reveal what the audience score was?

H2: Does the effect of smiling depend on professional status?

Linear regression:

$$Z = w_0 + w_1 x_A + w_2 x_B + \epsilon$$



	Estimate	95% CI
w_0	5.29	[4.86, 5.73]
w_1	1.19	[0.85, 1.53]
w_2	-1.69	[-2.14, -1.24]

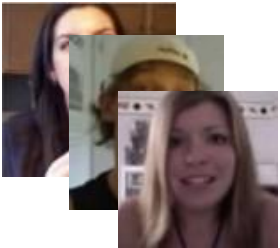
➡ Positive effect

➡ Negative effect

Linear Regression

Linear regression is used to test research hypotheses, over a whole dataset

300 book reviews



y : audience score

x_A : percentage of smiling

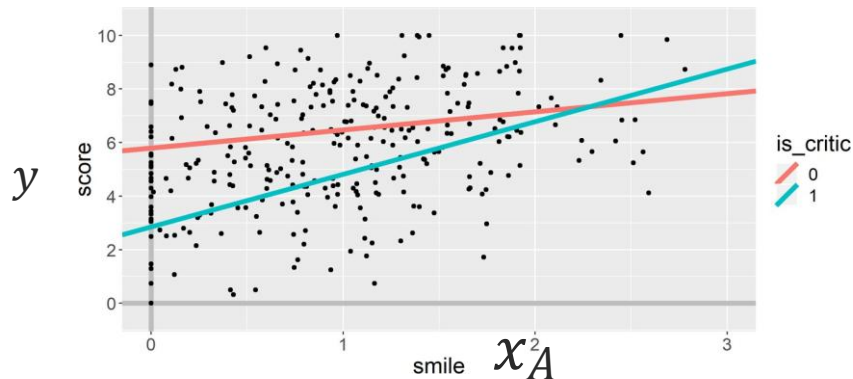
x_B : professional status
(0=non-critic, 1=critic)

H1: Does smiling reveal what the audience score was?

H2: Does the effect of smiling depend on professional status?

Linear regression:

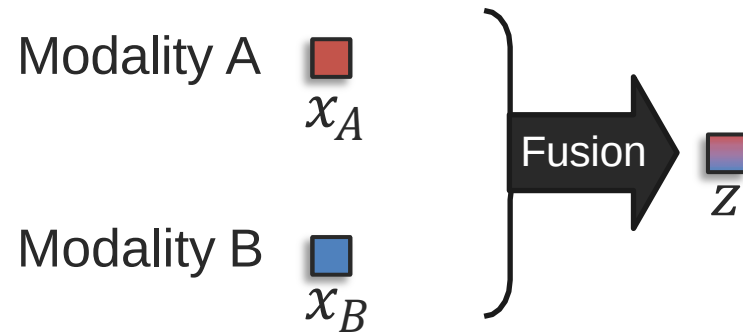
$$Z = w_0 + w_1 x_A + w_2 x_B + w_3 (x_A \times x_B) + \epsilon$$



	Estimate	95% CI
w_0	5.79	[5.29, 6.29]
w_1	0.68	[0.25, 1.11]
w_2	-2.94	[-3.73, -2.15]
w_3	1.29	[0.61, 1.97]

➔ **Multiplicative interaction!**

Basic Concepts for Representation Fusion (aka, Basic Fusion)



Goal: Model *cross-modal interactions* between the multimodal elements

Let's study the **univariate case first**
↳ (only 1-dimensional features)

Linear regression:

$$Z = w_0 + w_1x_A + w_2x_B + w_3(x_A \times x_B) + \epsilon$$

Diagram illustrating the components of the linear regression equation:

- w_0 : intercept (bias term)
- $w_1x_A + w_2x_B$: Additive terms
- $w_3(x_A \times x_B)$: Multiplicative term
- ϵ : error (residual term)

1 Additive terms:

$$Z = w_1x_A + w_2x_B + \epsilon$$

② Multiplicative “interaction” term:

$$Z = w_3(x_A \times x_B) + \epsilon$$

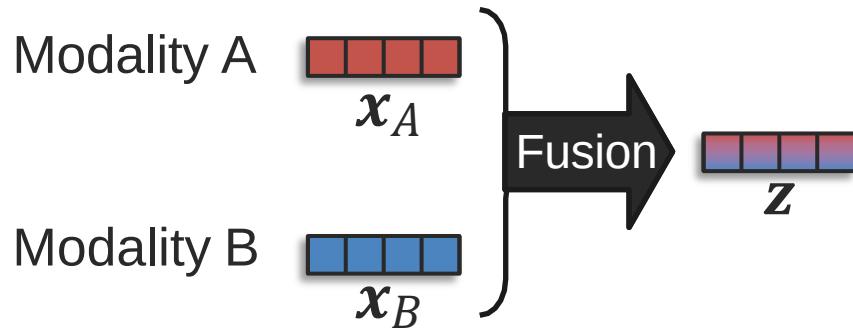
③ Additive and multiplicative terms:

$$Z = w_1x_A + w_2x_B + w_3(x_A \times x_B) + \epsilon$$

Additive Fusion

Back to multivariate case!

↳ (multi-dimensional features)

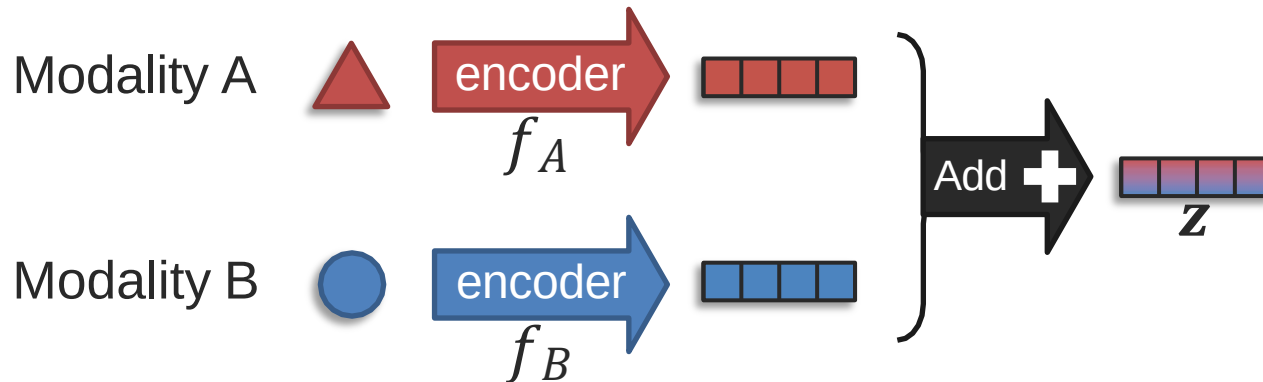


Additive fusion:

$$z = w_1 x_A + w_2 x_B$$

➡ 1-layer neural network
can be seen as additive

With unimodal encoders:

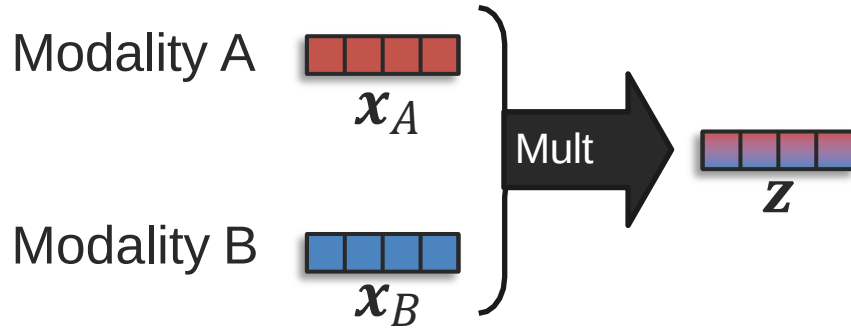


Additive fusion:

$$z = f_A(\triangle) + f_B(\circ)$$

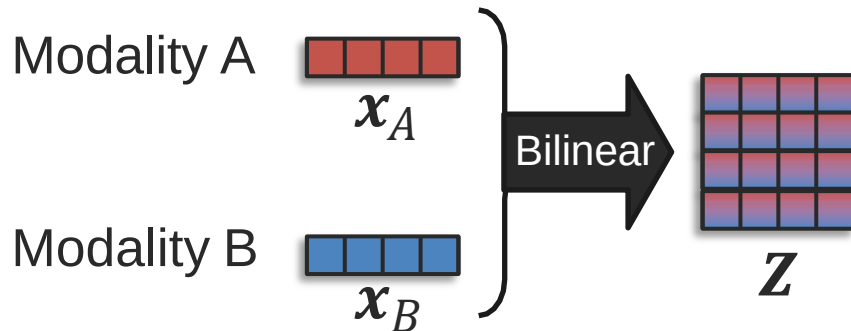
It could be seen as an
ensemble approach
(late fusion)

Multiplicative Fusion



Simple multiplicative fusion:

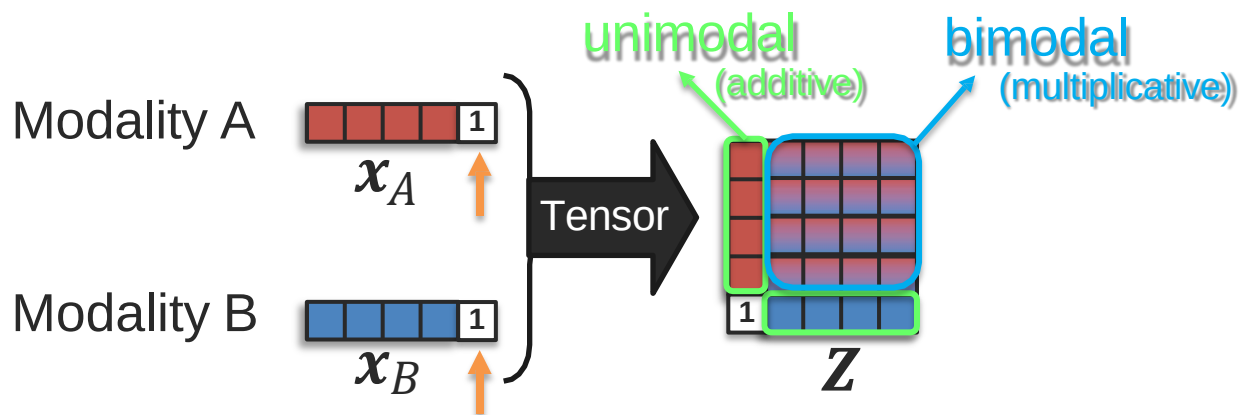
$$\mathbf{z} = \mathbf{w}(\mathbf{x}_A \times \mathbf{x}_B)$$



Bilinear Fusion:

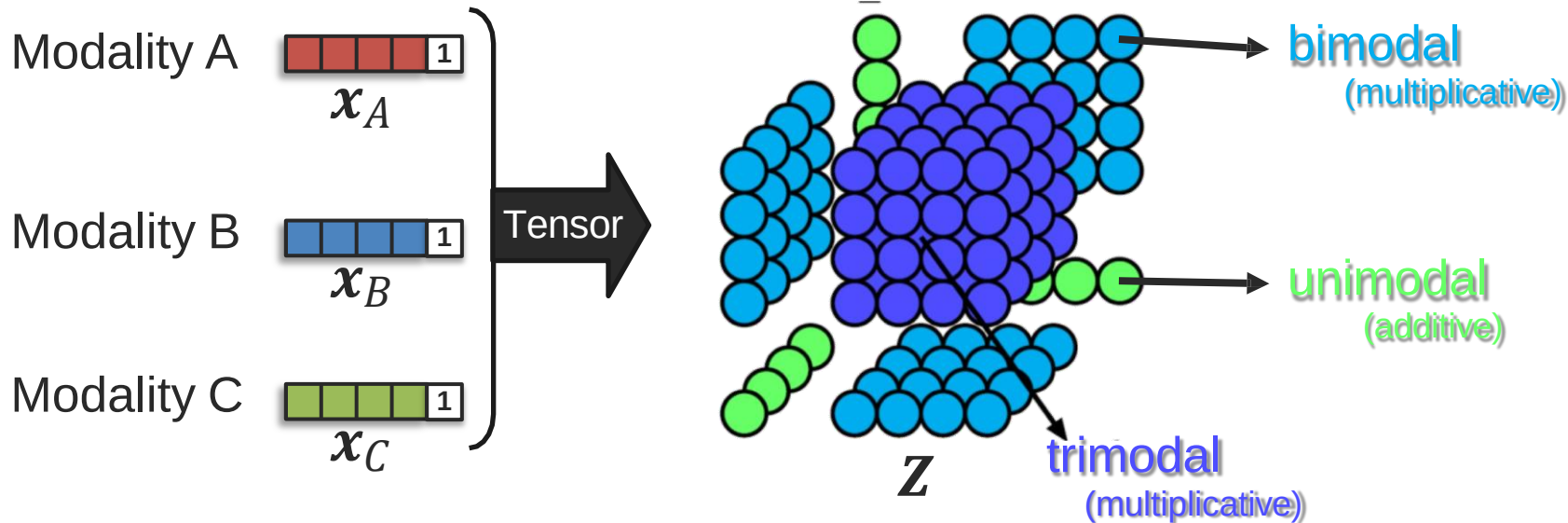
$$\mathbf{Z} = \mathbf{W}(\mathbf{x}_A^T / \mathbf{x}_B)$$

Tensor Fusion



Tensor Fusion (bimodal):

$$Z = w([x_A \ 1]^T * [x_B \ 1])$$



... but the weight matrix may end up quite large!

Low-rank Fusion

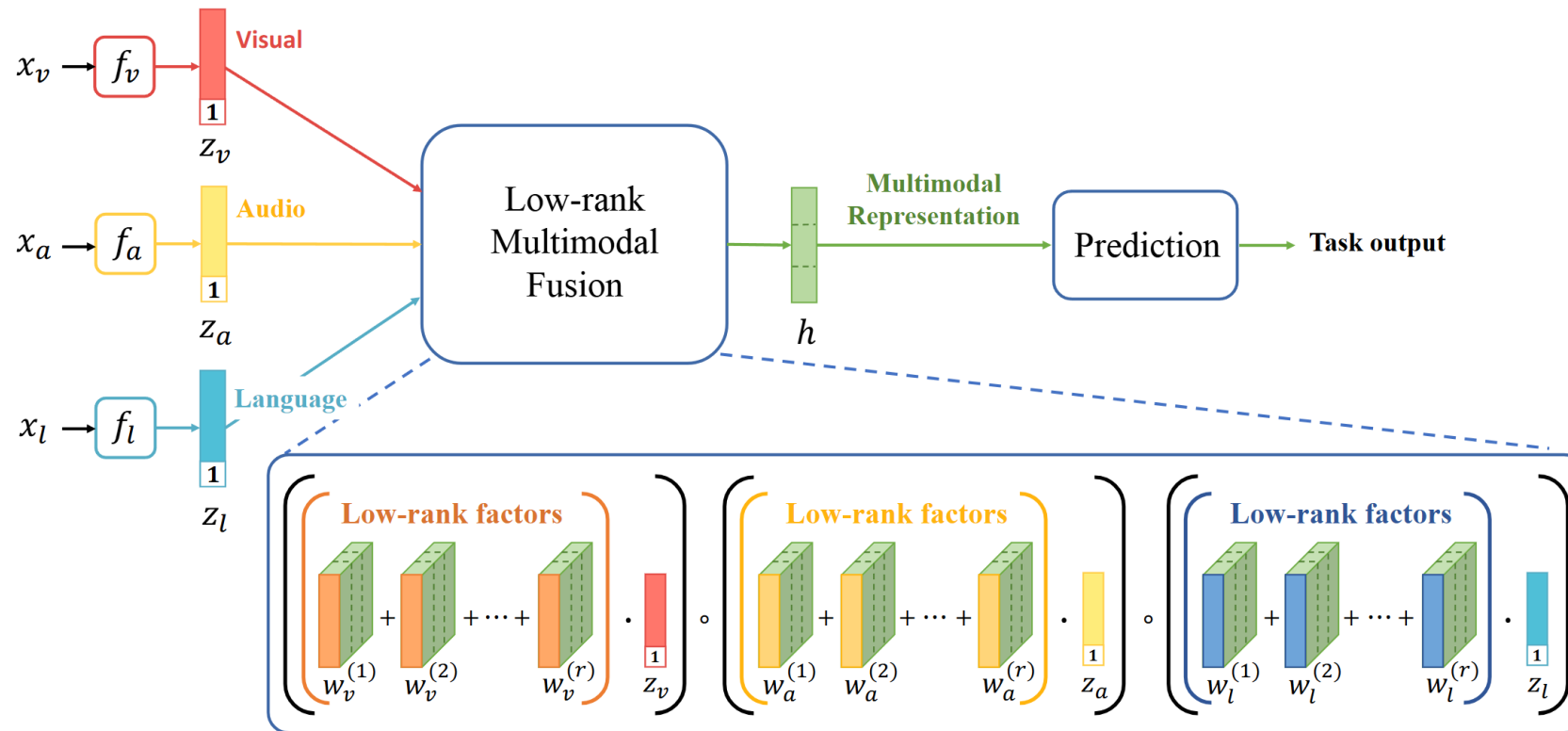
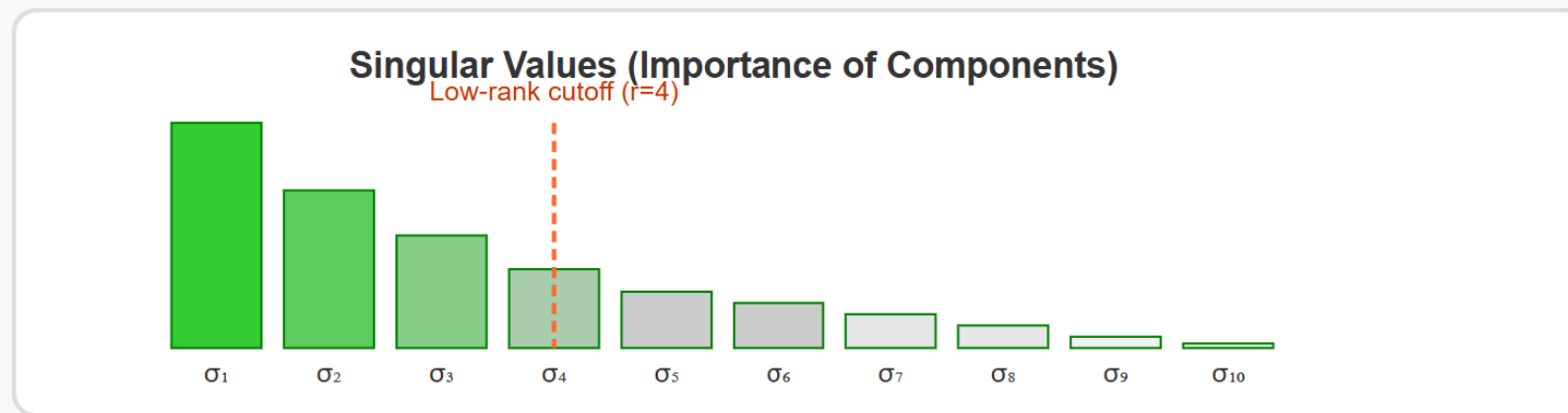
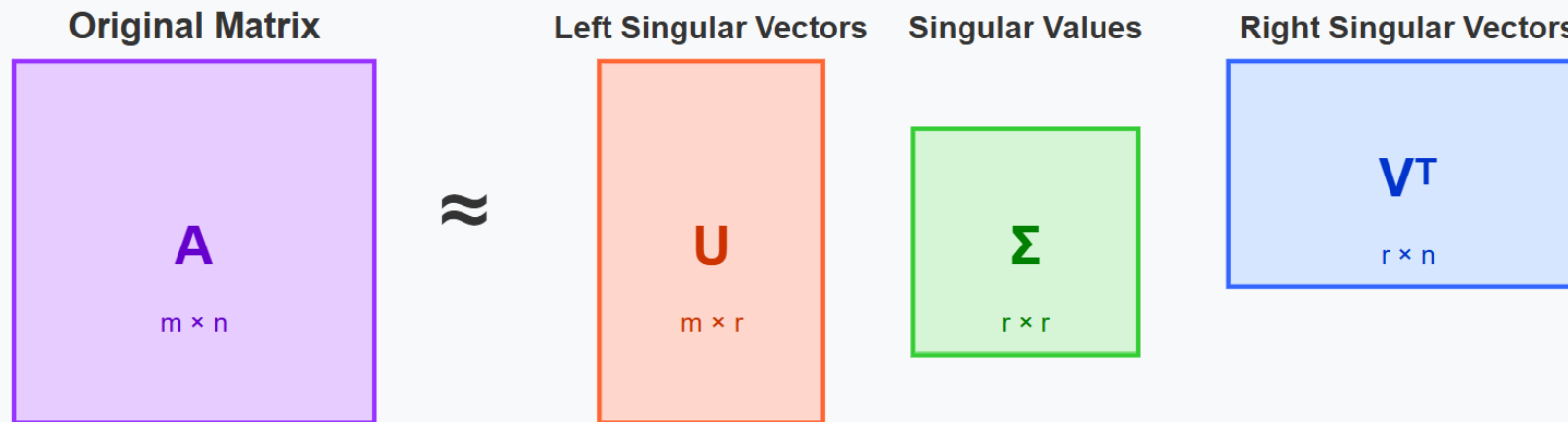


Figure 1: Overview of our Low-rank Multimodal Fusion model structure: LMF first obtains the unimodal representation z_a, z_v, z_l by passing the unimodal inputs x_a, x_v, x_l into three sub-embedding networks f_v, f_a, f_l respectively. LMF produces the multimodal output representation by performing low-rank multimodal fusion with modality-specific factors. The multimodal representation can be then used for generating prediction tasks.

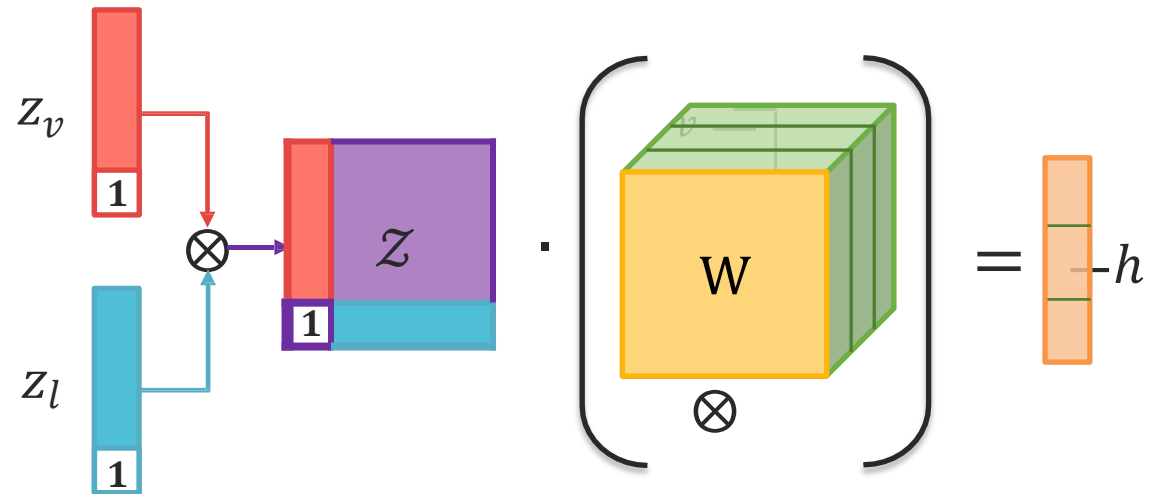
Low-rank Fusion

Low-Rank Matrix Decomposition

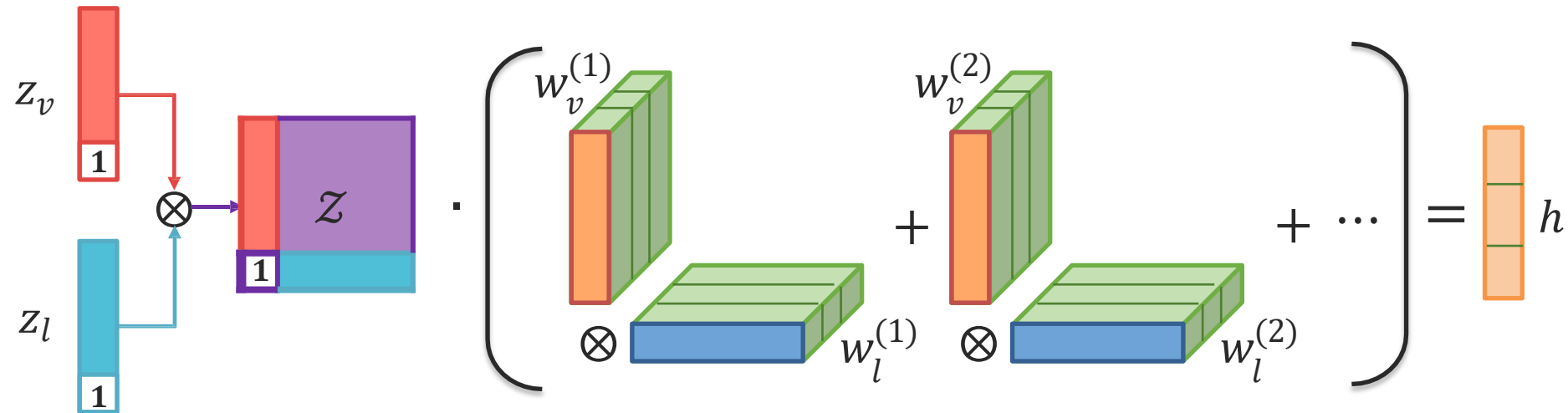
The Building Block of LMF



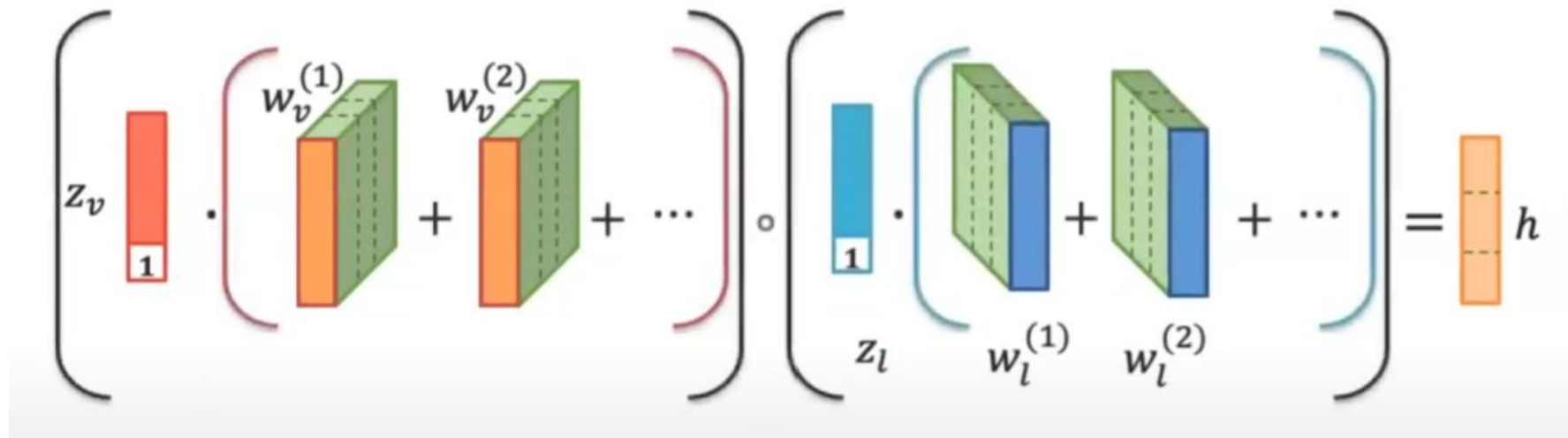
Low-rank Fusion



Low-rank Fusion



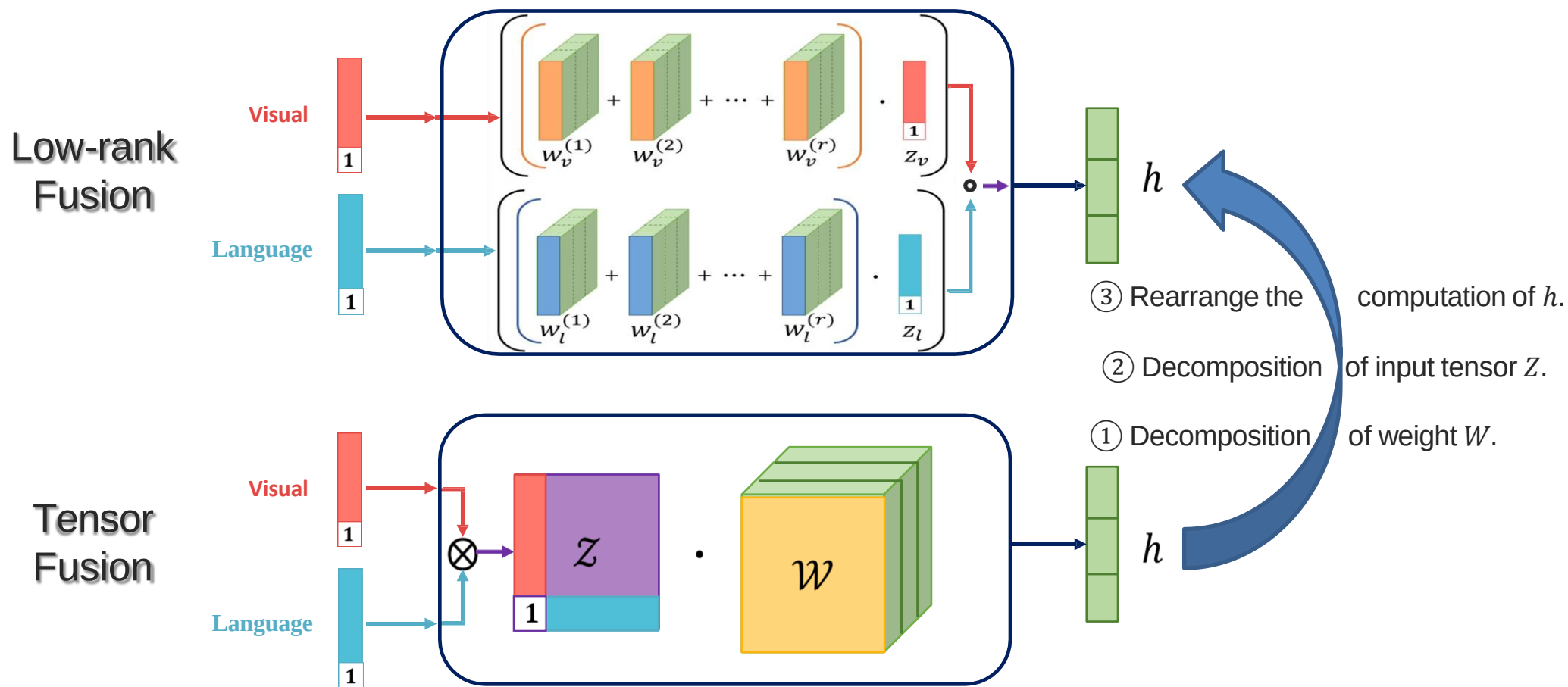
Low-rank Fusion



The diagram illustrates the Low-rank Fusion process. It shows two main components in large parentheses, separated by an element-wise multiplication operator ($\circ$). The first component is a vector z_v (represented by a red vertical bar with a '1' at the bottom) multiplied by a sum of weighted features $w_v^{(1)}, w_v^{(2)}, \dots$ (represented by green 3D blocks with orange bases). The second component is a vector z_l (represented by a blue vertical bar with a '1' at the bottom) multiplied by a sum of weighted features $w_l^{(1)}, w_l^{(2)}, \dots$ (represented by green 3D blocks with blue bases). The result of the multiplication is a fused representation h (represented by an orange vertical bar).

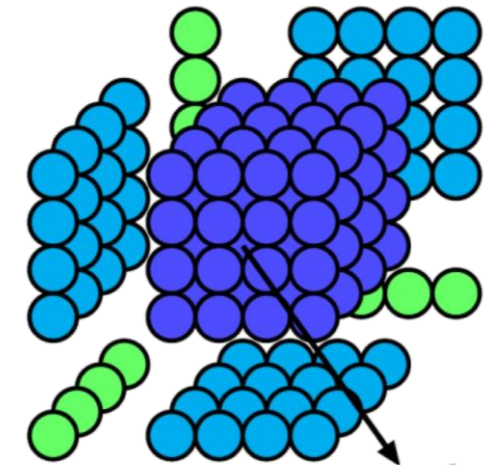
$$\left(z_v \cdot \left(w_v^{(1)} + w_v^{(2)} + \dots \right) \right) \circ \left(z_l \cdot \left(w_l^{(1)} + w_l^{(2)} + \dots \right) \right) = h$$

Low-rank Fusion

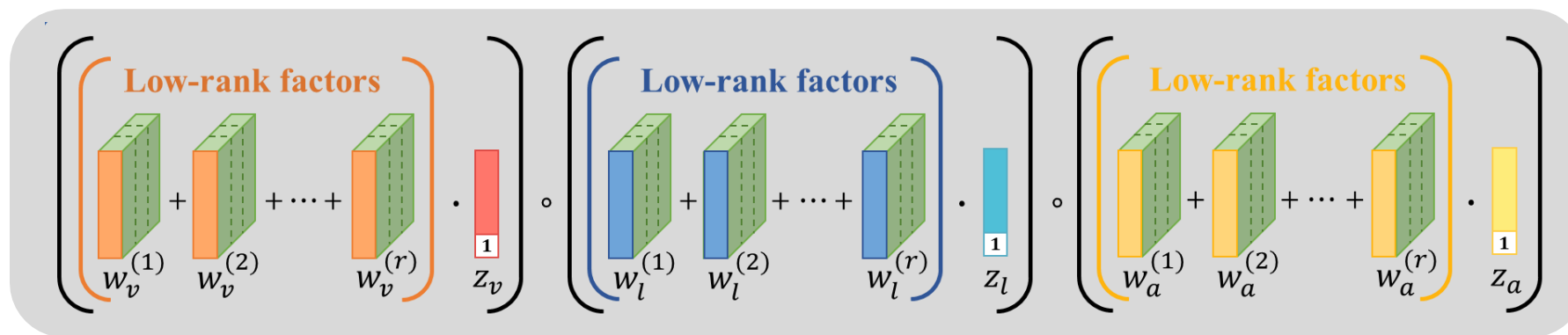


Low-rank Fusion with Trimodal Input

Tensor Fusion



Low-rank Fusion :



Canonical
Polyadic
Decomposition

Going Beyond Additive and Multiplicative Fusion

Additive interaction:

$$z = w_1x_A + w_2x_B$$

← First-order polynomial

Additive and multiplicative interaction:

$$z = w_1x_A + w_2x_B + w_3 x_A \times x_B$$

← Second-order polynomial

Trimodal fusion (e.g., tensor fusion):

$$z = w_1x_A + w_2x_B + w_3x_C + w_4 x_A \times x_C + w_5(x_A \times x_C) + w_6(x_B \times x_C) + w_7(x_A \times x_B \times x_C)$$

Unimodal terms
(first-order)

Bimodal terms
(second-order)

Trimodal terms
(third-order)

Can we add
higher-order
interaction terms?

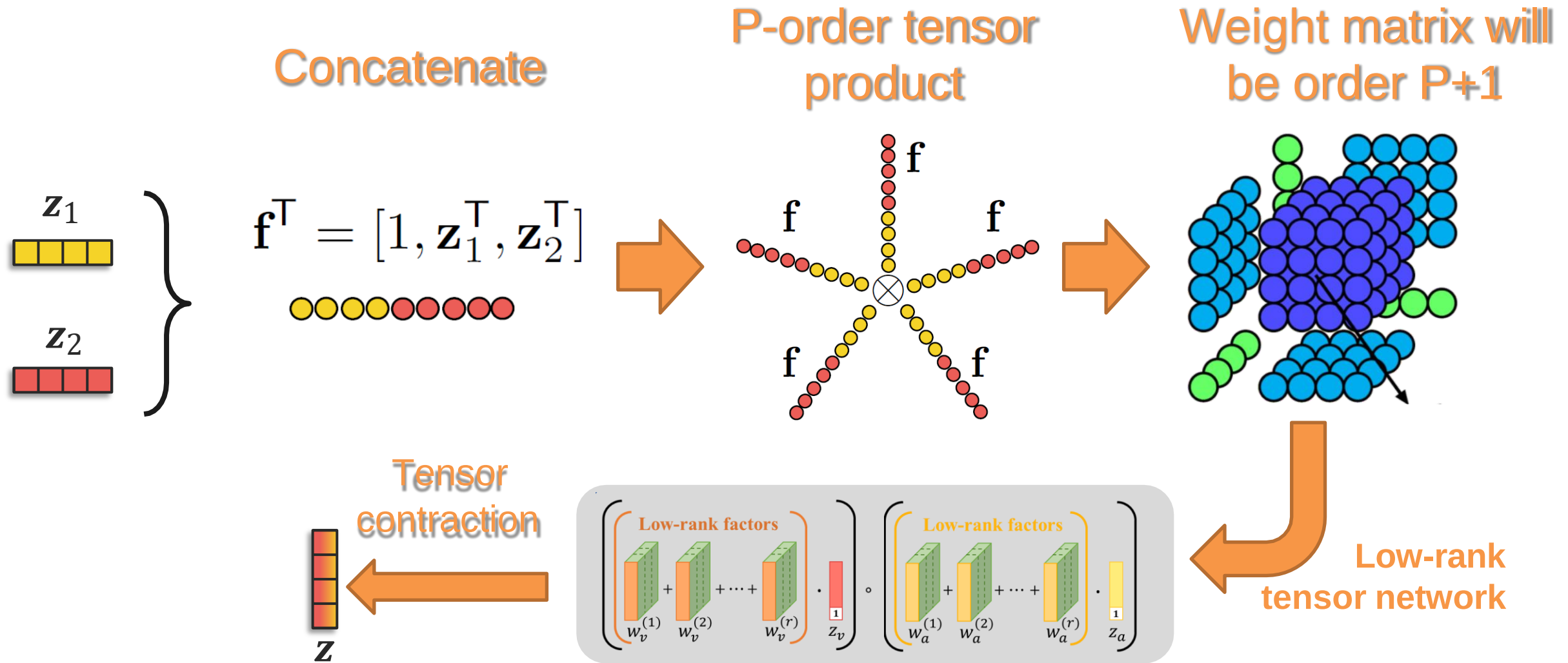
For example:

$$+w_8(x_A^2 \times x_B^2 \times x_C^2)$$

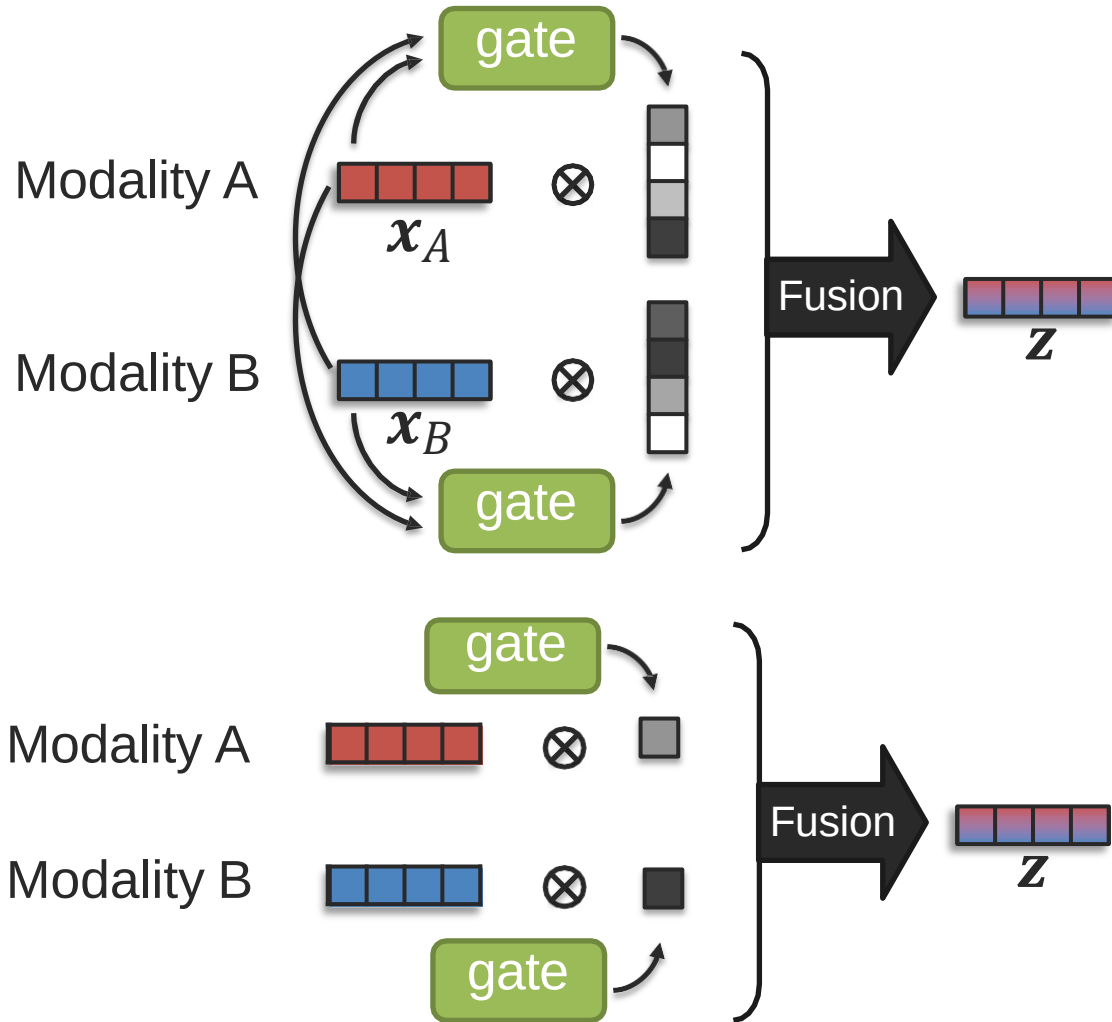
$$+w_9(x_A^3 \times x_B)$$

$$+w_{10}(x_B^3 \times x_C^3)$$

High-Order Polynomial Fusion



Gated Fusion



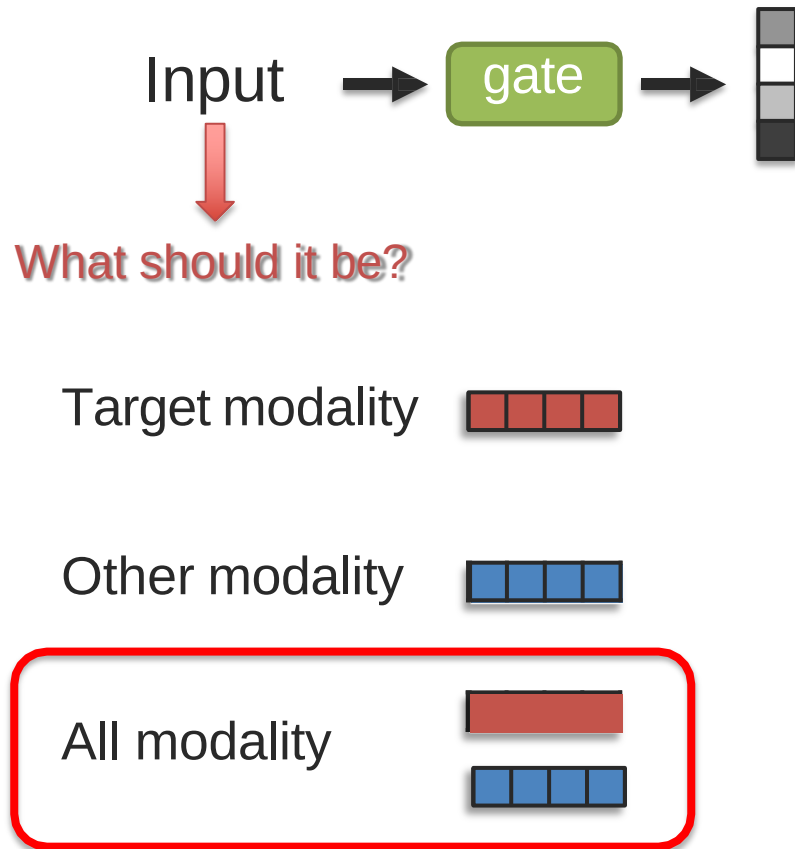
Example with additive fusion:

$$z = g_A(x_A, x_B) * x_A + g_B(x_A, x_B) * x_B$$

→ g_A and g_B can be seen as attention functions

Gating output can be one weight for the whole modality

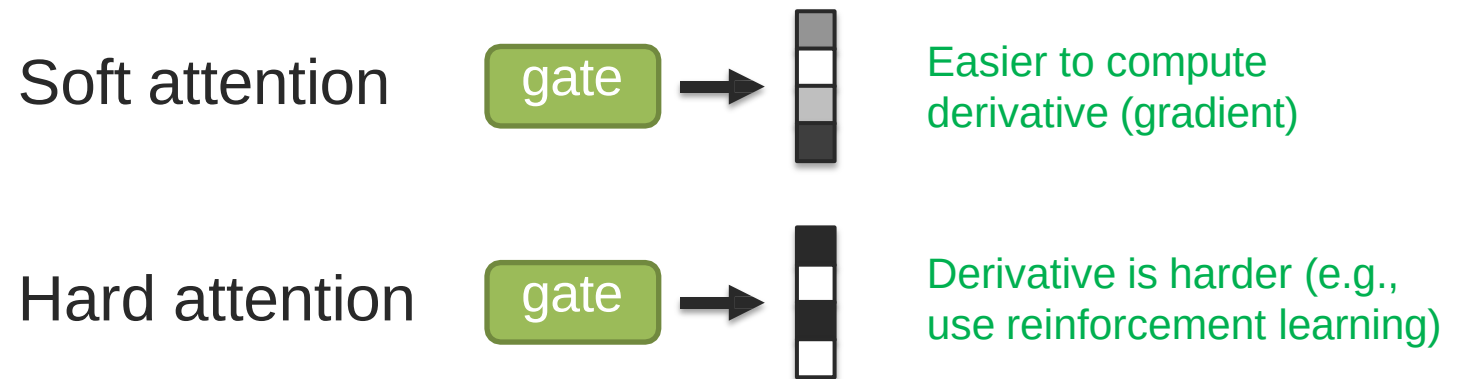
Gating Module (aka, attention module)



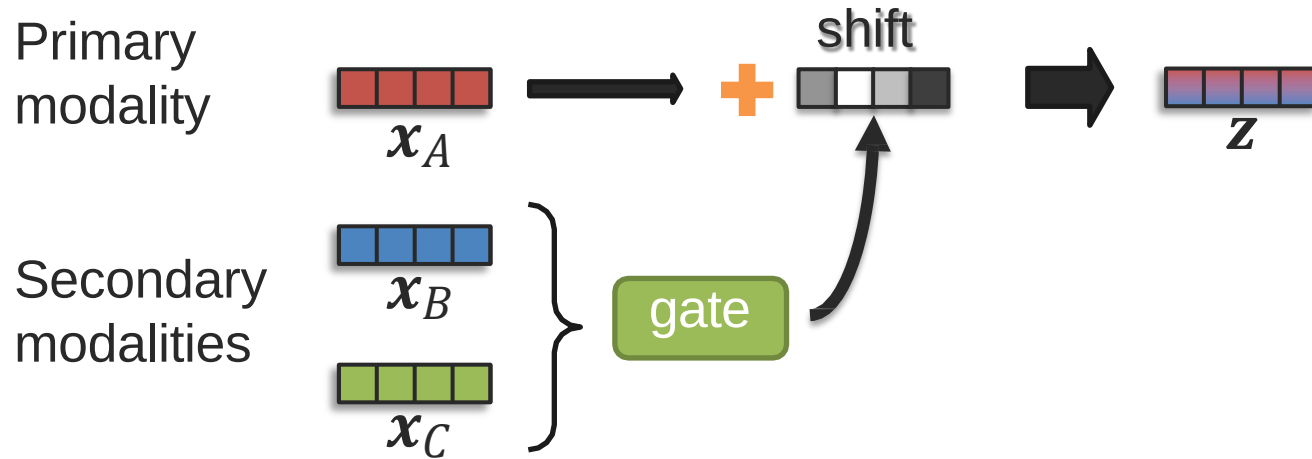
“Neural network designed to mask unwanted signal from propagating forward” (gating)

...or with a more positive view:

“Neural network designed to select preferable signal to move forward” (attention)



Modality-Shifting Fusion



Example with language modality:

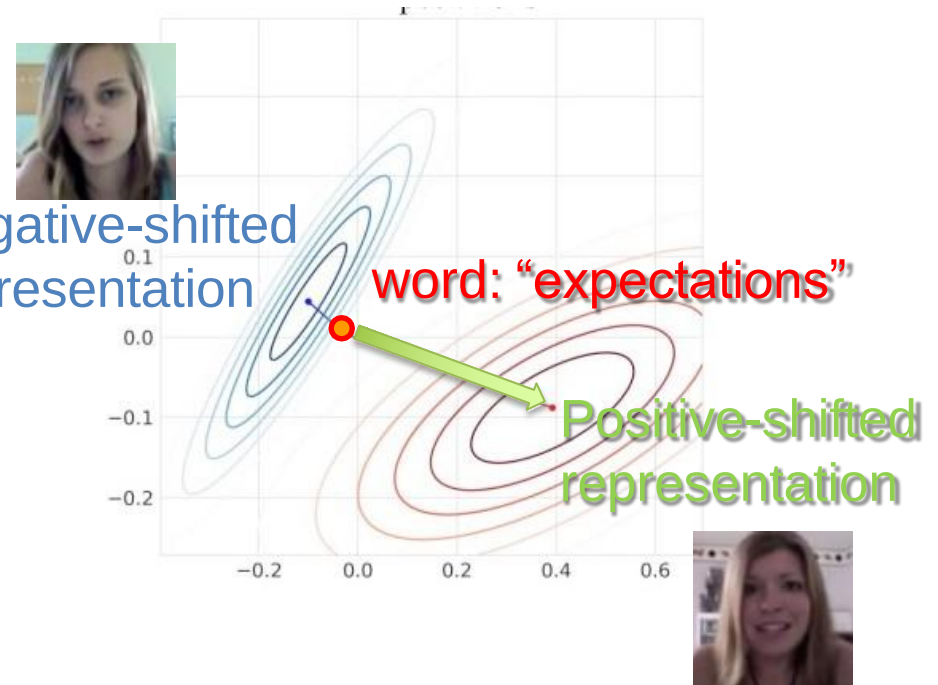
Primary modality: language

Secondary modalities: acoustic and visual

Negative-shifted
representation

word: "expectations"

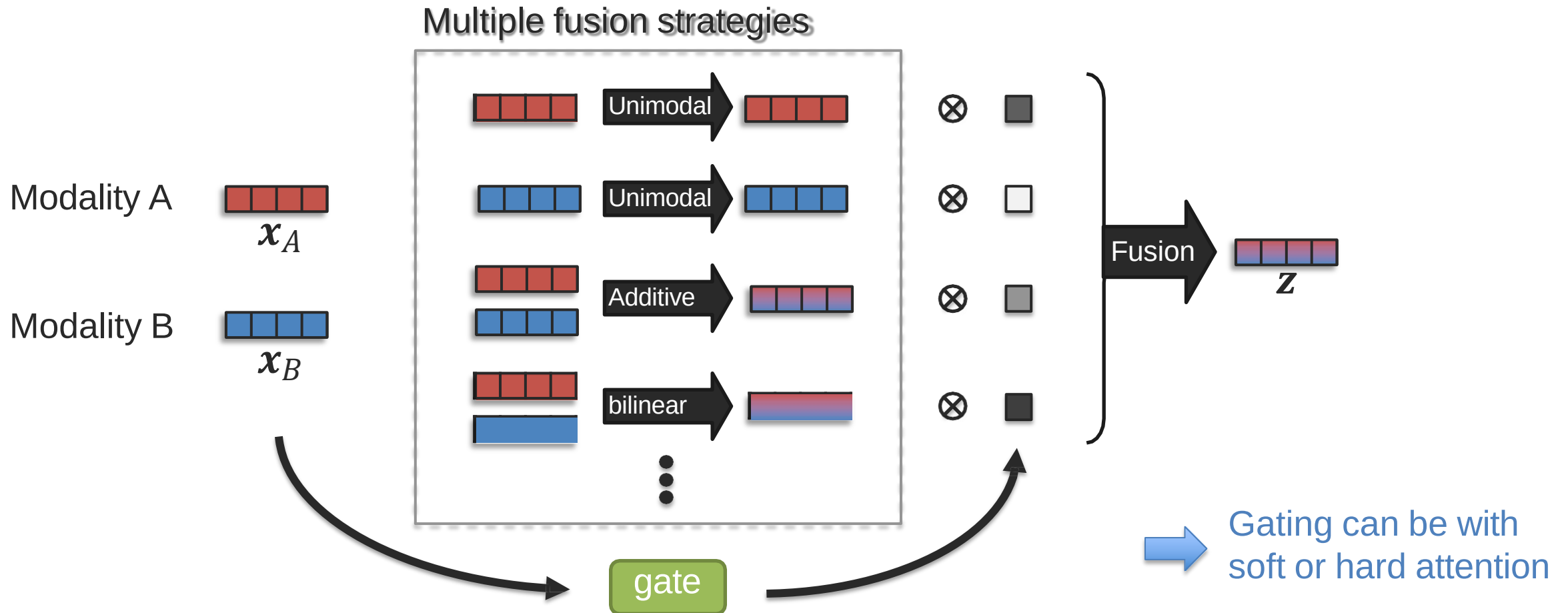
Positive-shifted
representation



[Wang et al., Words Can Shift: Dynamically Adjusting Word Representations Using Nonverbal Behaviors, AAAI 2019]

[Rahman et al., Integrating Multimodal Information in Large Pretrained Transformers, ACL 2020]

Mixture of Fusions



[Zadeh et al., Multimodal Language Analysis in the Wild: CMU-MOSEI Dataset and Interpretable Dynamic Fusion Graph, ACL 2018]

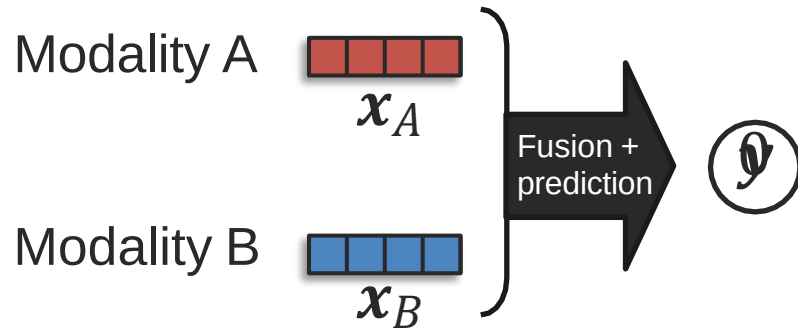
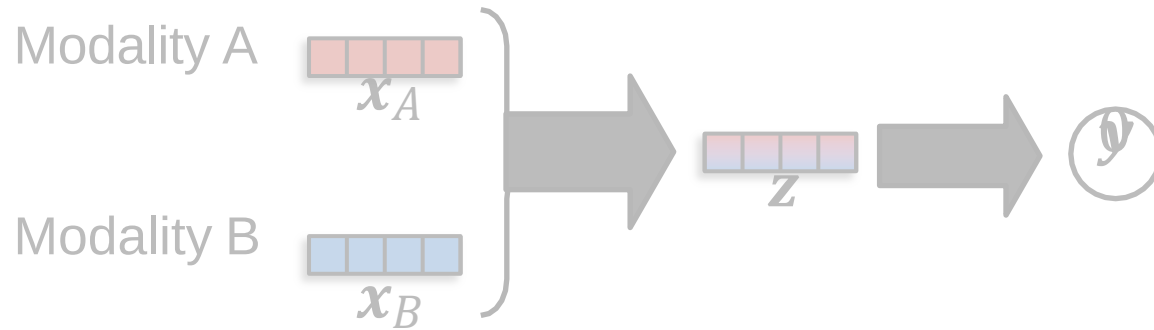
[Xu et al., MUFASA: Multimodal Fusion Architecture Search for Electronic Health Records, AAAI 2021]

Nonlinear Fusion

Nonlinear fusion:

$$\mathcal{Y} = f(\mathbf{x}_A, \mathbf{x}_B) \in \mathbb{R}^d$$

where f could be a multi-layer perceptron or any nonlinear model

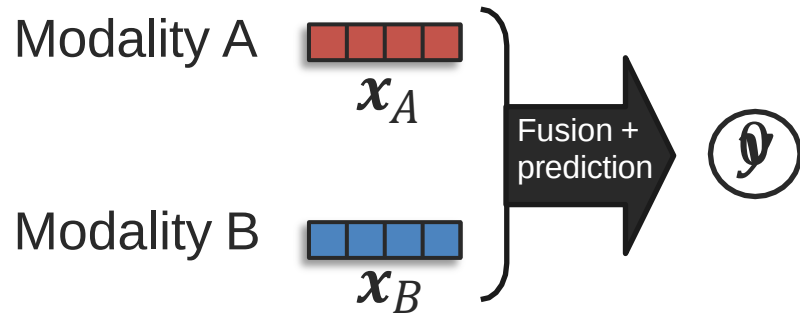


➡ This could be seen as *early fusion*:

$$\mathcal{Y} = f([\mathbf{x}_A, \mathbf{x}_B])$$

... but will our neural network learn the nonlinear interactions?

Measuring Non-Additive Interactions



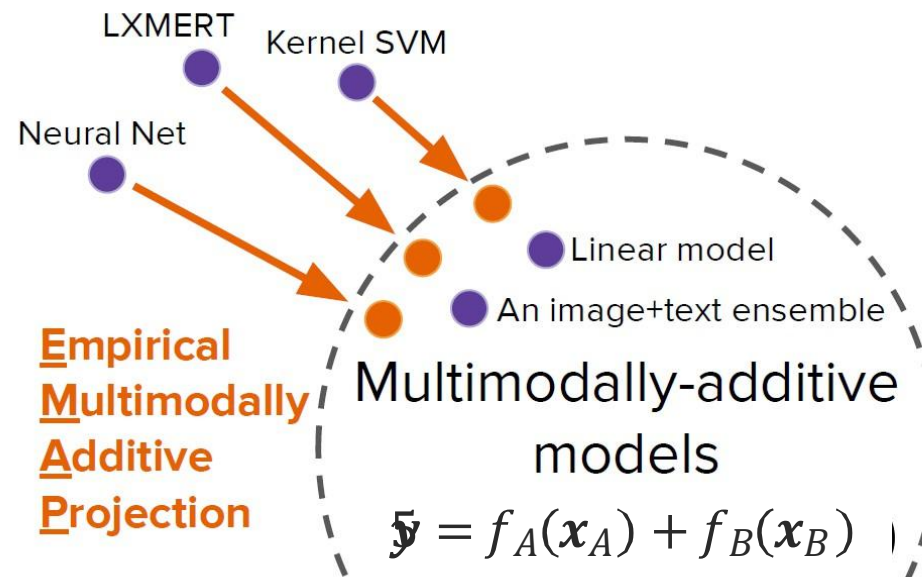
Nonlinear fusion:

$$\hat{y} = f(\mathbf{x}_A, \mathbf{x}_B)$$

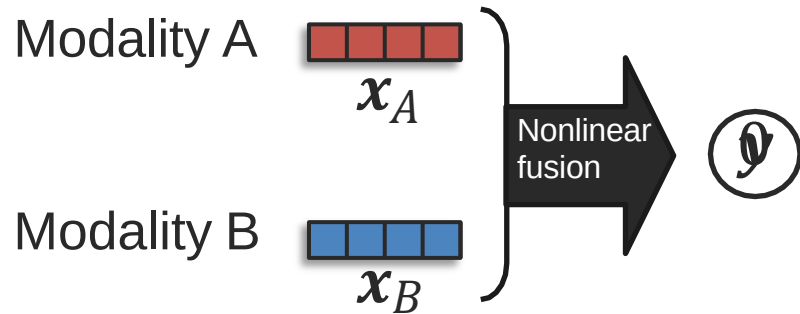
Projection?

Additive fusion:

$$\hat{y} = f_A(\mathbf{x}_A) + f_B(\mathbf{x}_B)$$



Measuring Non-Additive Interactions



Nonlinear fusion:

$$\vartheta = f(\mathbf{x}_A, \mathbf{x}_B)$$

Additive fusion:

$$\vartheta' = f_A(\mathbf{x}_A) + f_B(\mathbf{x}_B)$$

Projection?

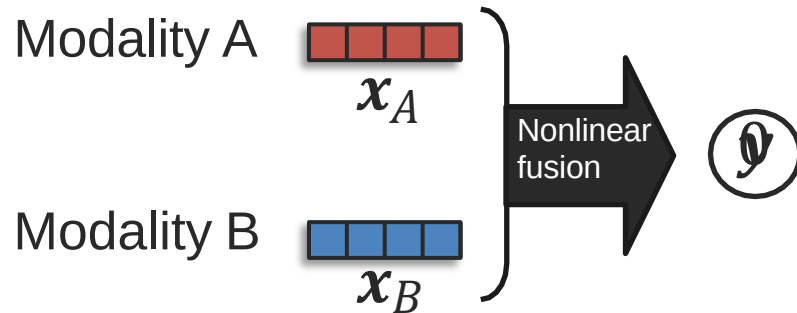
Projection from nonlinear to additive (using EMAP):

$$\hat{\beta}(\mathbf{x}_A, \mathbf{x}_B) = \underbrace{\mathbb{E}_{\mathbf{x}_B} [f(\mathbf{x}_A, \mathbf{x}_B)]}_{f_A(\mathbf{x}_A)} + \underbrace{\mathbb{E}_{\mathbf{x}_A} [f(\mathbf{x}_A, \mathbf{x}_B)]}_{f_B(\mathbf{x}_B)}$$

Modality A + Modality B

Additive fusion
(approximation)

Measuring Non-Additive Interactions



Nonlinear fusion:

$$\phi = f(x_A, x_B)$$

EMAP
projection

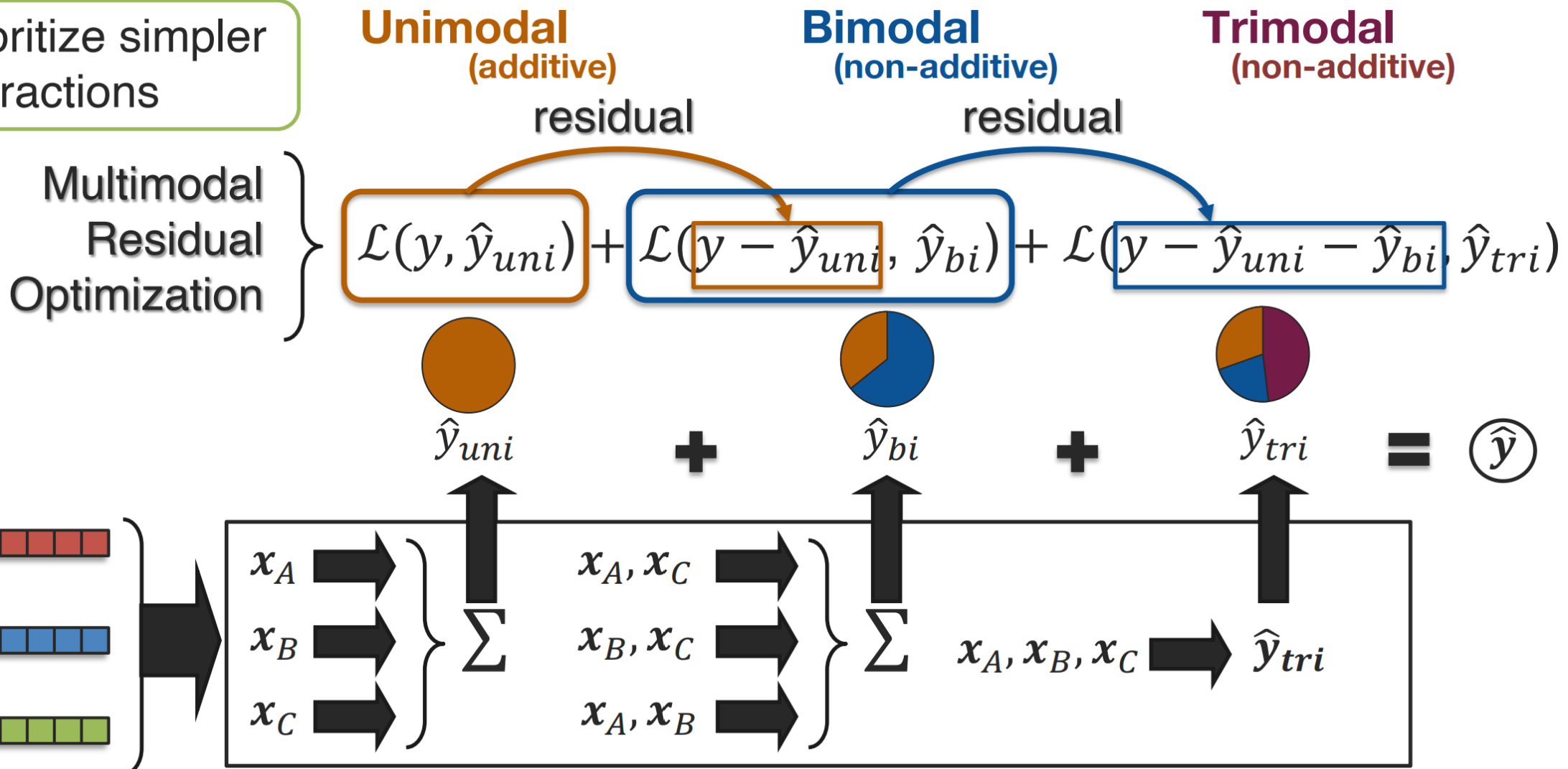
Additive fusion:

$$\phi^@ = f_A(x_A) + f_B(x_B) + \mu$$

		I-INT	I-SEM	I-CTX	T-VIS	R-POP	T-ST1	T-ST2	
Nonlinear	Neural Network	90.4	69.2	78.5	51.1	63.5	71.1	79.9	
Polynomial	Polykernel SVM	91.3	74.4	81.5	50.8	—	72.1	80.9	
Nonlinear	FT LXMERT	83.0	68.5	76.3	53.0	63.0	66.4	78.6	
Nonlinear	$\hookrightarrow$ + Linear Logits	89.9	73.0	80.7	53.4	64.1	75.5	80.3	
Additive	Linear Model	90.4	72.8	80.9	51.3	63.7	75.6	76.1	
	Best Model	91.3	74.4	81.5	53.4	64.2	75.5	80.9	
Additive	$\hookrightarrow$ + EMAP	91.1	74.2	81.3	51.0	64.1	75.9	80.7	Always a good baseline! Differences are small!!!!

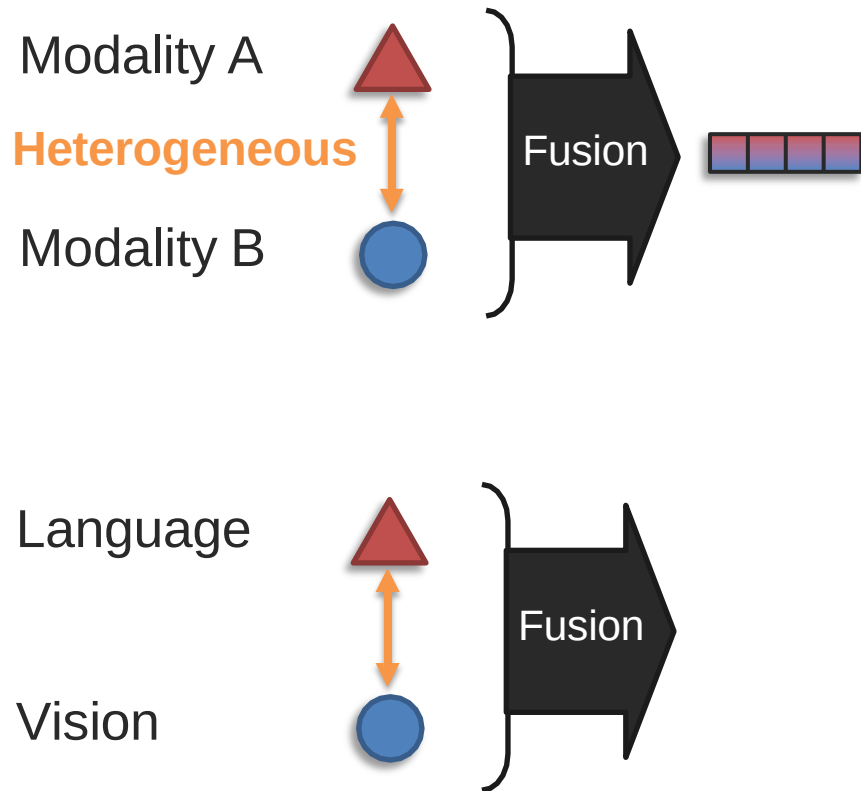
Learning Non-additive Bimodal and Trimodal Interactions

Idea: prioritize simpler interactions



Fusion with Heterogeneous Modalities

Goal: Fusion with raw modalities



Can the same fusion algorithm handle raw heterogeneous modalities?

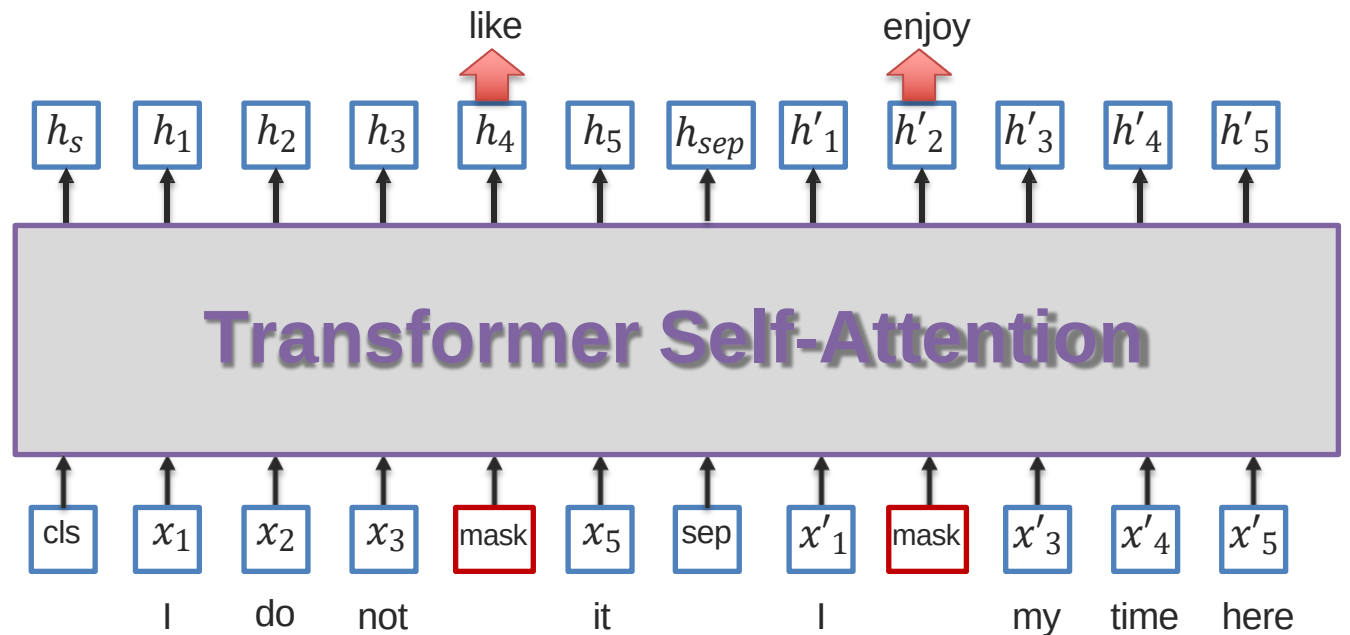
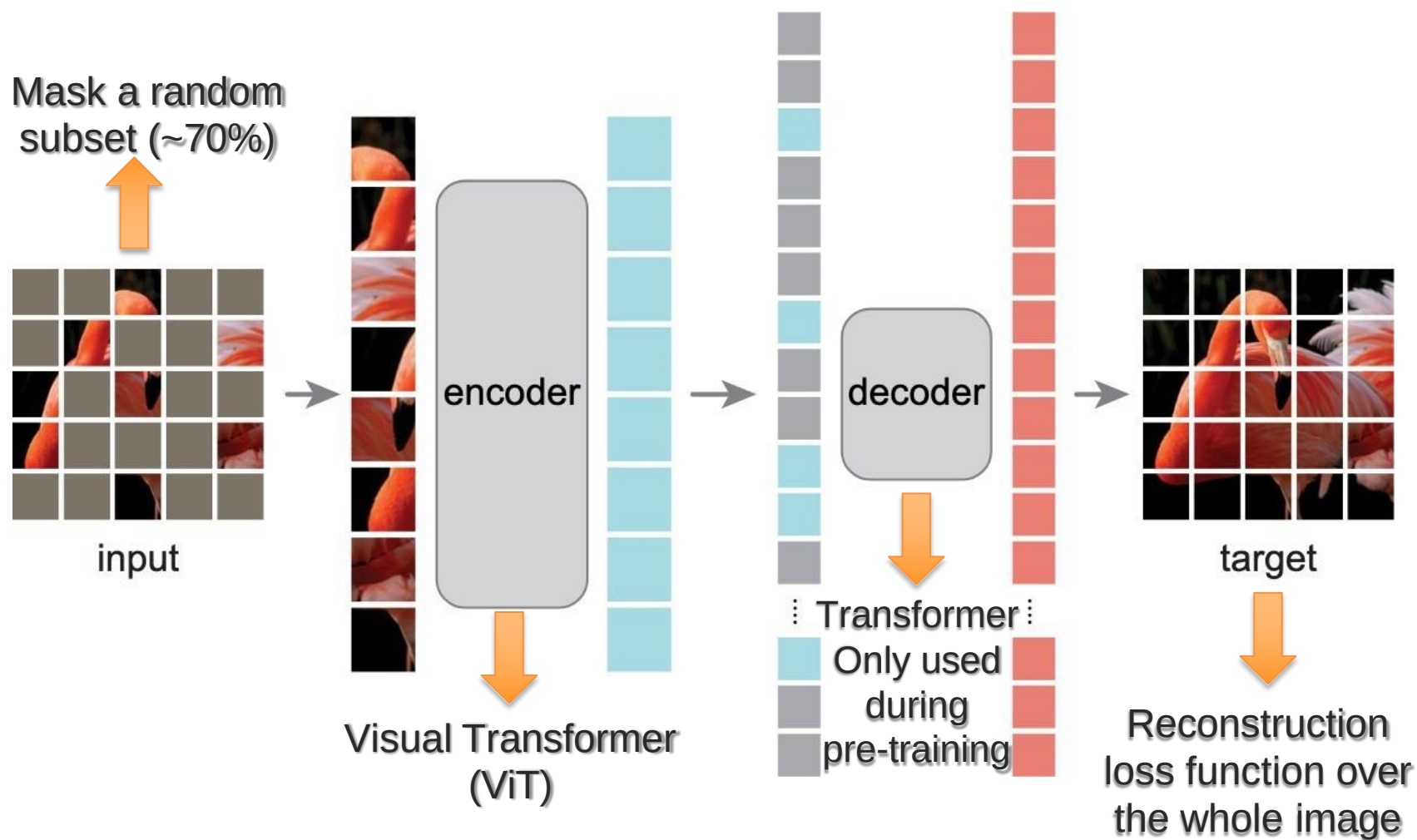
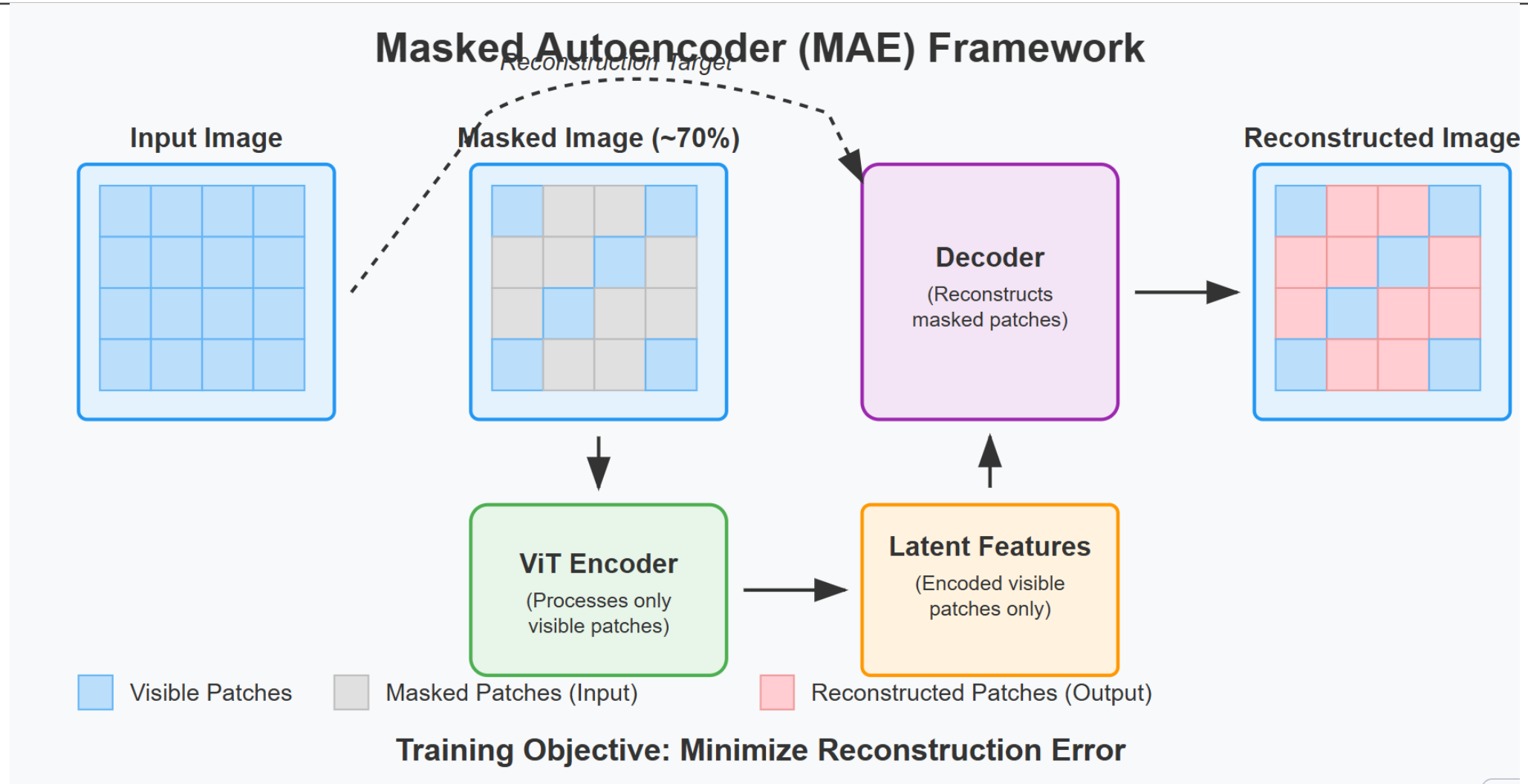


Image Representation Learning: Masked Auto-Encoder (MAE)

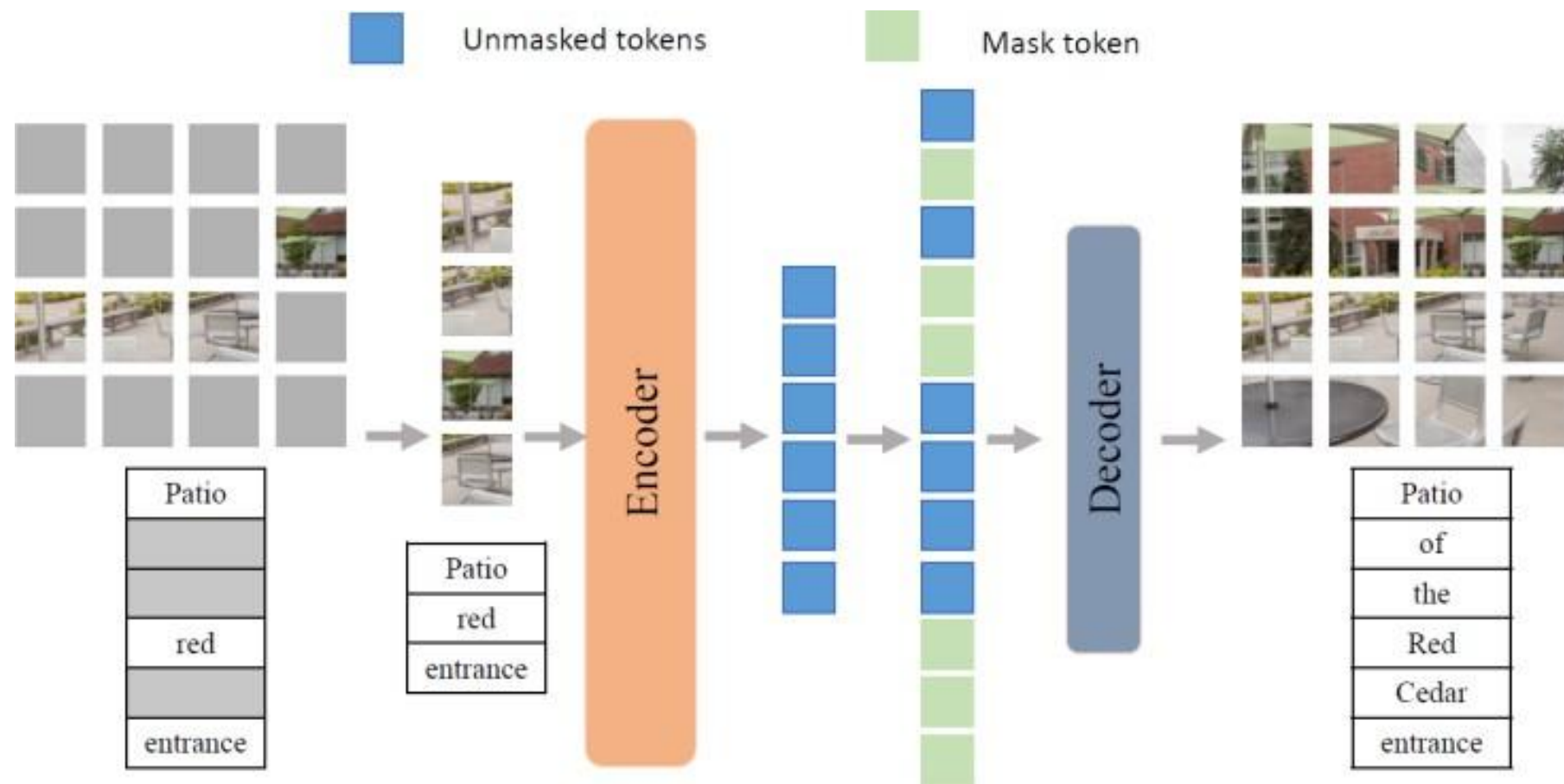


[He et al., Masked Autoencoders Are Scalable Vision Learners, CVPR 2022]

Image Representation Learning: Masked Auto-Encoder (MAE)



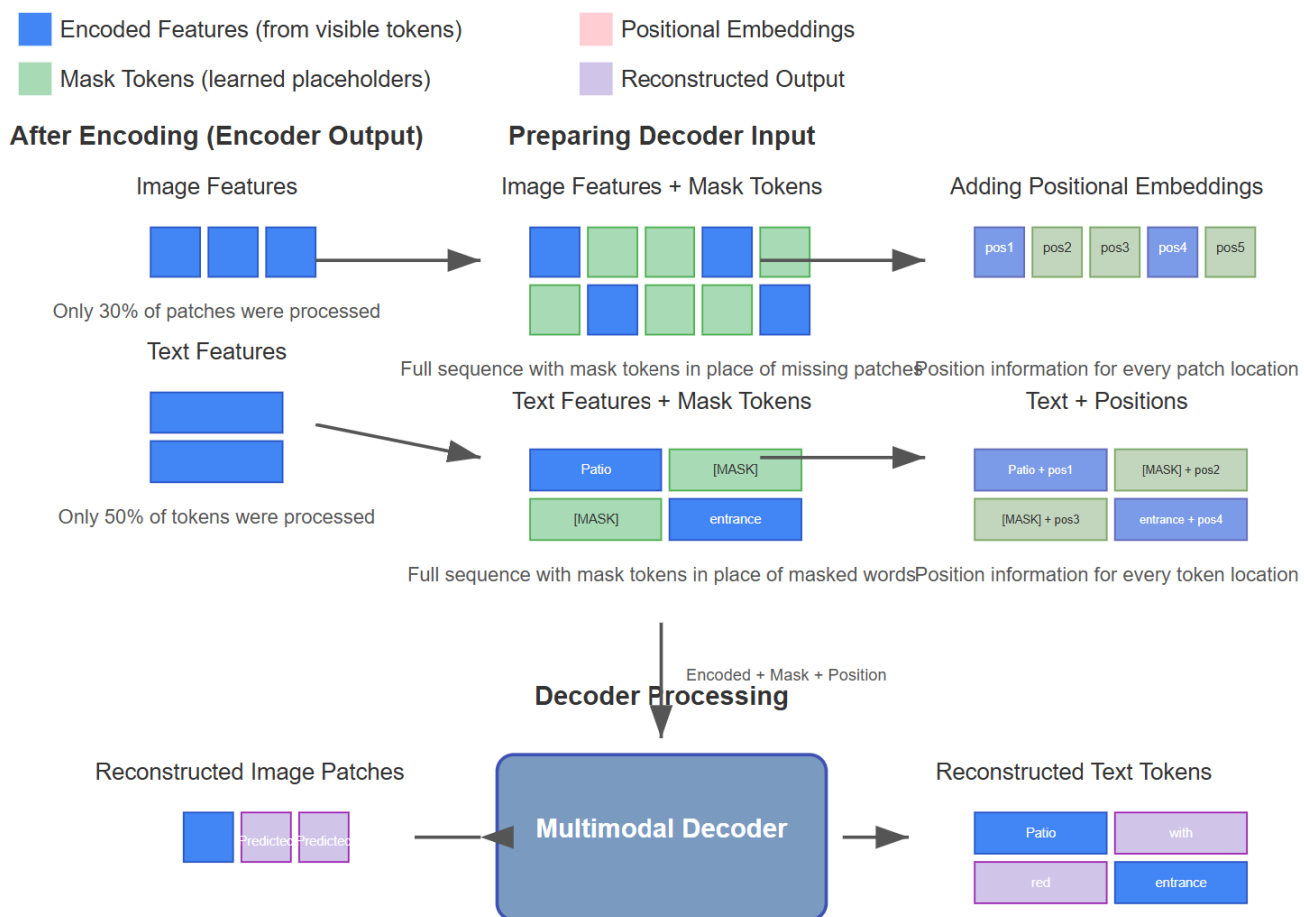
Multimodal Masked Autoencoder



Multimodal Masked Autoencoder

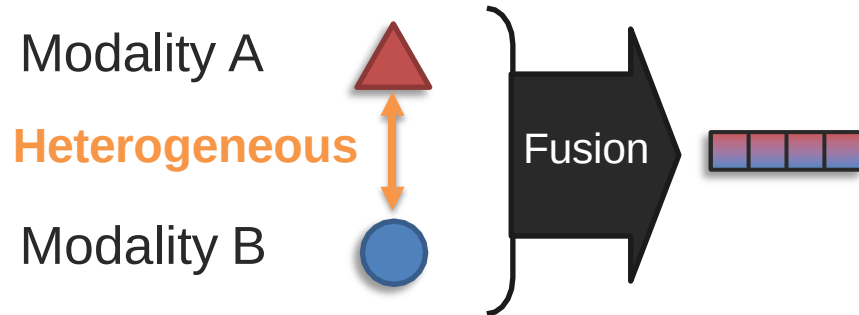
Mask Token Insertion in MMAE Decoder

Decoding Phase of Multimodal Masked Autoencoder

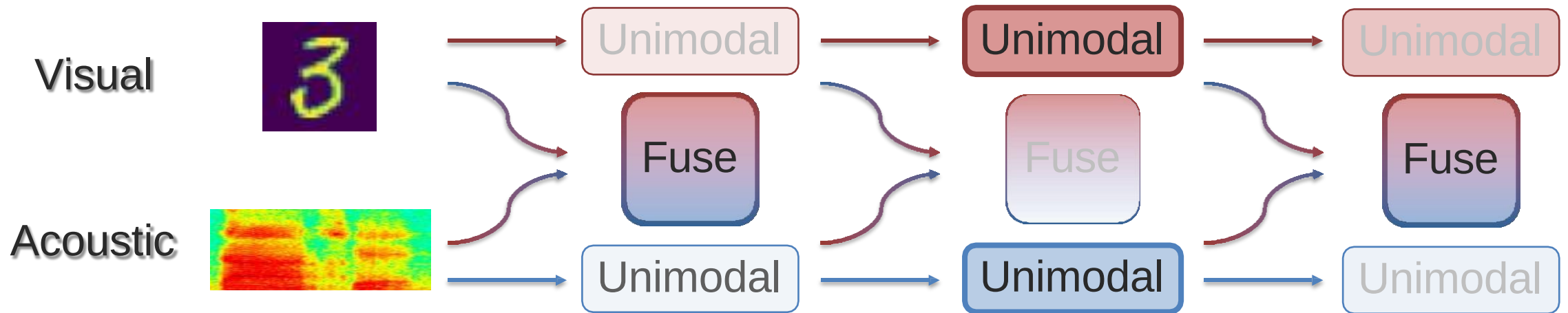


[Based on Geng et al., Multimodal Masked Autoencoders Learn Transferable Representations, 2022]

Dynamic Early Fusion

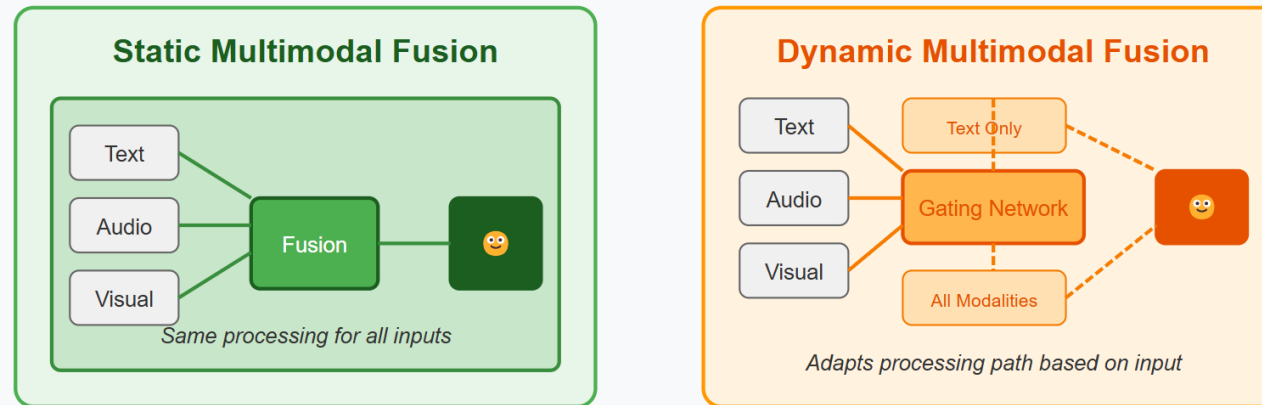


Idea: Deciding when to fuse in early fusion

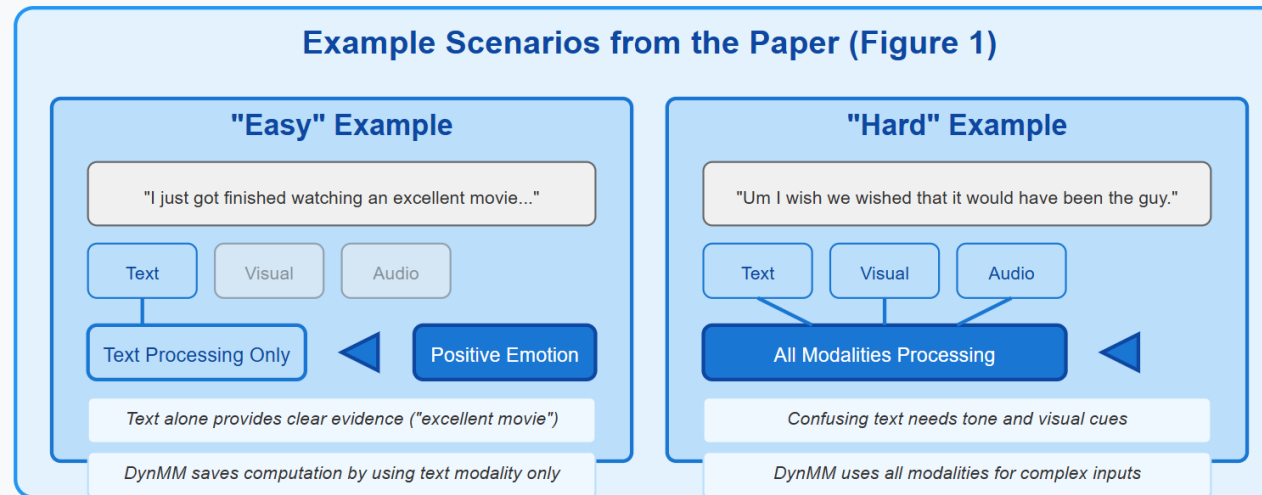


Dynamic Early Fusion

Dynamic Multimodal Fusion vs. Static Fusion



Example Scenarios from the Paper (Figure 1)



Dynamic Early Fusion

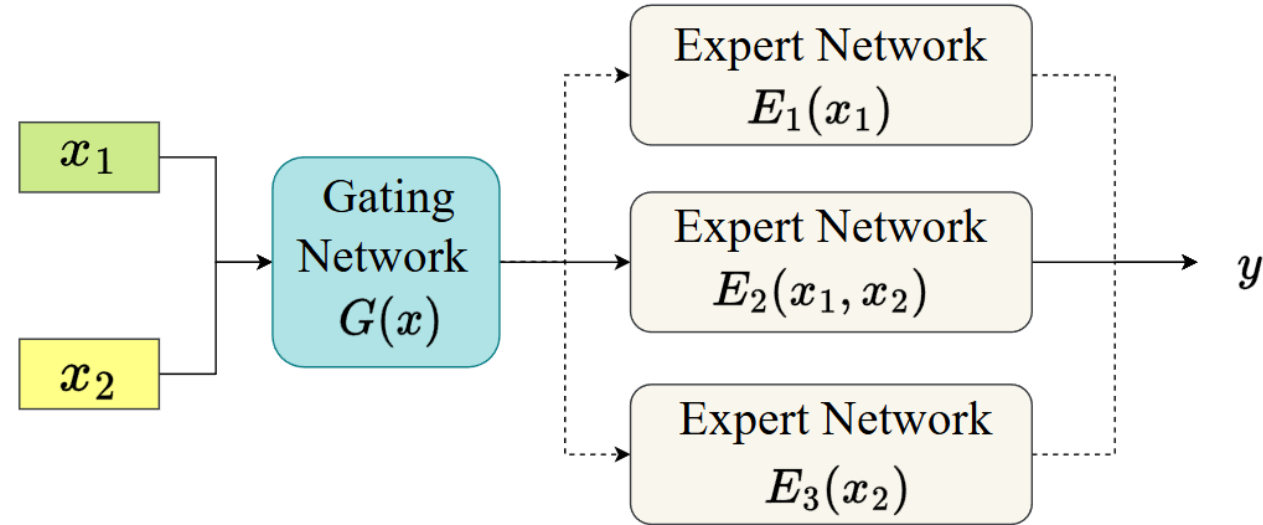


Figure 2. An illustration of modality-level DynMM, where input data has two modalities, denoted by x_1 and x_2 , and the output is denoted by y . We design a set of expert networks $\{E_i\}$ that specialize in different subsets of modalities and adopt a gating network $G(\mathbf{x})$ to generate data-dependent decisions on which expert network to select.

Dynamic Early Fusion

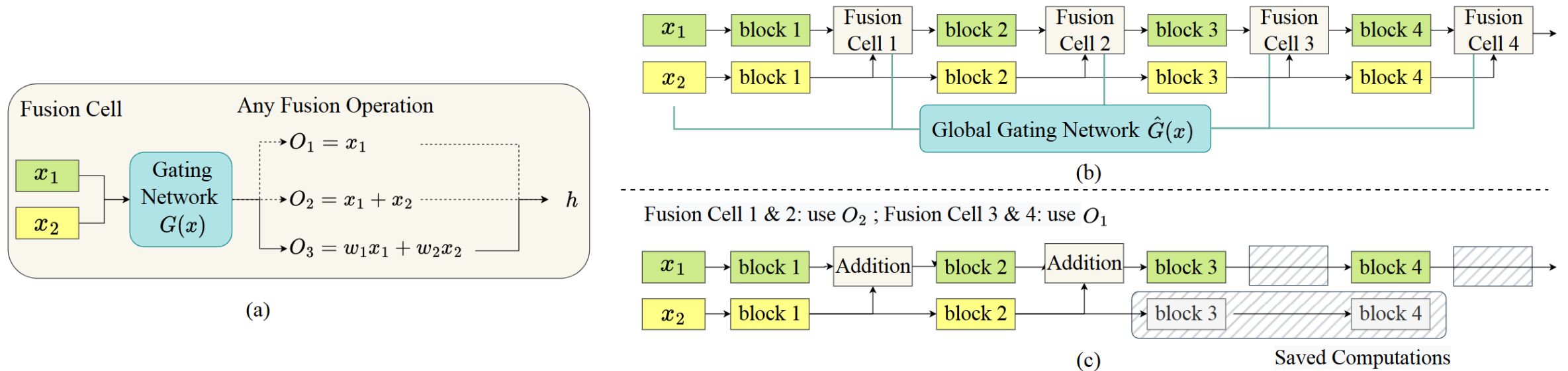
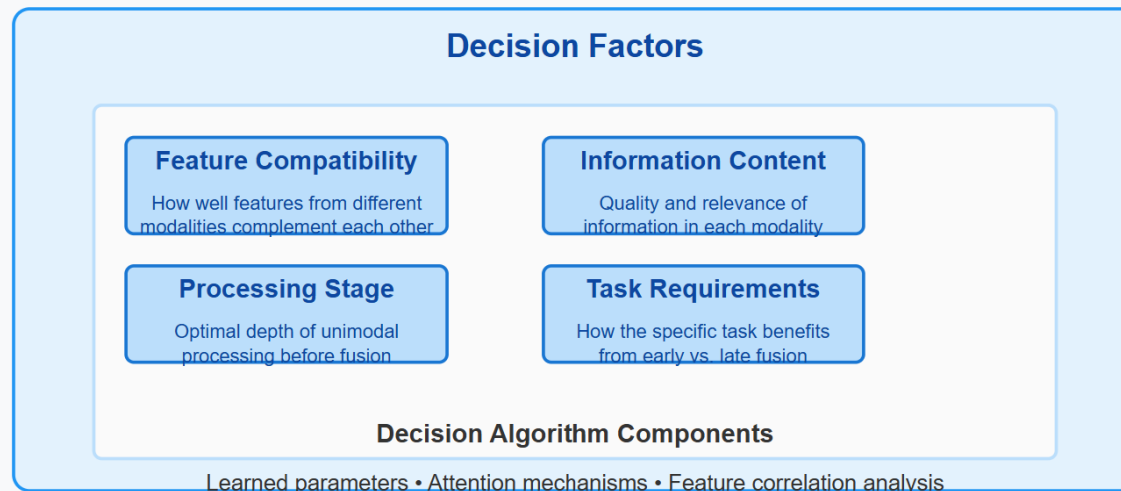
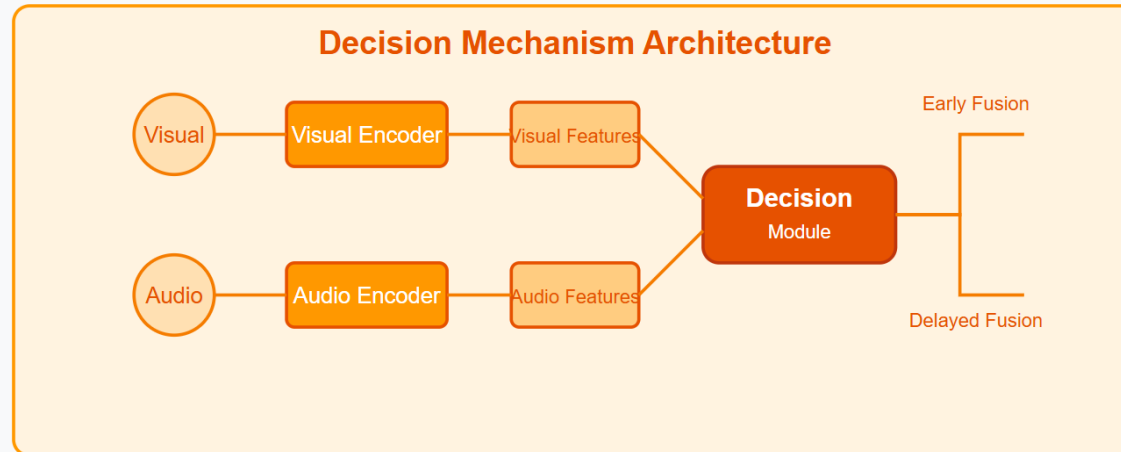


Figure 3. (a) An illustration of fusion-level DynMM, where input data has two modalities, denoted by x_1 and x_2 . We design a fusion cell with a set of candidate operations $\{O_i\}$ and a gating network $G(\mathbf{x})$. h represents output of the cell. (b) A dynamic multimodal architecture with stacked fusion cells, where we interlace static feature extraction blocks (colored with green and yellow) with dynamic fusion cells. Gating network $G(\mathbf{x})$ in four fusion cells are integrated as one global gating network $\hat{G}(\mathbf{x})$ that outputs decisions for four cells at once. (c) An example architecture when the gating network chooses O_2 for the first 2 fusion cells and O_1 for the last 2 cells. Consequently, computations of fusion cell 3 & 4 and feature extraction cell 3 & 4 for x_2 are saved.

Dynamic Early Fusion

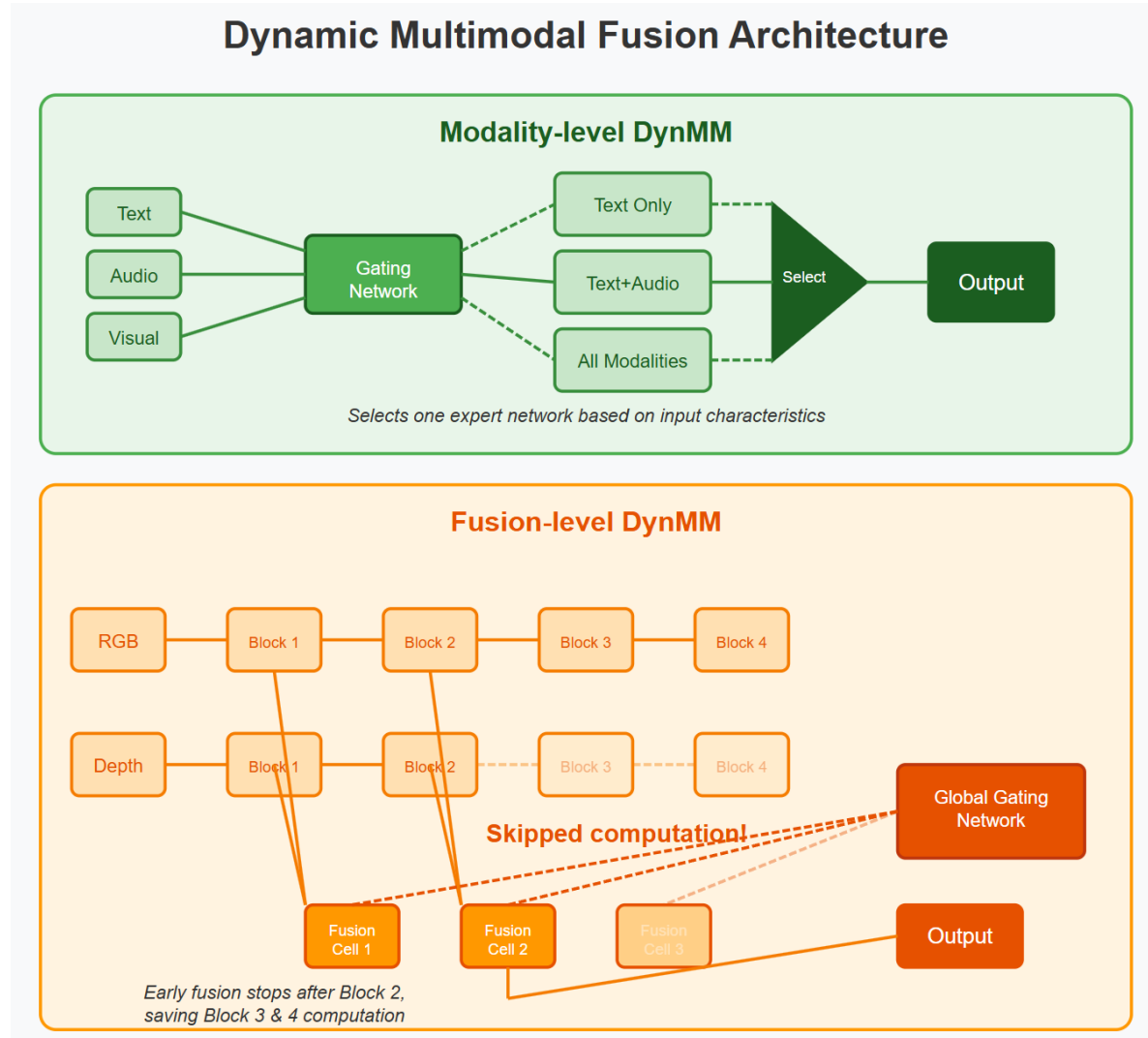
Dynamic Early Fusion - Decision Mechanism



$$\mathcal{L} = \mathcal{L}_{task} + \lambda \sum_{i=1}^B g_i C(E_i) \quad (\text{modality-level}) \quad (1)$$

$$\mathcal{L} = \mathcal{L}_{task} + \lambda \sum_{j=1}^F \sum_{i=1}^B g_i^{(j)} C(O_{i,j}) \quad (\text{fusion-level}) \quad (2)$$

Dynamic Early Fusion



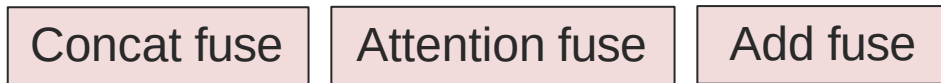
Dynamic Early Fusion

Fusion fully learned from optimization and data

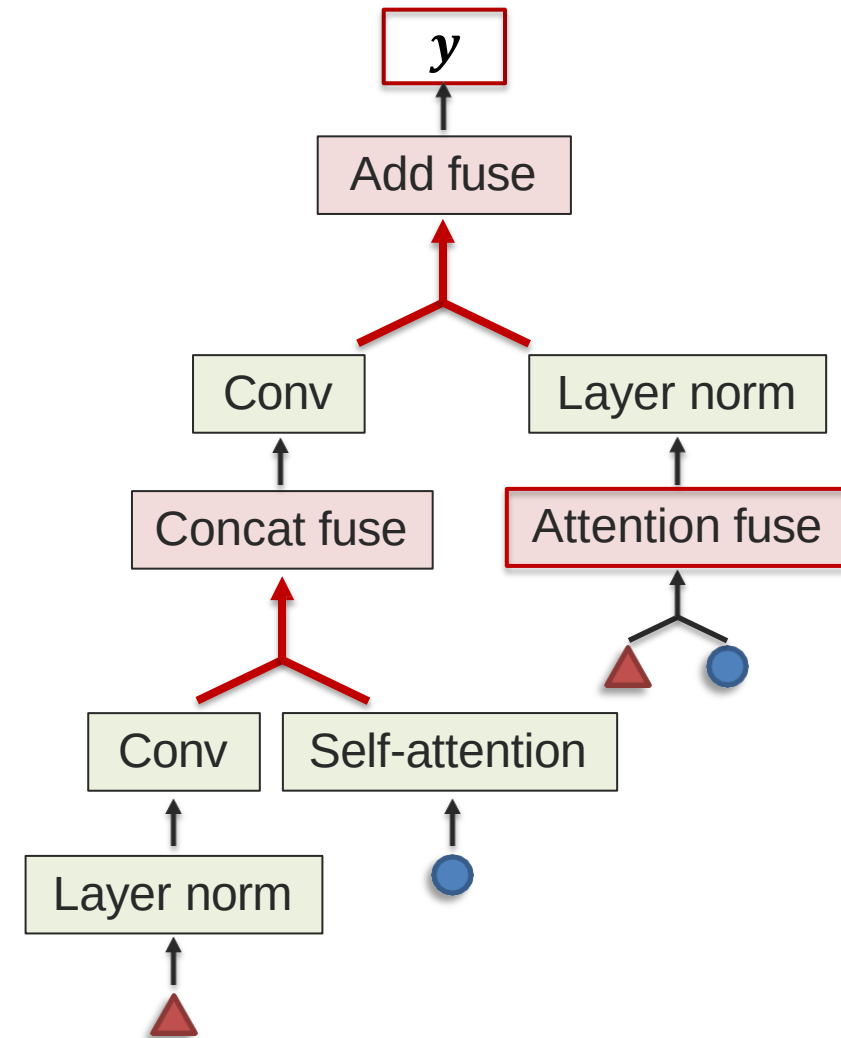
1. Define basic representation building blocks



2. Define basic fusion building blocks



3. Automatically search for composition using neural architecture search



[Xu et al., MUFASA: Multimodal Fusion Architecture Search for Electronic Health Records. AAAI 2021]

[Liu et al., DARTS: Differentiable Architecture Search. ICLR 2019]

Dynamic Early Fusion

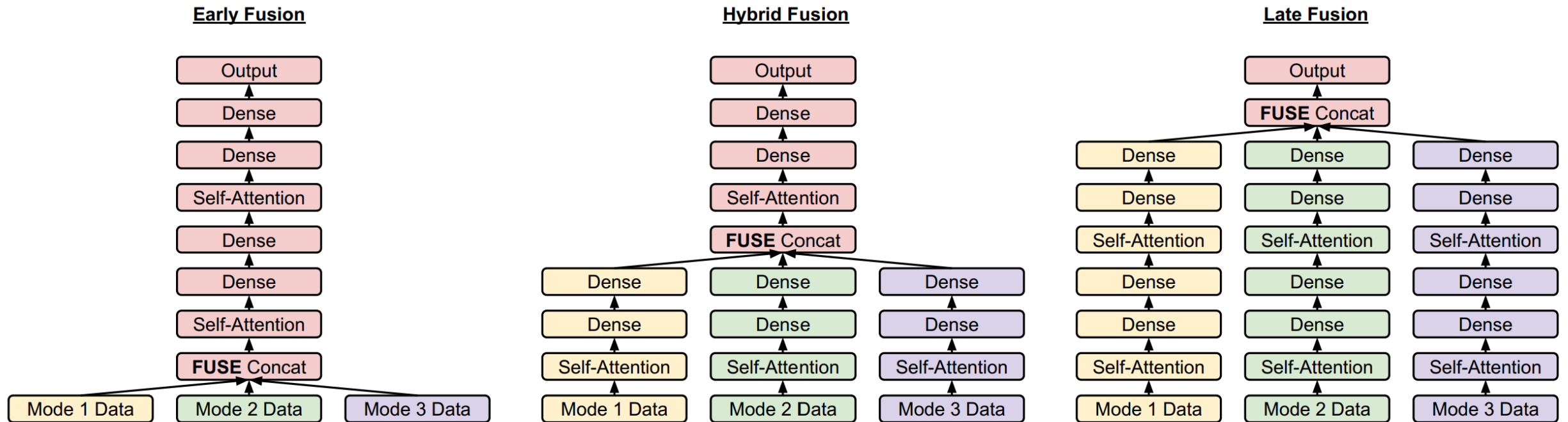


Figure 2: Example of different fusion strategies using a Transformer base architecture. In each depiction, yellow represents modeling only applied to Mode 1, green Mode 2, and purple Mode 3. Red represents modeling applied to all three modes jointly.

Dynamic Early Fusion

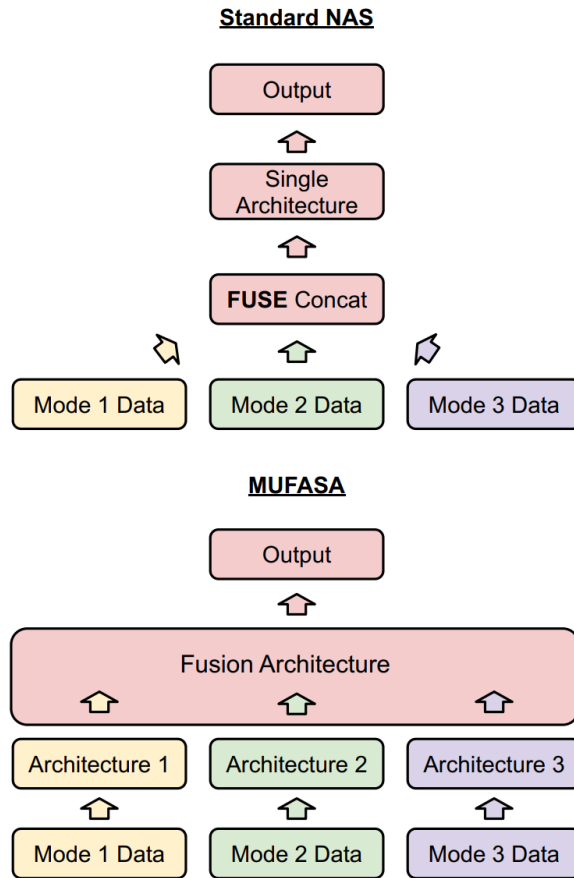


Figure 3: Standard NAS searches for a single unimodal architecture. MUFASA searches for independent architectures for each data modality, as well as a Fusion Architecture to tie those modal architectures together.

[Xue and Marculescu, Dynamic Multimodal Fusion, arxiv 2022]

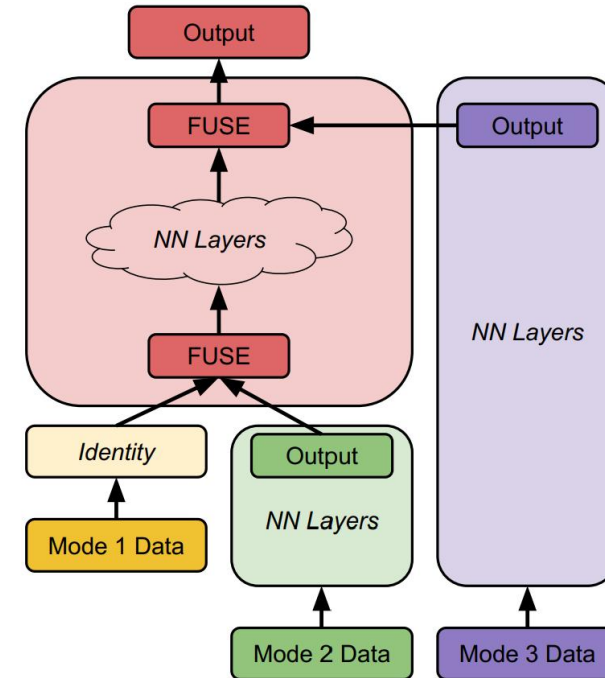
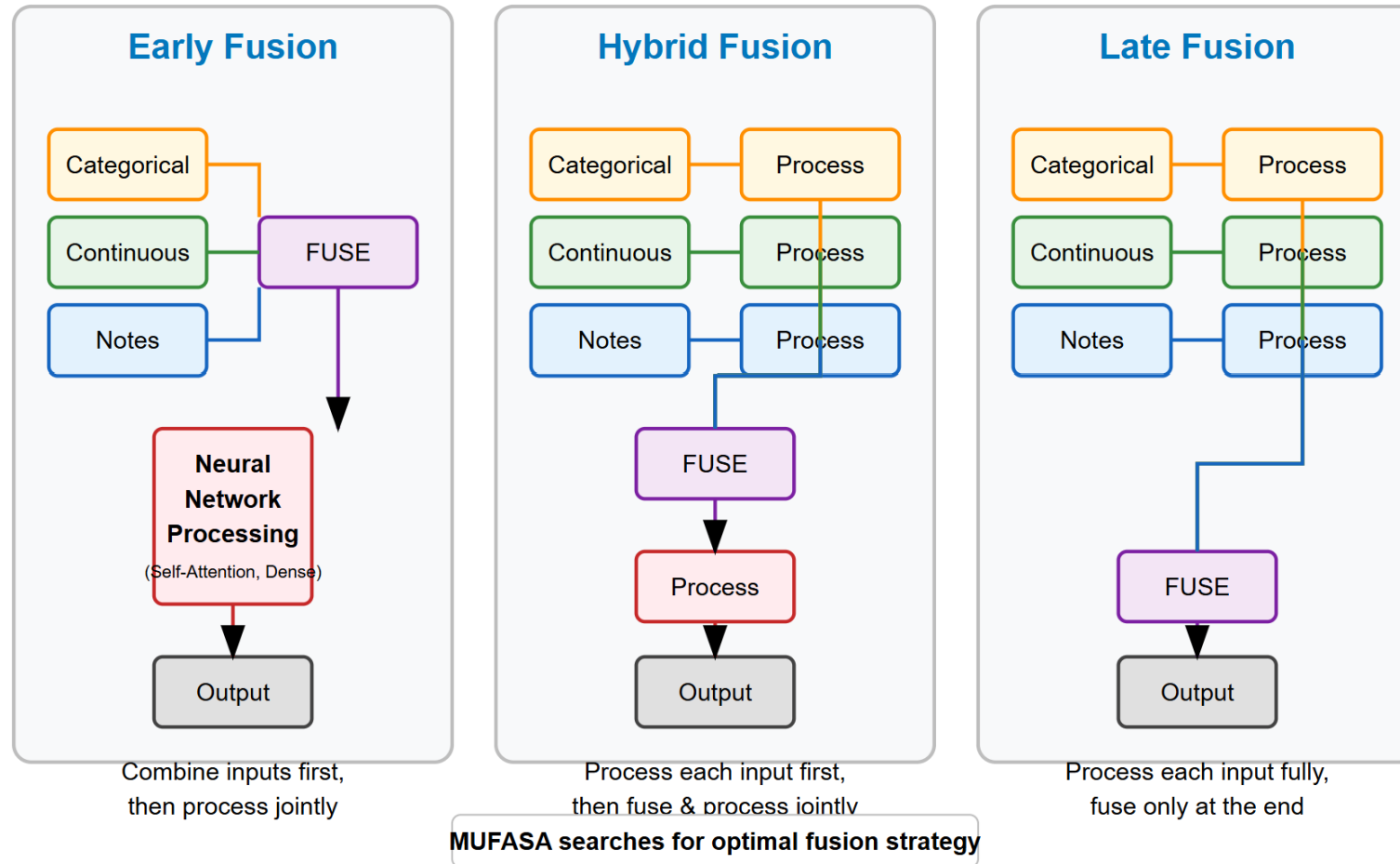


Figure 4: MUFASA example architecture that employs all three fusion types. Mode 1 utilizes early fusion, as its modality architecture is an identity transformation and thus it is fused with Mode 2 before it passes through any neural network (NN) layers. Mode 2 utilizes hybrid fusion, as it first passes through its own non-identity modality architecture before being fused with Mode 1 and transformed by the fusion architecture (in red). Mode 3 utilizes late fusion, as it only receives processing through its independent modality architecture, before being fused with the final output.

Dynamic Early Fusion

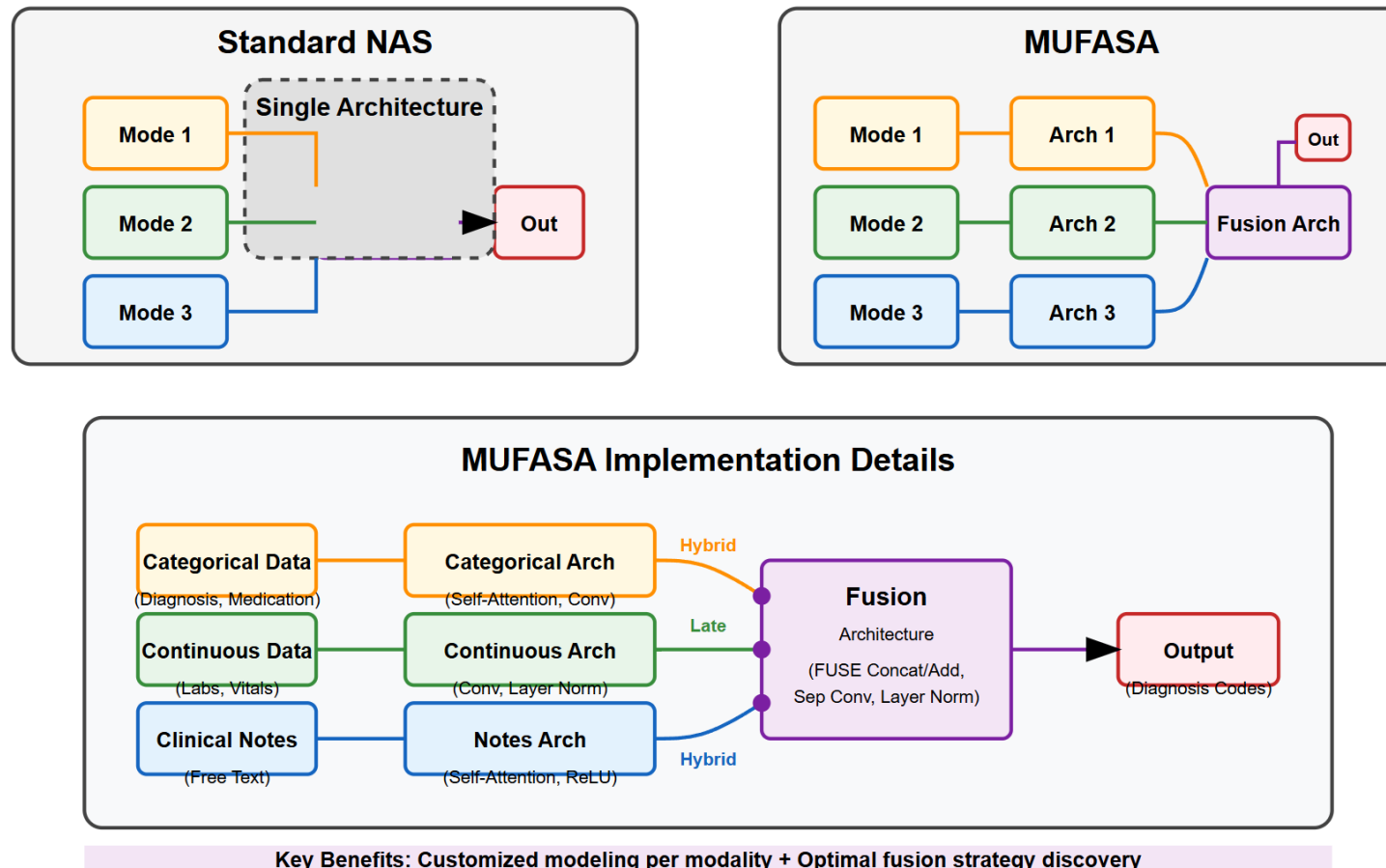
Multimodal Fusion Strategies for EHR Data



Dynamic Early Fusion

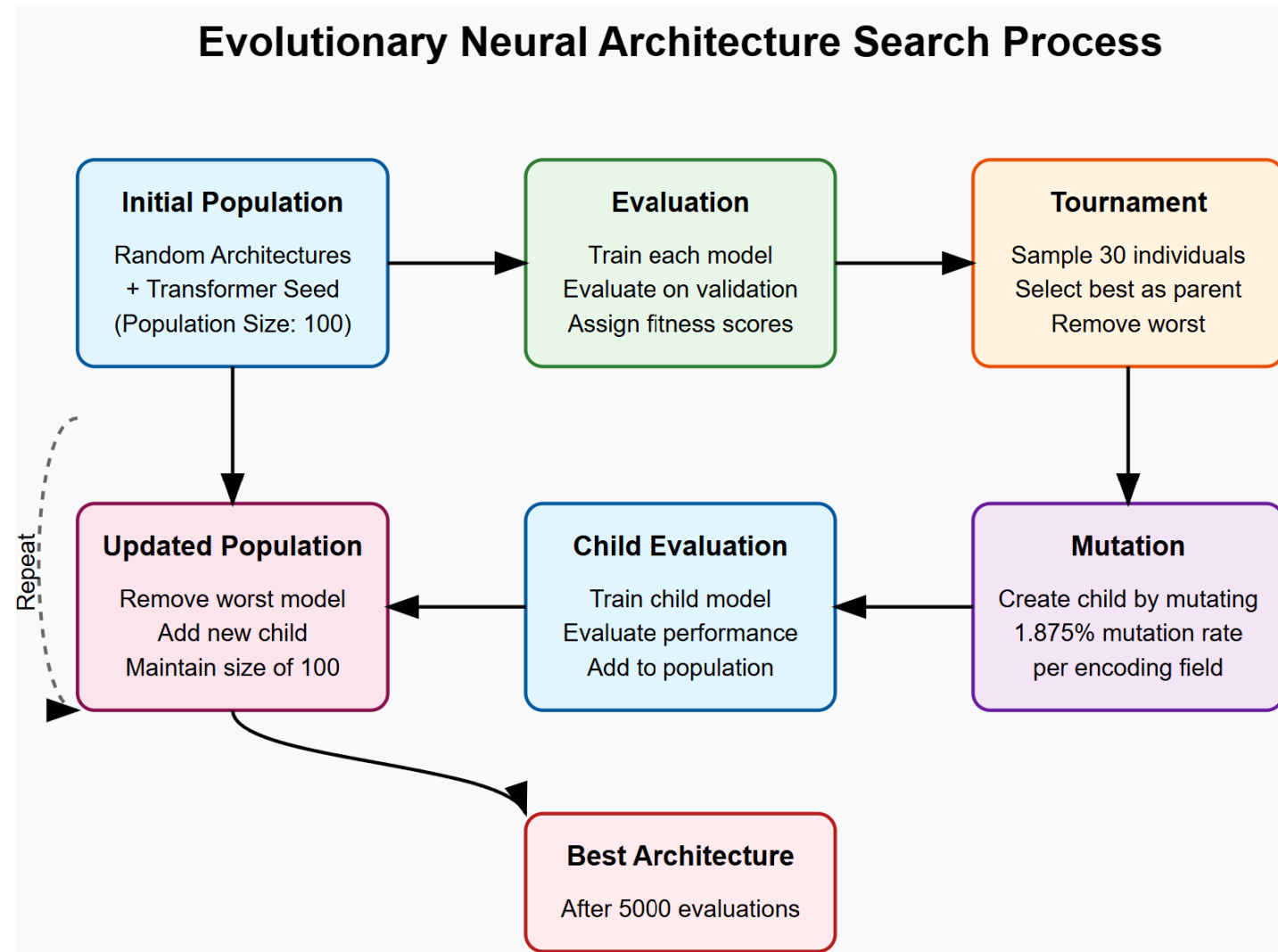
MUFASA: Multimodal Fusion Architecture Search

Automatically discovering optimal architectures for each modality and fusion strategy



[Xue and Marculescu, Dynamic Multimodal Fusion, arxiv 2022]

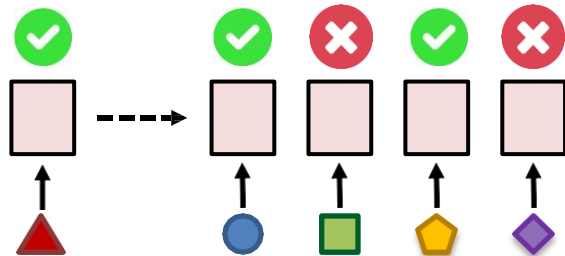
Dynamic Early Fusion



Heterogeneity-aware Fusion

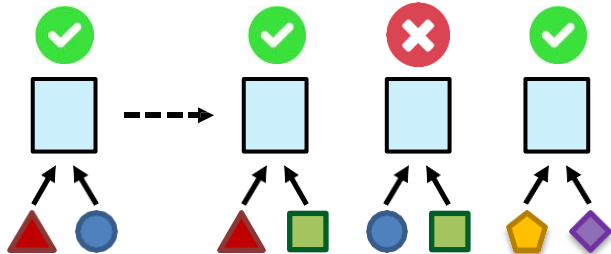
Information transfer, transfer learning perspective

1a. Estimate modality heterogeneity via transfer



(Implicitly captures heterogeneity)

1b. Estimate interaction heterogeneity via transfer



2a. Compute modality heterogeneity matrix

	0				
	1	0			
	3	2	0		
	1	2	3	0	
	5	4	6	3	0

2b. Compute interaction heterogeneity matrix

	0			
	1	0		
	3	2	0	
	1	2	4	0

3. Determine parameter clustering

$$\mathbb{U}_1 = \{U_1, U_2, U_4\}$$

$$\mathbb{U}_2 = \{U_3\}$$

$$\mathbb{U}_3 = \{U_5\}$$

$$\mathbb{C}_1 = \{C_{12}, C_{13}, C_{45}\}$$

$$\mathbb{C}_2 = \{C_{23}\}$$

[Zamir et al., Taskonomy: Disentangling Task Transfer Learning. CVPR 2018]

[Liang et al., HighMMT: Quantifying Modality & Interaction Heterogeneity for High-Modality Learning. TMLR 2022]

Dynamic Early Fusion

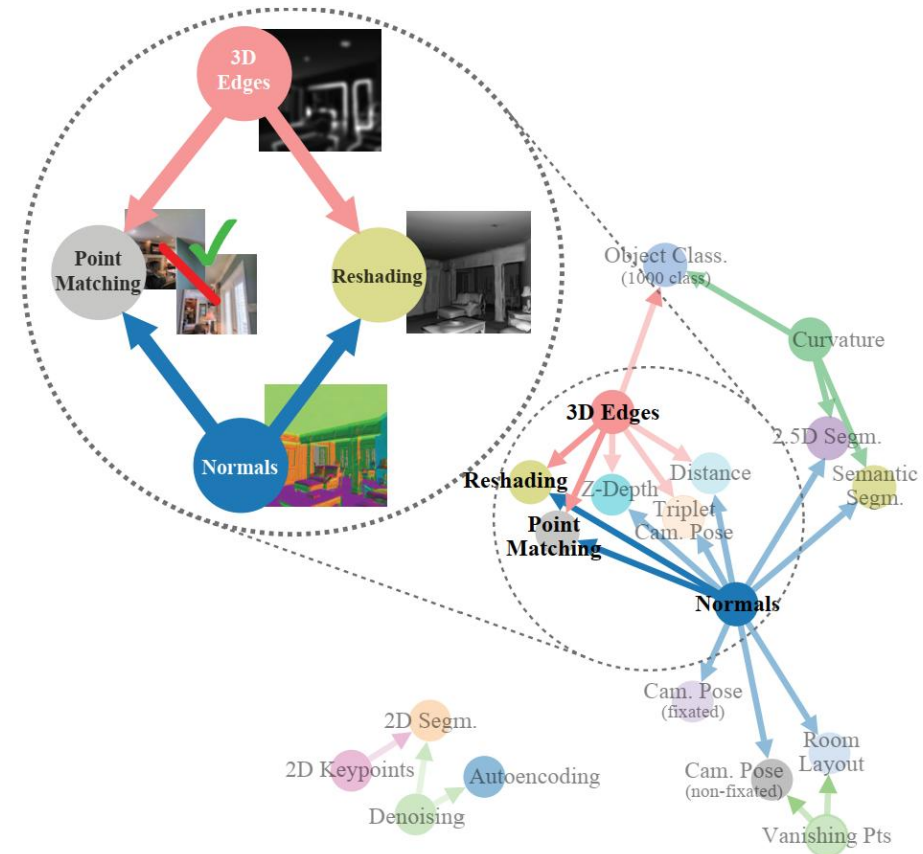


Figure 1: A sample task structure discovered by the computational task taxonomy (*taskonomy*). It found that, for instance, by combining the learned features of a surface normal estimator and occlusion edge detector, good networks for reshading and point matching can be rapidly trained with little labeled data.

Dynamic Early Fusion

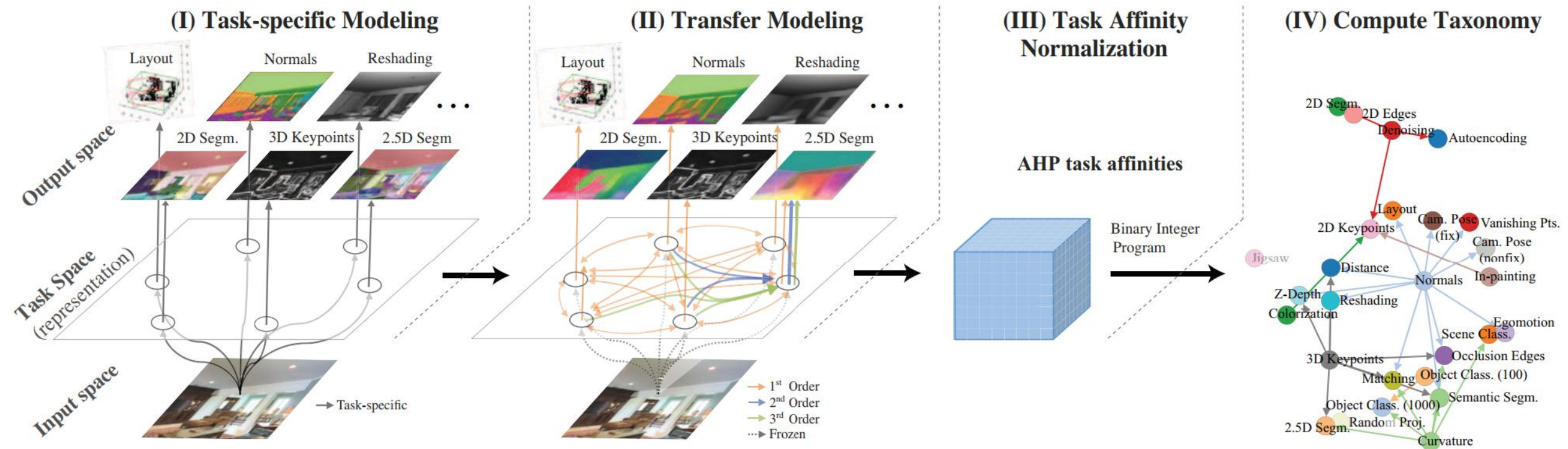


Figure 2: **Computational modeling of task relations and creating the taxonomy.** From left to right: I. Train task-specific networks. II. Train (first order and higher) transfer functions among tasks in a latent space. III. Get normalized transfer affinities using AHP (Analytic Hierarchy Process). IV. Find global transfer taxonomy using BIP (Binary Integer Program).

Dynamic Early Fusion

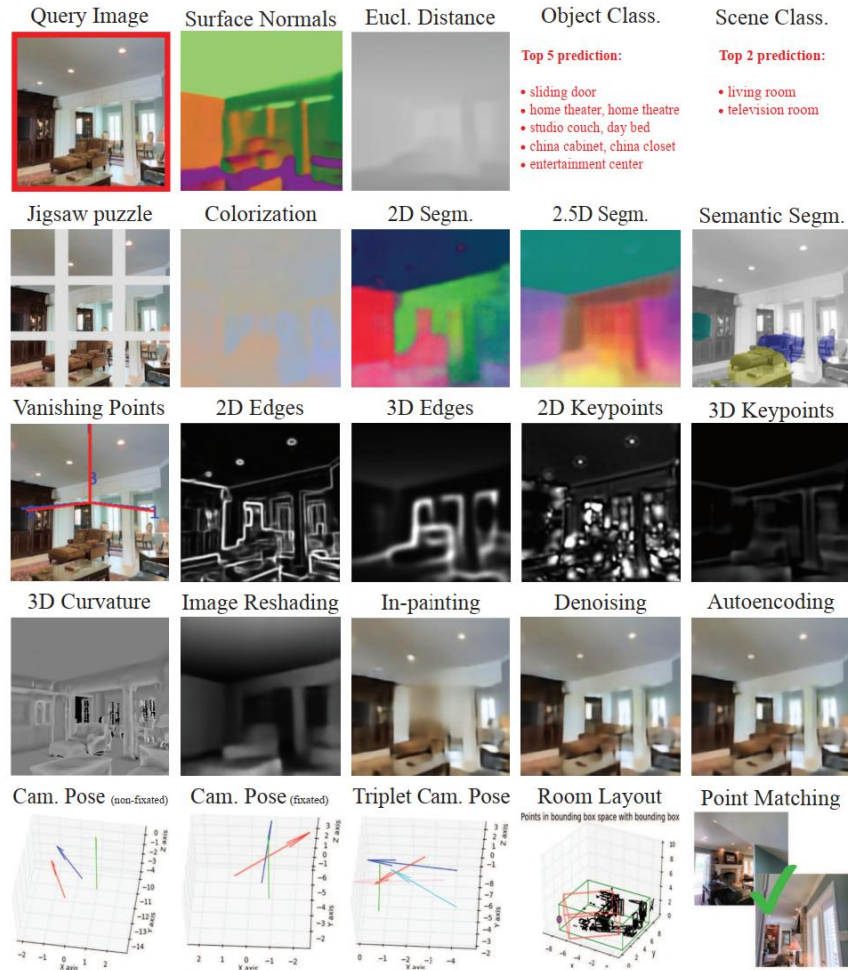


Figure 3: Task Dictionary. Outputs of 24 (of 26) task-specific networks for a query (top left). See results of applying frame-wise on a video [here](#).

Dynamic Early Fusion

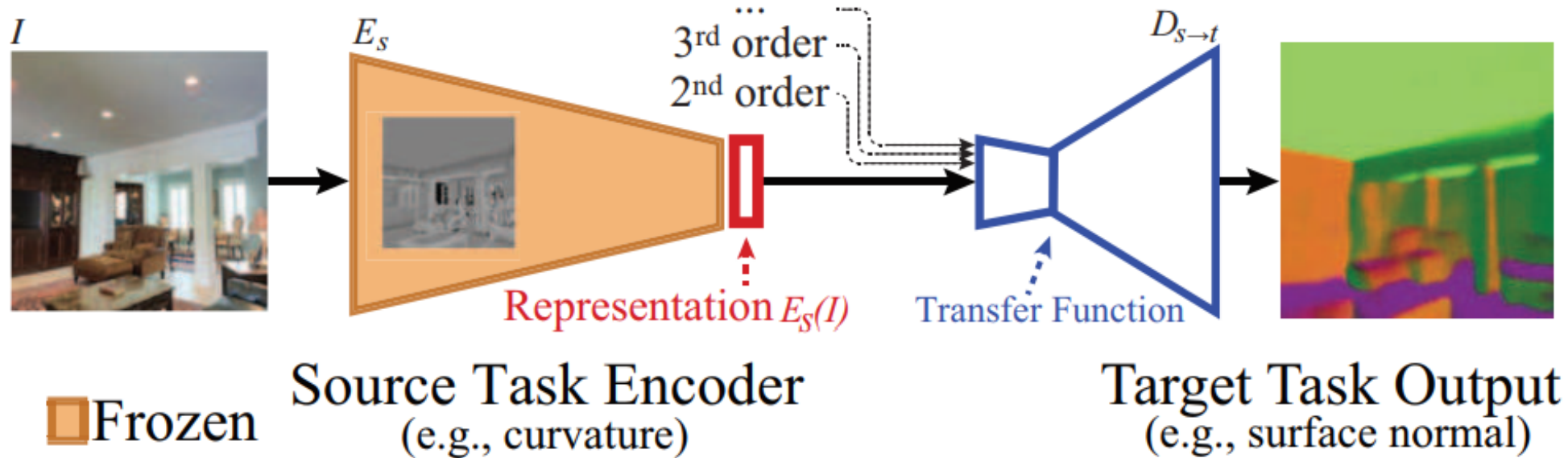


Figure 4: Transfer Function. A small readout function is trained to map representations of source task's frozen encoder to target task's labels. If $\text{order} > 1$, transfer function receives representations from multiple sources.

Dynamic Early Fusion

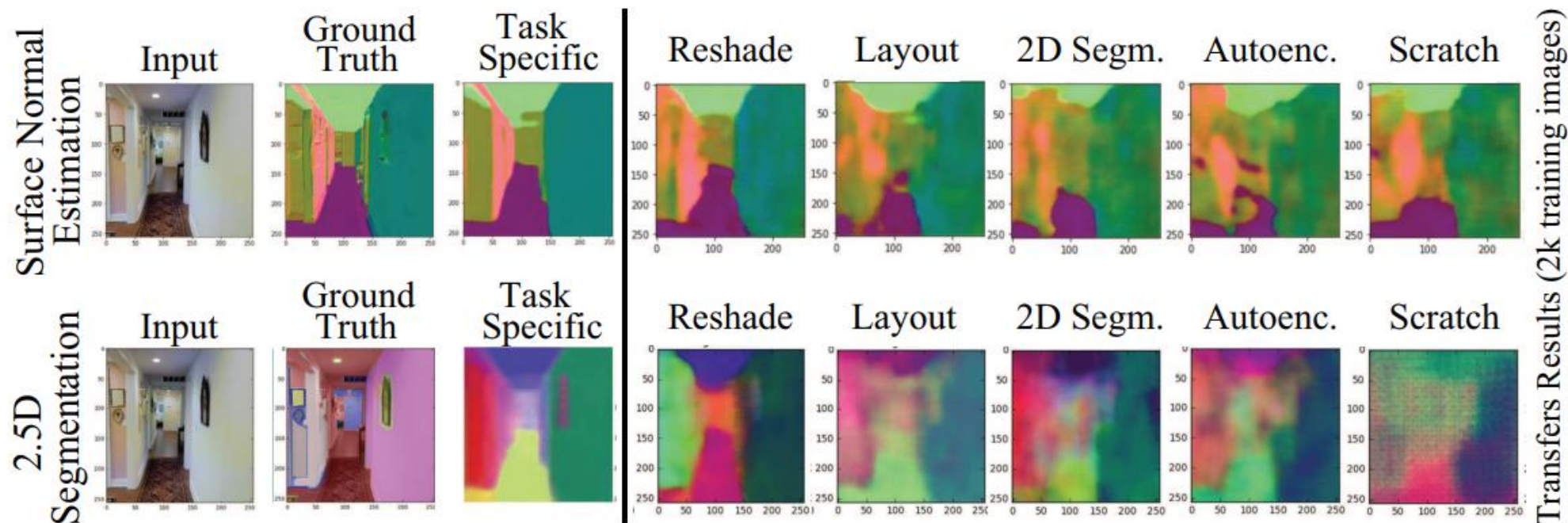


Figure 5: Transfer results to normals and 2.5D Segmentation from 5 different source tasks. The spread in transferability among sources is apparent. “Scratch” was trained from scratch without transfer learning.

Dynamic Early Fusion

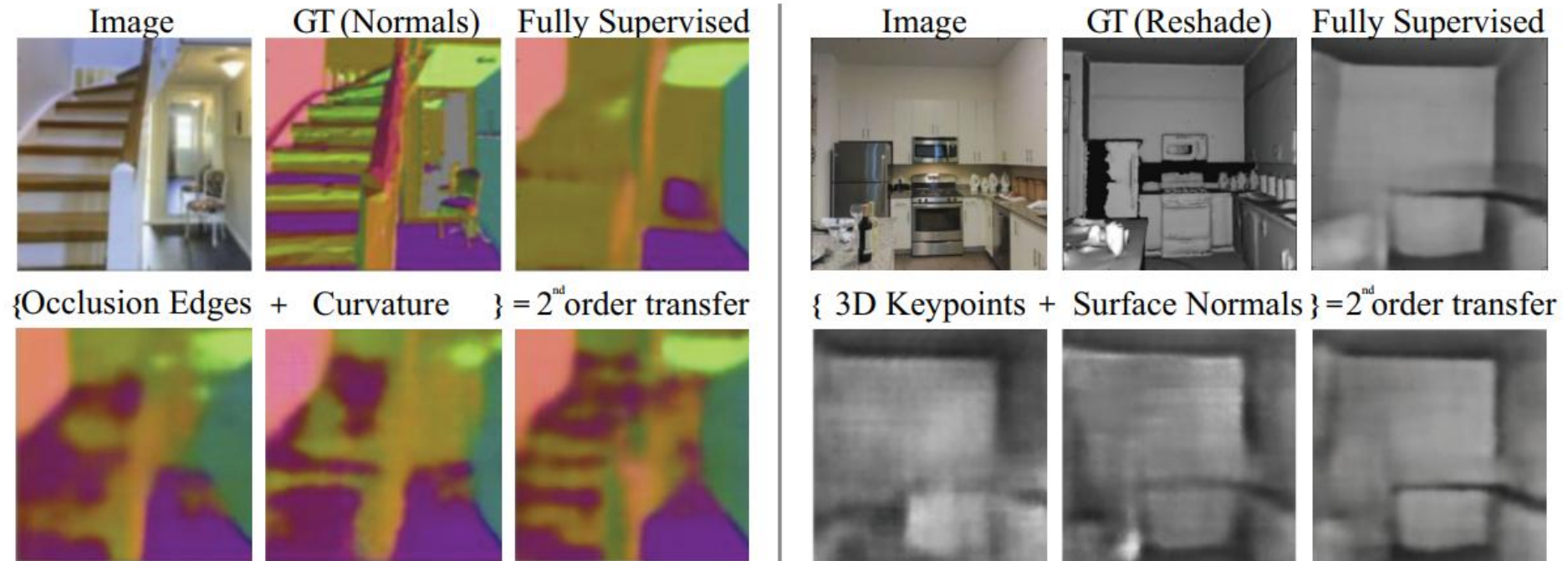


Figure 6: Higher-Order Transfers. Representations can contain complementary information. E.g. by transferring simultaneously from 3D Edges and Curvature individual stairs were brought out. See our publicly available interactive [transfer visualization page](#) for more examples.

Dynamic Early Fusion

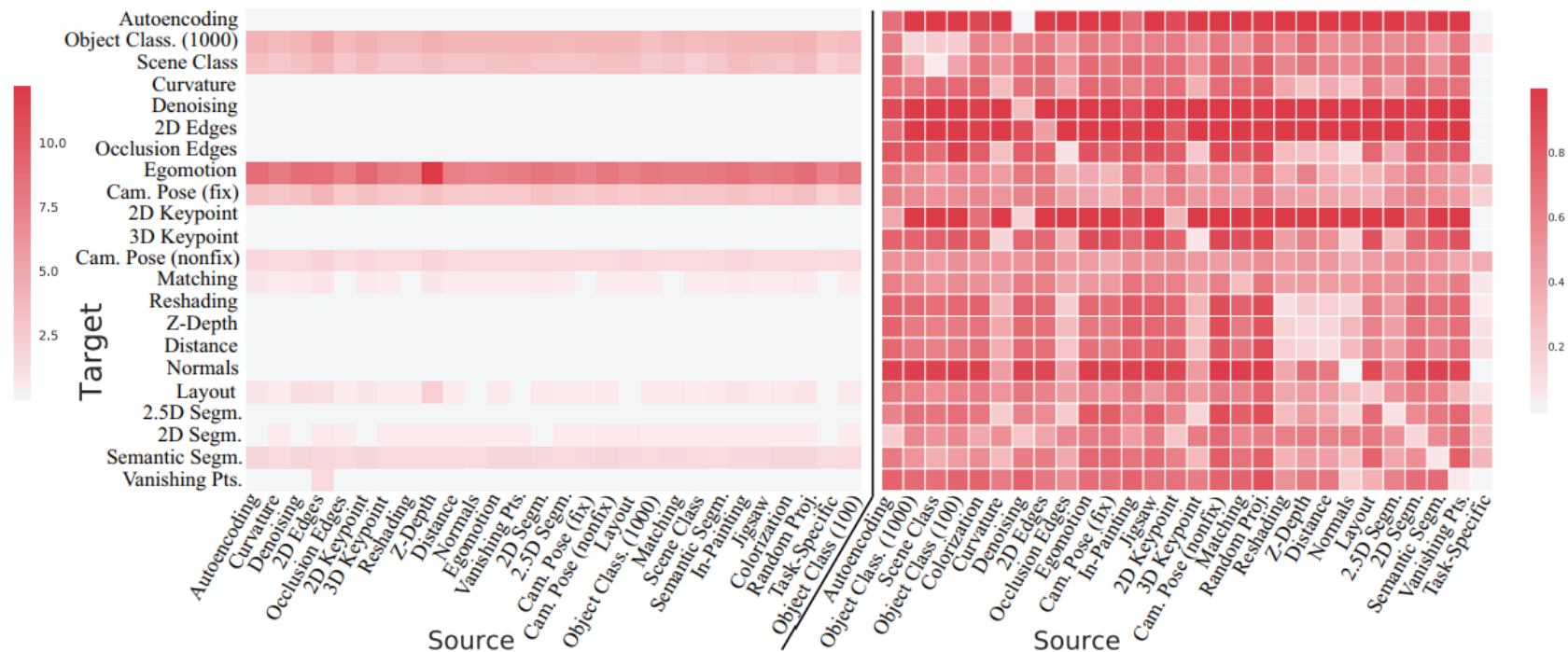


Figure 7: First-order task affinity matrix before (left) and after (right) Analytic Hierarchy Process (AHP) normalization. Lower means better transferred. For visualization, we use standard affinity-distance method $dist = e^{-\beta \cdot P}$ (where $\beta = 20$ and e is element-wise matrix exponential). See [supplementary material](#) for the full matrix with higher-order transfers.

Dynamic Early Fusion

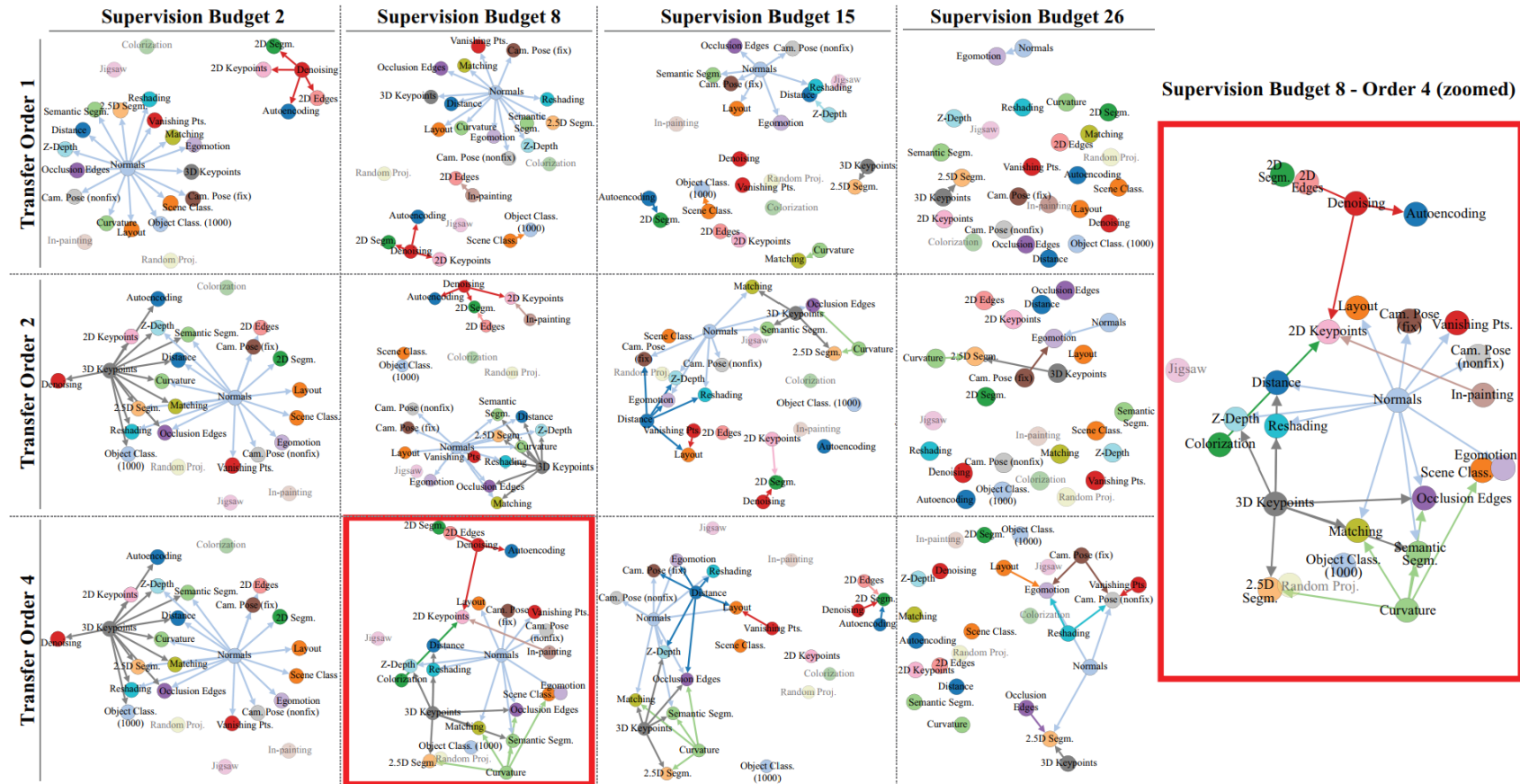


Figure 8: **Computed taxonomies** for solving 22 tasks given various supervision budgets (x-axes), and maximum allowed transfer orders (y-axes). One is magnified for better visibility. Nodes with incoming edges are target tasks, and the number of their incoming edges is the order of their chosen transfer function. Still transferring to some targets when the budget is 26 (full budget) means certain transfers started performing better than their fully supervised task-specific counterpart. See the interactive [solver website](#) for color coding of the nodes by *Gain* and *Quality* metrics. Dimmed nodes are the source-only tasks, and thus, only participate in the taxonomy if found worthwhile by the BIP optimization to be one of the sources.

Dynamic Early Fusion

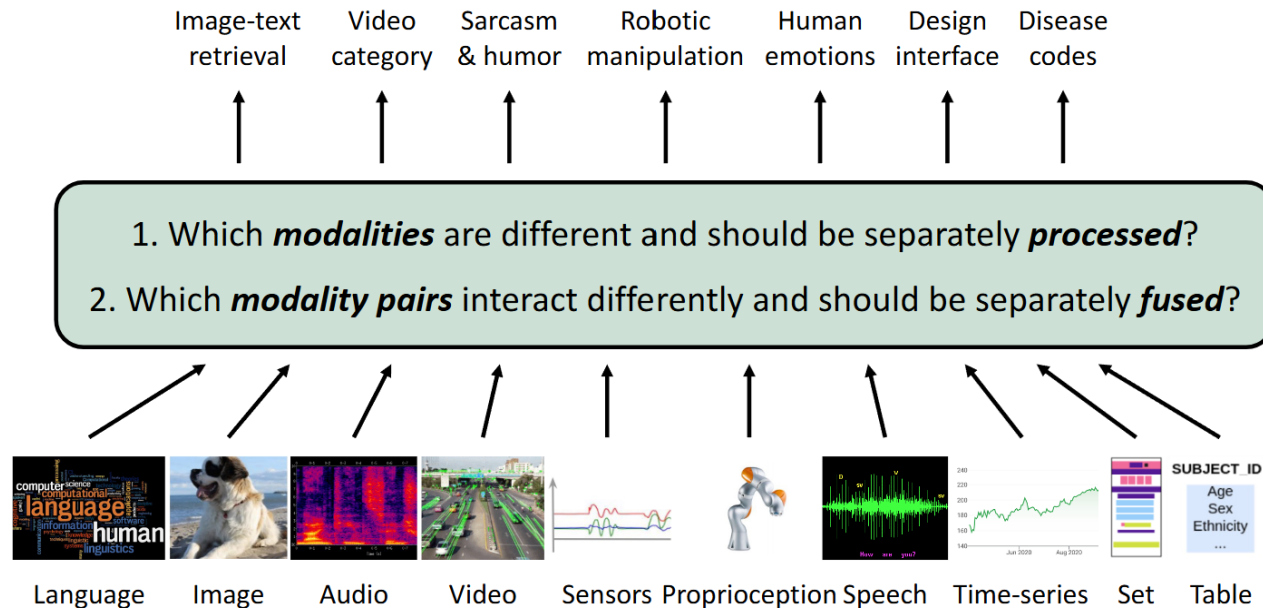
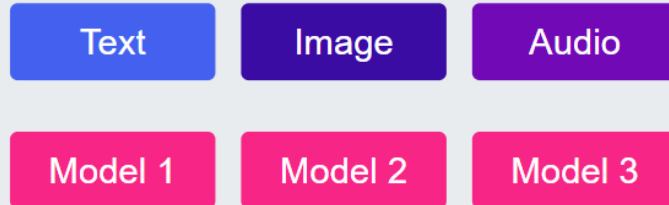


Figure 1: **Heterogeneity quantification**: Efficiently learning from many modalities requires measuring (1) *modality heterogeneity*: which modalities are different and should be separately processed, and (2) *interaction heterogeneity*: which modality pairs interact differently and should be separately fused. HIGHMMT uses these measurements to dynamically group parameters balancing performance and efficiency.

The High-Modality Challenge

Traditional Approach

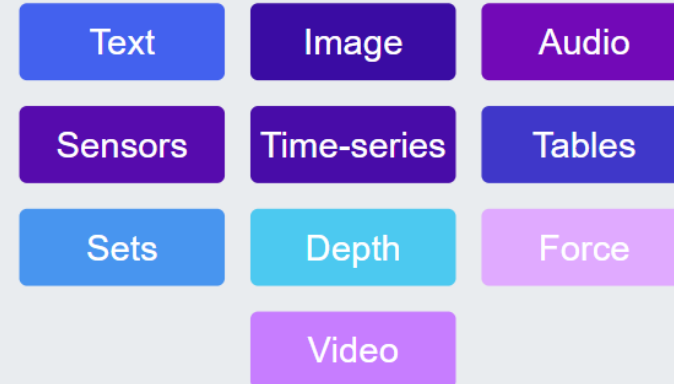


Parameters grow linearly with modalities

Key Challenges:

1. How to determine which modalities should be processed separately?
2. How to determine which modality pairs should be fused separately?

HighMMT Approach



Shared Model

Parameter sharing based on
modality & interaction heterogeneity

Dynamic Early Fusion

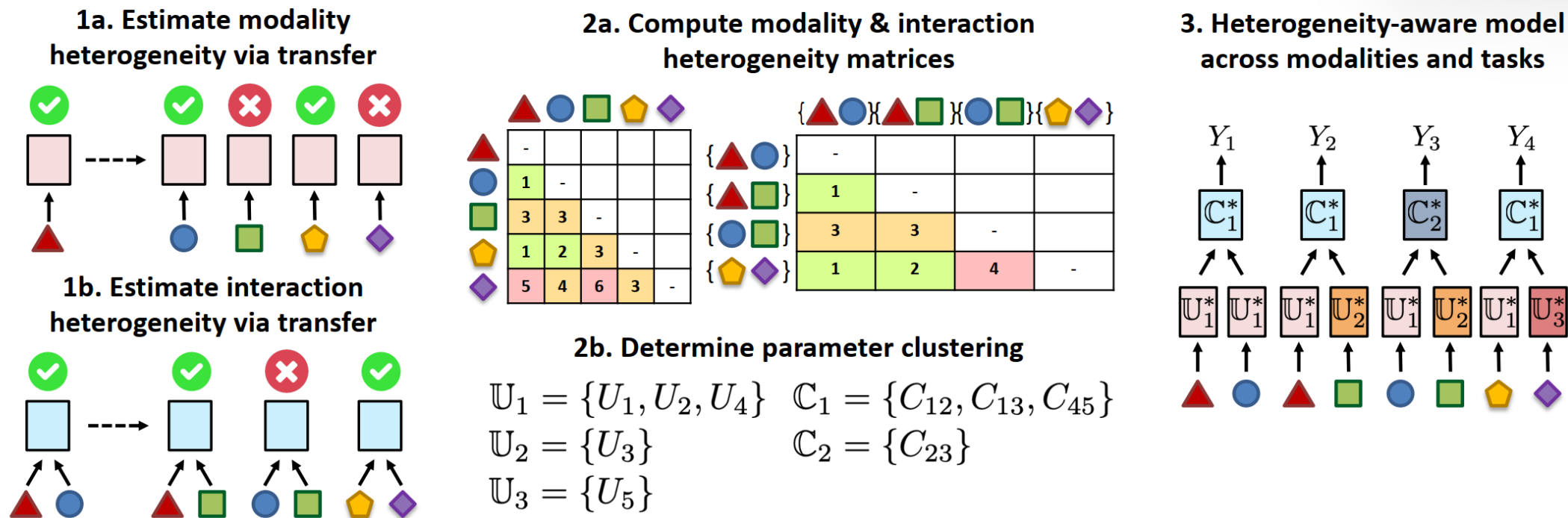


Figure 2: **HighMMT workflow:** (1) We estimate modality and interaction heterogeneity via modality transfer to determine which modalities should be processed and fused differently. (2) Using the inferred heterogeneity, we determine the optimal grouping of parameters balancing both total performance and parameter efficiency, which (3) informs our design of a heterogeneity-aware model with dynamic parameter sharing across many modalities and tasks. HIGHMMT enables statistical strength sharing, efficiency, and generalization to new modalities and tasks.

Dynamic Early Fusion

- Modality Transfer Loss:

$$T(X_1 \rightarrow X_2; Y) = \mathcal{L}_{1 \rightarrow 2}^* - \mathcal{L}_2^*$$

This equation measures the difficulty of transferring a model trained on modality X_1 to modality X_2 for task Y . $\mathcal{L}_{1 \rightarrow 2}^*$ is the loss after fine-tuning a model initialized with parameters from X_1 on X_2 , while $\mathcal{L}_2^*$ is the baseline loss of a model trained directly on X_2 .

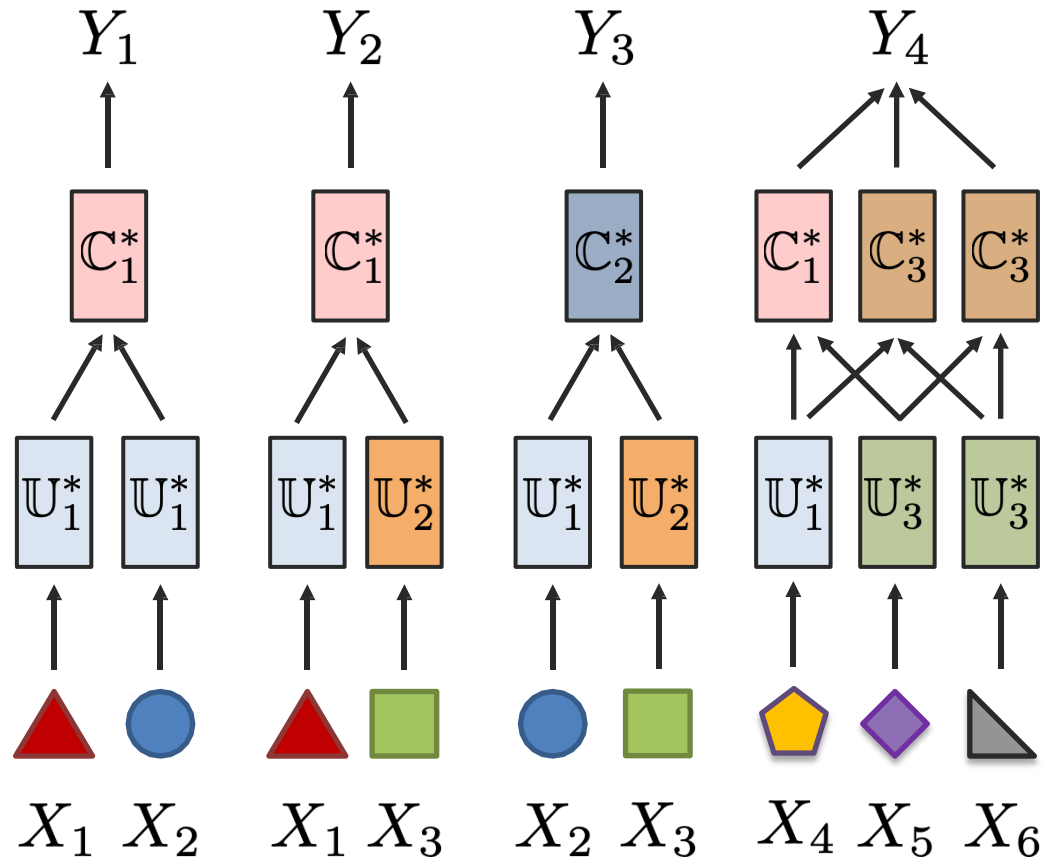
- Modality Heterogeneity Measure:

$$d(X_1; X_2) = T(X_1 \rightarrow X_2; Y)_{\geq 0} + T(X_2 \rightarrow X_1; Y)_{\geq 0}$$

This equation aggregates the non-negative transfer difficulties in both directions between two modalities to quantify their heterogeneity.

Heterogeneity-aware Fusion

Information transfer, transfer learning perspective



Improving Optimization

Kinetics dataset



(a) headbanging



(c) shaking hands



(e) robot dancing



(g) riding a bike



Adding more modalities should always help?

Modalities: RGB (video clips)

A (Audio features)

OF (optical flow - motion)

Dataset	Multi-modal	V@1	Best Uni	V@1	Drop
Kinetics	A + RGB	71.4	RGB	72.6	-1.2
	RGB + OF	71.3	RGB	72.6	-1.3
	A + OF	58.3	OF	62.1	-3.8
	A + RGB + OF	70.0	RGB	72.6	-2.6

But sometimes multimodal doesn't help! **Why?**

[Wang et al., What Makes Training Multi-modal Classification Networks Hard? CVPR 2020]

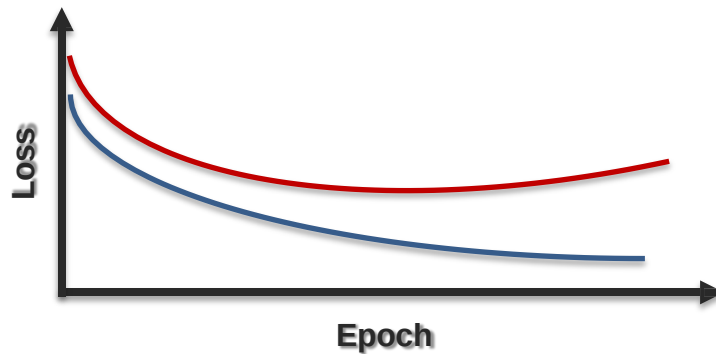
[Wu et al., Characterizing and Overcoming the Greedy Nature of Learning in Multi-modal Deep Neural Networks. ICML 2022]

Improving Optimization

Relevance heterogeneity

2 explanations for drop in performance:

1. Multimodal networks are more prone to overfitting due to **increased complexity**
2. Different modalities overfit and generalize at **different rates**



Key idea 1: compute overfitting-to-generalization ratio (OGR)



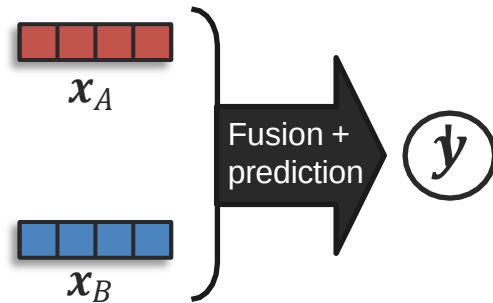
Gap between training and valid loss

OGR wrt each modality tells us
how much to train that modality

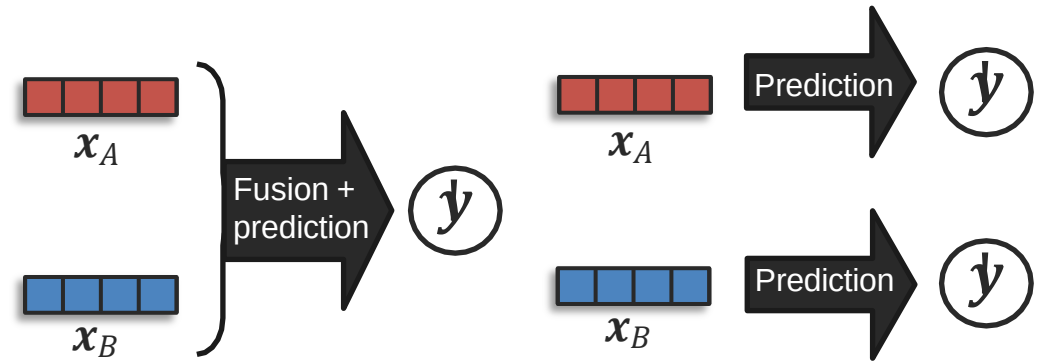
Improving Optimization

Relevance heterogeneity

Conventional approach



Proposed approach



Key idea 2: Simultaneously train unimodal networks to estimate OGR wrt each modality



Reweight multimodal loss using unimodal OGR values



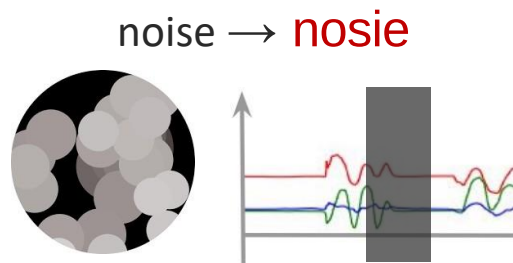
Allows to better balance generalization & overfitting rate of different modalities

[Wang et al., What Makes Training Multi-modal Classification Networks Hard? CVPR 2020]

[Wu et al., Characterizing and Overcoming the Greedy Nature of Learning in Multi-modal Deep Neural Networks. ICML 2022]

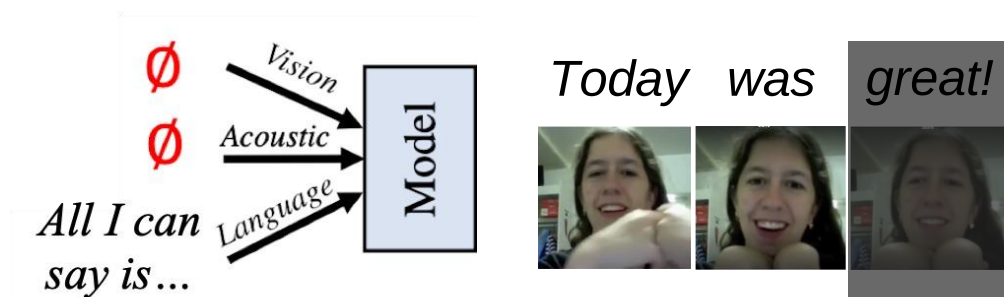
Heterogeneity in Noise: Studying Robustness

Noise within Modality



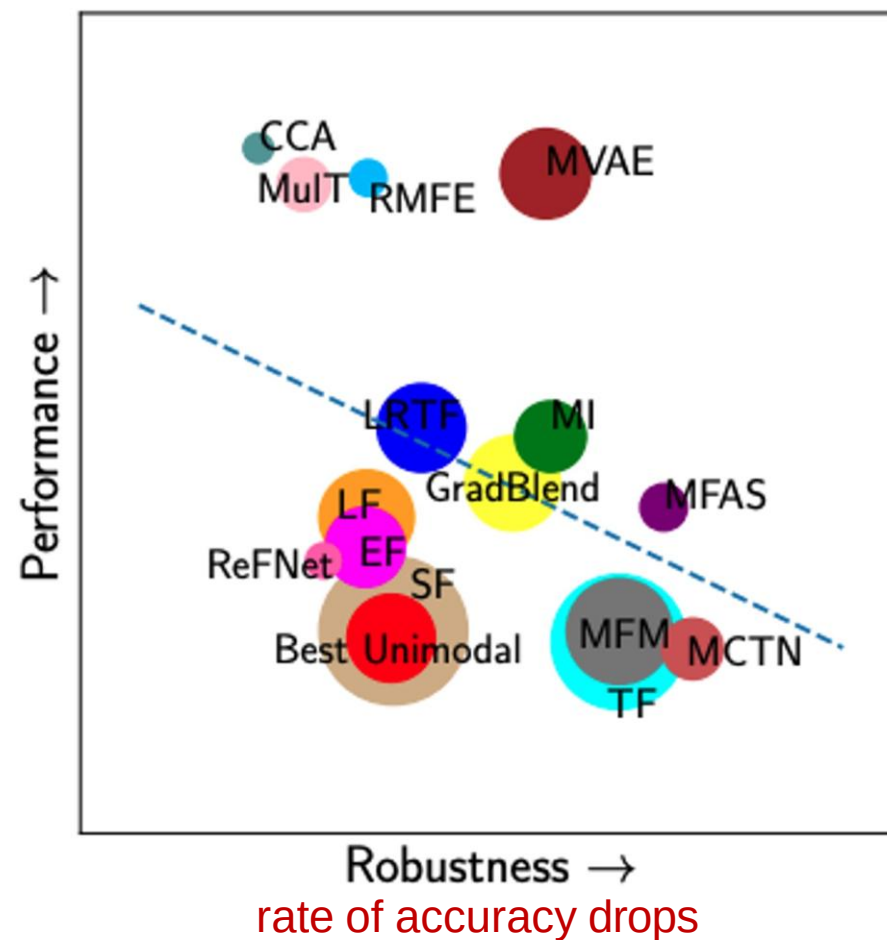
[Belinkov & Bisk, 2018; Subramaniam et al., 2009; Boyat & Joshi, 2015]

Missing Modalities



[Zadeh et al., 2020]

Strong tradeoffs between performance and robustness

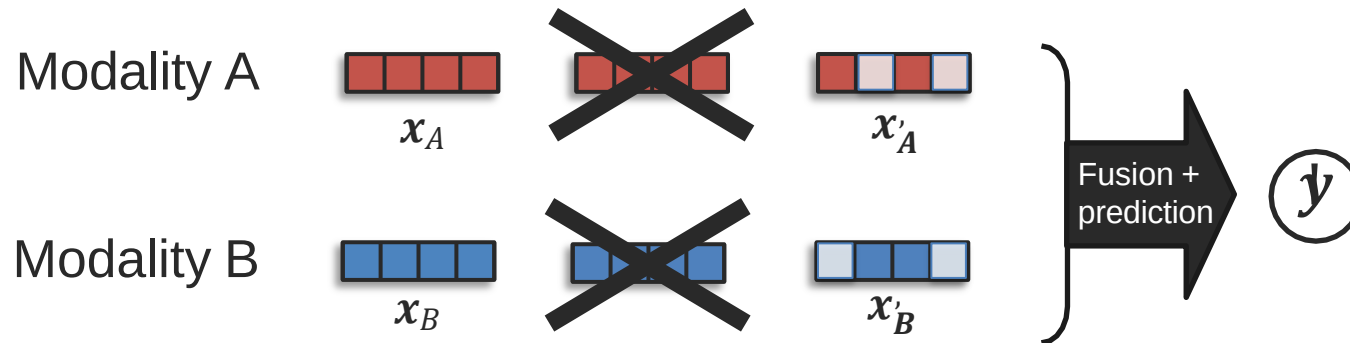


[Liang et al., MultiBench: Multiscale Benchmarks for Multimodal Representation Learning. NeurIPS 2021]

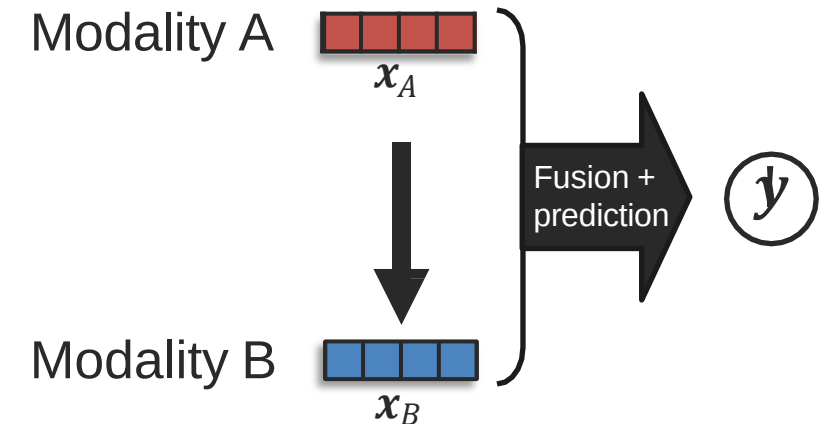
Heterogeneity in Noise: Studying Robustness

Several approaches towards more robust models

Robust data + training



Infer missing modalities



Translation model
Joint probabilistic model

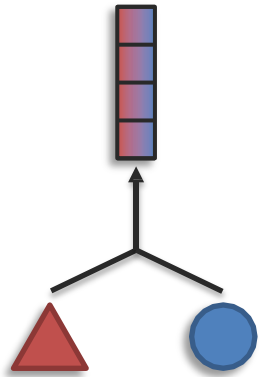
[Ngiam et al., Multimodal Deep Learning. ICML 2011]

[Srivastava and Salakhutdinov, Multimodal Learning with Deep Boltzmann Machines. JMLR 2014]

[Tran et al., Missing Modalities Imputation via Cascaded Residual Autoencoder. CVPR 2017]

[Pham et al., Found in Translation: Learning Robust Joint Representations by Cyclic Translations Between Modalities. AAAI 2019]

Sub-Challenge 1a: Representation Fusion



Definition: Learn a joint representation that models cross-modal interactions between individual elements of different modalities

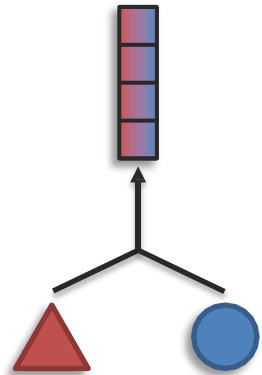


Challenge 1: Representation

Definition: Learning representations that reflect cross-modal interactions between individual elements, across different modalities

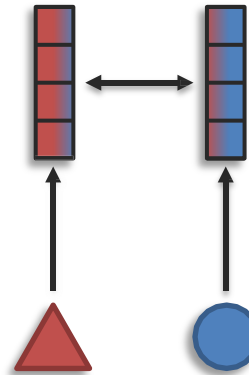
Sub-challenges:

Fusion



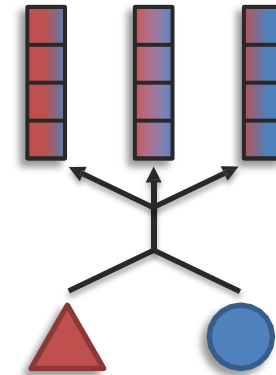
modalities $>$ # representations

Coordination



modalities $=$ # representations

Fission



modalities $<$ # representations

Next week